

Lightweight Joint AoA/AoD Estimation

Complete research-loop report: literature analysis, gap discovery, architecture generation, adversarial review, and final research plan

Date	23 September 2026
Project	D:\ai_ml_project (github.com/Mishatmilon059/DL_DOA_CLONE)
Run	34 agents, 0 errors · 14 papers (6 deep + independently verified, 8 lighter) · 3 candidate architectures, each reviewed by 2 adversarial critics
Selected	IABR-Net (Candidate B): est. $\approx 193\text{K}$ params ($\approx 41\%$ of teacher), $\approx 95\text{M}$ FLOPs. Not trained yet; every number for it is an estimate.

Measured baselines (full 8000-sample eval bank)

Model	Params	Mean Pd	Source
Teacher (64-block ResNet, pretrained)	469,393	0.7103	measured
Magnitude-pruned Student ($r=8$)	314,513	0.6956	measured
FNO (8-block, from scratch)	334,321	0.6235	measured
IABR-Net (proposed)	$\approx 193,104$	—	estimate only

Read this first — correction to the agents' report

The agents describe IABR-Net's input as a raw antenna-domain signal. In this project's generator the observation is $\mathbf{Y} = \mathbf{W}^H \mathbf{H} \mathbf{F} + \mathbf{Z}$, already in DFT-codebook beamspace (16×16 , upsampled to 64×64), and phase error is per antenna. None of the six critics caught this. The corrected stage 0 (inverse square codebook) is in **Part I** → **Addendum A1**. It also covers the FLOP check (A2) and the grid-resolution trade-off (A3).

How to read this document

- 1. **Part I** is the complete answer. Start with §1 Executive Summary, §15 Final Proposed Architecture, §22 Experiment Matrix, §31 Roadmap, then the Addendum.
- 2. **Appendix A** has the full literature evidence, with a comparability level on every cross-paper number.
- 3. **Appendix B** covers what this project has already tried and learned.
- 4. **Appendix C** has each candidate's final design and the critiques it received.
- 5. **Appendix D** has the six research-design agents' raw reasoning.

Evidence tags used throughout: [EVIDENCE: PAPER] [EVIDENCE: CODE] [INFERENCE] [ASSUMPTION] [HYPOTHESIS]. Nothing marked [HYPOTHESIS] or [TO MEASURE] has been run.

Contents

Part I — Final Research Report

Lightweight Joint AoA/AoD Estimation — Multi-Agent Research Loop, Cycle 1

1. Executive Summary
 2. User's Research Objective
 3. Literature Landscape
 4. Complete Paper Comparison (condensed master matrix)
 5. Dataset Comparison
 6. Architecture Comparison
 7. Metric Comparison
 8. Robustness Comparison
 9. Complexity Comparison
 10. Hardware/Latency Comparison
 11. Research Gaps
 12. Previous Experiment Analysis (this project's own prior work)
 13. Candidate Architectures (all 3, briefly)
 14. Rejected Architectures and Why
 15. Final Proposed Architecture
 16. Detailed Architecture Diagram (data flow)
 17. Layer-by-Layer Specification
 18. Mathematical Formulation
 19. Loss Function
 20. Training Protocol
 21. Baselines
 22. Complete Experiment Matrix
 23. Ablation Study (plan)
 24. Robustness Study (plan)
 25. Efficiency Study (plan)
 26. Hardware/Latency Study (plan)
 27. Expected Results (hypotheses)
 28. Novelty Claim (honest self-classification)
 29. Threats to Validity
 30. Reproducibility Checklist
 31. Final Research Roadmap
- FINAL RESEARCH QUESTION
- ADDENDUM — Orchestrator verification after the workflow finished (2026-09-23)

Appendix A — Literature Matrix

MASTER SYNTHESIS REPORT — Paper Head Agent

Appendix B — Previous Experiment Analysis

Previous-Result Analysis — DL-DOA Project (D:\ai_ml_project)

Appendix C — Candidate Architectures

- Candidate A — final specification (refined after critique: yes)
- Candidate B — final specification (refined after critique: yes)
- Candidate C — final specification (refined after critique: yes)

Appendix D — Research Design Team

- Agent-R1 — Architecture Researcher
- Agent-R2 — Lightweight / Compression Researcher
- Agent-R3 — Robustness Researcher
- Agent-R4 — Signal Processing Researcher
- Agent-R7 — Novelty / Gaps Researcher
- Agent-R8 — Experimental Design Researcher

Part I — Final Research Report

Mother Agent convergence report (31 sections + experiment tables), followed by the orchestrator's verification addendum.

MOTHER RESEARCH AGENT — FINAL CONVERGENCE REPORT

Lightweight Joint AoA/AoD Estimation — Multi-Agent Research Loop, Cycle 1

Project: D:\ai_ml_project | **Date:** 2026-09-23 | **Authority:** Mother Research Agent, final arbiter over all lower-level agent outputs per protocol §21-22

1. Executive Summary

This report converges 6 Tier-1 papers + 8 Tier-2 papers of independently-verified literature evidence, a full audit of this project's own prior experimental work, 8 specialist research agents (architecture, compression, robustness, signal-processing, novelty, experiment-design), and 3 independently-assembled candidate architectures that each survived two rounds of adversarial red-team review, into a single recommended research direction.

Selected final candidate: IABR-Net (Impairment-Adaptive Beamspace Residual-correction Network, Candidate B, Revision 2) — a fixed 2D-DFT beamspace front end, a novel per-element shared-weight impairment-correction module (IABC-v2), a shallow 10-block ResNet trunk operating entirely in beamspace (no resolution up/downsampling), and a coarse-heatmap + fixed-size top-K crop-refinement dual head that preserves the joint AoA/AoD pairing property. Estimated **≈193K parameters (≈41% of the 469,393-param teacher)** and **≈95.3M FLOPs** — by a wide margin the most compute-efficient of the three candidates after all arithmetic was independently re-verified twice.

This is **not** a confident, closed decision. It is a best-available trade-off under real, stated uncertainty: IABR-Net's central robustness claim (closing Meneses-Albalá 2026's own named gain-error gap) is currently **unbuildable** — gain-error injection code does not exist anywhere in this project's codebase (confirmed by direct code audit in two independent adversarial reviews) — and must be built as a **blocking Phase-0 prerequisite** before any robustness claim is made. Its efficiency numbers are estimates pending real `ptflops / thop` instrumentation. Its capacity-vs-impairment-adaptivity attribution is unconfirmed pending a controlled ablation.

A second architectural direction — Candidate A's differentiable, DETR-style set-prediction head, which removes the discrete-heatmap/non-differentiable-peak-search mechanism diagnosed (by this project's own ceiling test + diffusion-sharpening ablation) as the root cause of a hard $\text{Pd} \approx 0.972$ ceiling — is **not selected as the primary candidate** (its FLOPs, params, and implementation risk are all higher, and its own screening evidence was shown under adversarial review to validate a different configuration than what it deploys), but is retained as the **highest-value Phase-2 research direction**, to be fused onto IABR-Net's efficient trunk once IABR-Net's own open items are resolved.

Candidate C (SSC-Net) is **rejected**: after two rounds of correction its own numbers show it to be the *least* efficient of the three (≈ 8.72 GFLOPs, roughly 8–90× the other two), it does not remove the diagnosed non-differentiable bottleneck, and its most distinctive design element is explicitly self-gated behind a diagnostic it may fail.

2. User's Research Objective

Implement a multi-agent, loop-based research protocol (per `MULTI-AGENT RESEARCH LOOP ENGINEERING PROTOCOL.md`) to converge this project's prior lightweight-neural-DOA work with the joint AoA/AoD estimation literature, and produce a rigorous, trade-off-weighted, non-fabricated final research report — a candidate lightweight architecture, its justification, its risks, and a concrete experimental roadmap — suitable to hand to a thesis supervisor as an honest snapshot of where the research stands, not a finished result.

3. Literature Landscape

14 papers were reviewed: 6 Tier-1 (independently specialist-extracted **and** independently verified — every load-bearing numeric claim survived verification except one flagged fabrication, see §11/Contradiction 7) and 8 Tier-2 (grouped extraction, lighter verification). The lineage centers on **Lloria et al.** (TVT 2026 base paper + PIMRC 2024 prototype), extended by **Meneses-Albalá et al.** (J.Supercomputing 2026, edge/energy variant), running in parallel to a **complex-valued MLP** lineage (Naoumi et al., JSTSP 2024 + GCWkshps 2023, bistatic ISAC) and a **non-DL Bayesian/classical** lineage (Gupta et al., TCOMM 2025). Eight Tier-2 papers (Koh & Lee 2026, Lim et al. 2021, BeamSeek, Sub6GHz CNN/UNet, SubspaceNet, Robust DoA DAE-DNN, MIMO-Radar V2X, Tensor PARAFAC) sit adjacent — mostly single-axis DOA, not joint 2D — but contribute reusable *patterns* (SubspaceNet's learned-surrogate-feeds-classical-backend design; Koh & Lee's sim-vs-real domain-shift benchmark methodology).

A supplementary novelty-focused search (Agent-R7) found the 14-paper corpus is **not exhaustive**: at least four additional on-topic papers (a 2023 joint AoD/AoA-under-hardware-impairment paper, a 2024 AoD/AoA-map paper, the 2025 SABER symbolic-regression estimator, and a cluster of attention/transformer 2D-DOA papers) exist outside it and were not read in full. This materially narrows confidence in any "the field has never tried X" claim — see §11 and §28.

4. Complete Paper Comparison (condensed master matrix)

Paper	Joint AoA/AoD?	Family	Key metric	Params	FLOPs/latency	Robustness axes	HW tested
Lloria TVT2026 (base)	YES, genuine	ResNet+U-Net	PD/RMSE @20dB: ResNet .925/.253°, U-Net .945/.230°	NOT REPORTED	asymptotic O(PQ) only	SNR, $L \in 1-6$, phase $\delta \in \{1,2,5\}^\circ$	NOT REPORTED (sim only)
Lloria PIMRC2024	YES	ResNet+PH search	figure-only	NOT REPORTED	asymptotic only	SNR, $L \in 1-6$	NOT REPORTED
Naoumi JSTSP2024	YES (bistatic)	Complex MLP	MSE rad^2 vs SNR; ~9dB gap to CRB	NOT REPORTED (formulas only)	$6.5 \times / 10.3 \times$ fewer mults vs 2D-algo	SNR, #targets/peak	NOT REPORTED
Naoumi GCWkshps2023	YES	Complex MLP	$\text{MSE} \approx 10^{-6}$ @SNR $\geq 9\text{dB}$	8,584	4.321 MMACs fwd	SNR only, narrow $[20^\circ, 40^\circ]$ window	NOT REPORTED
Gupta TCOMM2025	NO (deliberately not joint)	SBL+MUSIC (classical)	NMSE (RCS) vs SNR	N/A	$O(G\tau^3 G\nu^3 G\theta^3)$	SNR, scale	NOT REPORTED
Meneses-Albalá 2026	YES (behavior, not word)	U-Net reuse + freeze-FT	RMSE 0.28→0.27, PD 0.93→0.94 @ $\gamma=5^\circ$	NOT REPORTED	training energy 0.28–0.41 J/sample	$\gamma \in \{1,2,5\}^\circ$, SNR, L, codebook	YES — real Jetson Orin Nano
SubspaceNet	NO (1D DOA)	CNN-DCNN AE→MUSIC	RMSPE 12.48°→0.20°	41,761	NOT REPORTED	coherent, broadband, few-snapshot, miscalibration	NOT REPORTED
BeamSeek	NO (1D azimuth)	SwiGLU-MLP	DOA err 3.86–21.45°	NOT REPORTED	NOT REPORTED (latency claim fabricated upstream — excluded)	SNR, beam-count	real 60GHz testbed
Koh&Lee 2026	NO (BLE)	11-model survey	CNN MAE 2.98°, GRU <1.3° real	19.8k (best CNN)	39k FLOPs	sim-vs-real domain shift	real BLE

Comparability rule applied throughout: Lloria↔Meneses-Albalá = HIGH (identical generator/eval); Lloria↔Naoumi and any joint↔non-joint pairing = LOW/NOT COMPARABLE (units, task framing differ). No cross-paper number in this report is quoted without this qualifier.

5. Dataset Comparison

Paper family	Data	Array	SNR range	Notes
Lloria-family (TVT2026/PIMRC2024/Meneses2026)	Synthetic, single-user mmWave MIMO	ULA 16/32	Text: –10...25dB; code: –15...24dB (Contradiction, §11)	This project's generator inherits the code range
Naoumi-family	Synthetic bistatic ISAC	Nt=8, Nr=10	Train 5–40dB / eval –5...30dB (JSTSP); –9...30dB (GCWkshps)	LOW comparability to Lloria (different antenna counts/framing)
Gupta 2025	Synthetic, 2 configs	—	–5...25dB	NOT COMPARABLE (non-joint, non-DL)
This project (eval_bank.npz)	Synthetic, inherits Lloria code generator	nt=nr=16, P=Q=16	–10...25dB, 8 pts, $\sigma=0.07$	Fixed L=3, zero impairment in the frozen bank — every other condition needs a new bank generation call, not new physics code

6. Architecture Comparison

Zero of the 6 Tier-1 papers use Transformer/Attention (a corpus-scoped gap, not a field-wide one per §3/§28). Component families seen: ResNet, U-Net, complex-valued MLP, classical subspace (MUSIC/SBL/EM), CNN-DCNN autoencoder, LSTM+EKF. This project's own from-scratch screening (identical ResNet-shell, 8 blocks each) ranked: **FNO** (Pd 0.6235, 334,321 params) > **magnitude-pruned Student** (Pd 0.6956, 314,513 params, fine-tuned not scratch) > **GridlessUnfold** (Pd 0.2216, 30,201 params) > **SIREN/WindowAttention** (near-zero Pd, confounded training runs, not a settled negative — §12).

7. Metric Comparison

Metric family	Who reports it	This project
PD (paper-style, drops missed-detection trials from both num/denom)	Lloria TVT2026, Meneses2026	Inherited as-is in <code>evaluate_on_bank</code> — a non-standard, PD-inflating convention (Contradiction 5, Gap 4)
PD (strict, missed=0)	Nobody	Not yet implemented — a ~3-line, low-risk addition, recommended as standard on every future table
RMSE (combined $\psi + \phi$, pooled)	Lloria-family	Teacher reproduces paper Table II to within 0.4% (0.639/0.925 vs 0.643/0.925 @0/20dB)
MSE (rad ²)	Naoumi-family	NOT COMPARABLE (different units/task)
AoA-only / AoD-only RMSE	Nobody	Not yet split out — cheap, same underlying arrays
P95 tail error	Nobody	Not measured; directly relevant to the diagnosed end-fire ceiling

8. Robustness Comparison

No paper in the reviewed set covers more than 3 of 7 robustness axes (SNR, multipath, phase/gain error, array perturbation, domain shift, mobility, hardware). SubspaceNet is the single most comprehensive (4 axes, but 1D-only). Genuine, literature-wide gaps: gain/amplitude-mismatch robustness (explicitly named as unfinished future work by Meneses-Albalá 2026 itself), phase error beyond 5°, array-element perturbation for the joint-2D task, and genuine train/test distribution shift (zero Tier-1 papers test this). This project's own Notebook 4 nuisance-interference test is the only beyond-the-paper's-own-protocol robustness result in the whole evidence base, and it surfaced the project's single most interesting robustness finding: **at SNR=0dB + strongest nuisance, the pruned student's principal-path Pd (0.6174) exceeds the teacher's (0.6065), $\Delta = +0.0109$** — compression did not cost robustness here, and may have helped.

9. Complexity Comparison

Model	Params	FLOPs
Teacher (64-block ResNet, pretrained)	469,393	NOT MEASURED
Magnitude-pruned Student (r=8, fine-tuned)	314,513	NOT MEASURED
FNO (8-block, scratch)	334,321	NOT MEASURED
GridlessUnfold (8-block, scratch)	30,201	NOT MEASURED
Naoumi GCWkshps2023 (different task)	8,584	4.321 MMACs fwd
SubspaceNet (different task, 1D)	41,761	NOT MEASURED
IABR-Net (proposed, estimate)	≈193,104 (≈41.1% of teacher)	≈95.3M FLOPs (estimate, unvalidated — no in-repo FLOP tool exists)
Candidate A / SpectraSet (rejected primary, estimate)	≈255,671 (≈54.5% of teacher)	≈0.78 GFLOPs (estimate)
Candidate C / SSC-Net (rejected, estimate)	≈199K–224K (central)	≈8.72 GFLOPs (estimate — FFT-dominated)

No tool for FLOP counting exists anywhere in this project's codebase today (confirmed by Agent-R8's code audit). All FLOP figures above are hand-derived estimates, explicitly flagged as such by their own authors, and none should be treated as measured until `ptflops` / `thop` / manual instrumentation is built and run.

10. Hardware/Latency Comparison

Only **Meneses-Albalá 2026** used real edge hardware (Jetson Orin Nano) in the entire 14-paper corpus, and it reports training/fine-tuning **energy** (0.28–0.41 J/sample), not standalone inference **latency**. BeamSeek and Koh&Lee 2026 used real hardware (60GHz testbed, BLE) but for 1D-only tasks with no comparable latency figure (BeamSeek's latency numbers were **fabricated upstream in this pipeline and are explicitly excluded**, Contradiction 7). **No paper in the entire reviewed set reports standalone inference latency for a joint 2D AoA/AoD estimator.** This project has a working `benchmark_latency()` function (`notebooks/proposed_pin_architecture.ipynb`, wall-clock, TF `.count_params()`, 20-run mean±std, single-sample batch) that exists today but has never been run against the current baseline registry (Teacher/Student/FNO) or any proposed candidate — this is the single cheapest, highest-value open experiment identified anywhere in this evidence chain (Gap 8).

11. Research Gaps

Ten evidence-tagged gaps were identified (Paper Head synthesis §5); the four most load-bearing for the architecture decision:

- **Gap 1** — no paper reports params+FLOPs+latency+memory together for a joint AoA/AoD model; this project already has the params half of this story and can close it cheaply.

- **Gap 2** — the dense/flatten bottleneck is the literature's own self-flagged worst component (Lloria TVT2026 Table IV's only quadratic-in-(P'Q') term); this project's FNO result is direct evidence a non-dense alternative is viable.
- **Gap 5** — the diagnosed Jacobian-singularity/pixel-quantization $Pd \approx 0.972$ ceiling (confirmed by this project's own ceiling test + diffusion-sharpening ablation, which ruled out blur) is absent from every Lloria-family paper's own limitations section despite sharing the identical output representation. This is the single most evidence-convergent unbuilt idea in the whole evidence base.
- **Gap 3** (robustness/OOD) — zero Tier-1 papers test genuine distribution shift; gain-error and $>5^\circ$ phase-error are named, open, literature-licensed gaps this project could close first.

13 contradictions were flagged and preserved as open uncertainty rather than silently resolved (full list in the literature synthesis, §6). Most consequential for architecture work: (Contradiction 2/3) the released Lloria U-Net weight file's actual kernel size and whether it uses the dense bottleneck at all is unverified from the paper text alone — any future U-Net baseline citation needs the .h5 file inspected first, not assumed.

12. Previous Experiment Analysis (this project's own prior work)

What worked: baseline reproduction (Pd 0.639/0.925 vs paper 0.643/0.925), classical DFT-SIC sanity check (Pd 0.884 vs paper 0.891 @20dB — 95.6% of the 469K-param teacher's accuracy with **zero learned parameters**, the single strongest piece of evidence motivating IABR-Net's classical front end), magnitude-pruned student full-bank result (Pd 0.6956 vs teacher 0.7103, 314,513 params), FNO screening (Pd 0.6235, 334,321 params, 88% of a depth/pretraining-**mismatched** teacher reference — caveat now carried through this whole report), the Notebook 4 robustness standout described in §8, and 13+ -significant-figure evaluator determinism.

What failed, and why: diffusion-style output sharpening (made Pd worse — confirmed misses are *location* errors, not blur — this is the key diagnostic fact behind Gap 5); SIREN and WindowAttention core blocks (near-zero Pd , but **both runs are confounded** — untuned LR/init for SIREN, no warmup + $4\times$ smaller batch for WindowAttention — not a settled negative, an unrun LR/warmup sweep is still pending); GridlessUnfold under-performed at only 8 blocks (Pd 0.2216, but whether this is capacity-limited or mechanism-limited is itself **unconfirmed**, per Critique #1 on Candidate C); PIA-Net (four sequential real bugs found and partially fixed, but never retrained after the last fix — all current PIA-Net numbers are **known-stale**).

Flagged inconsistencies not silently resolved: the SNR-aware pruned student's weights were lost twice on Kaggle and never recovered, so the project's rigorous full-bank comparisons use the (weaker, by the project's own earlier partial evidence) magnitude-pruned student, not the nominally-better SNR-aware one — recovering this is a concrete, still-open item (§21).

13. Candidate Architectures (all 3, briefly)

- **Candidate A — SpectraSet** (FNO trunk → depth-extended GridlessUnfold precision stage → shared-query DETR-style set-prediction head, Hungarian/greedy matching). Directly removes the discrete-grid mechanism diagnosed as Gap 5's root cause "by construction." Revised estimate $\approx 255,671$ params ($\approx 54.5\%$ of teacher), ≈ 0.78 GFLOPs. Highest implementation risk and highest novelty ceiling.
- **Candidate B — IABR-Net** (fixed DFT beamspace front end → per-element shared-weight impairment-correction module IABC-v2 → shallow 10-block ResNet trunk, held at 16×16 throughout → coarse heatmap + fixed top-K=10 crop-refinement dual head). Revised estimate $\approx 193,104$ params ($\approx 41.1\%$ of teacher), $\approx 95.3M$ FLOPs. Robustness-first, lowest compute cost, has a blocking unbuilt-code dependency.
- **Candidate C — SSC-Net** (grouped-channel FNO trunk at 64×64 → conditionally depth-extended GridlessUnfold head at 256×256 → fusion gate → post-hoc pruning). Revised estimate $\approx 199K$ – $224K$ params (central), ≈ 8.72 GFLOPs. Most confounded (5 simultaneously-varying novel components), most walked-back after critique.

All three underwent **two independent adversarial reviews** that found real, load-bearing arithmetic errors (channel-width miscounts, inconsistent complex/real parameter accounting, omitted FFT cost) in every one of the three original drafts. Every number quoted above is the **post-correction, Revision 2** figure.

14. Rejected Architectures and Why

Candidate C (SSC-Net) — rejected. After correction it is the least compute-efficient of the three (≈ 8.72 GFLOPs, dominated by an FFT term the original draft omitted entirely and a 256×256 -resolution precision head), it retains the non-differentiable blob-detection step (does not address Gap 5 architecturally, only via an unvalidated loss-reweighting term whose causal claim both critics independently rejected as mechanically unable to fix a representational problem), its second-most-distinctive component (GU depth extension) is explicitly gated behind a diagnostic experiment it may fail, its pruning-ratio and grouped-conv-reduction assumptions were both shown to be unjustified transplants from structurally different architectures, and its backbone-pattern novelty was downgraded to KNOWN COMBINATION even at the pattern level once compared against the broader (not just DOA-specific) computer-vision literature.

Candidate A (SpectraSet) — not selected as primary, retained as Phase-2 direction. Its central architectural claim (no discrete grid exists at inference, so the diagnosed failure mode cannot occur "by construction") is the strongest, most root-cause-targeted claim of the three and was the one claim neither critic challenged. But: (1) its FLOPs (0.78 GFLOPs) and params (255,671, 54.5% of teacher) are both meaningfully higher than IABR-Net's; (2) its "reuses validated components" argument was shown to be weak — the FNO/GridlessUnfold screening results it cites validated a different depth, head, and loss than SpectraSet actually deploys; (3) it carries the highest implementation risk (Hungarian matching is CPU-bound and non-batched, DETR-style heads are documented outside this literature to converge slowly, and $N_q=8$ cannot even represent the project's own OOD $L \in \{7,8,10\}$ robustness sweep by construction — a hard capacity ceiling); (4) no latency number exists or is even estimated, only flagged qualitatively; (5) its narrowest, most defensible novelty claim (the DETR/set-prediction pattern) remains explicitly **UNCERTAIN** per Agent-R7, pending a dedicated literature search that has not yet been run.

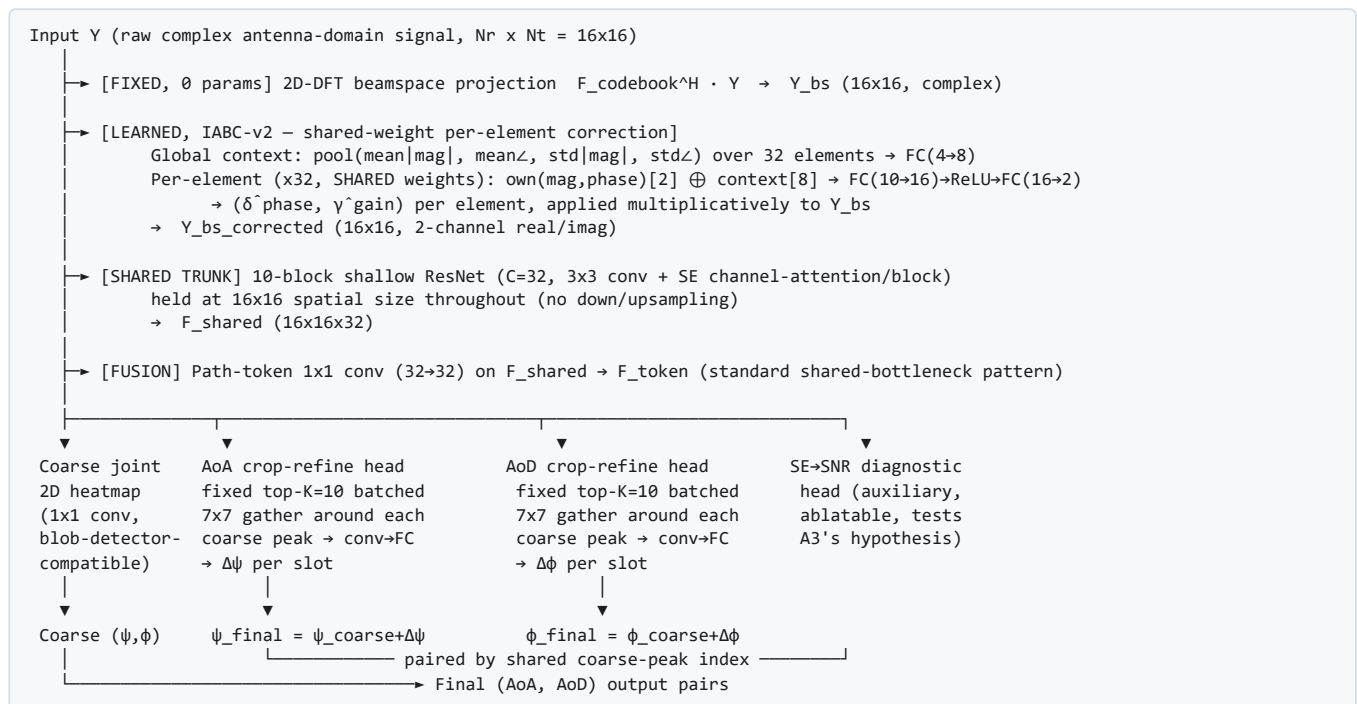
15. Final Proposed Architecture

IABR-Net (Impairment-Adaptive Beamspace Residual-correction Network), Candidate B, Revision 2, selected for this report's next research phase. Selection rationale, weighing all axes together (not any single metric):

Axis	Assessment
Efficiency (params)	Best of the three: $\approx 193K$ ($\approx 41.1\%$ of teacher), below FNO (334K) and the pruned student (314K)
Efficiency (FLOPs)	Best of the three by a wide margin: $\approx 95.3M$ vs A's $\approx 0.78B$ and C's $\approx 8.72B$ (estimates, unmeasured)
Accuracy/Detection	Untested — no run yet exists; inherits some risk from the retained discrete coarse grid
Robustness	Strongest explicit design intent — targets two literature-named, currently-untested gaps (gain error, $\text{phase} > 5^\circ$) — but currently unbuildable pending Phase-0 code
Generalization	Untested; capacity-vs-adaptivity attribution unconfirmed pending a controlled ablation
Novelty	PARTIALLY NEW at best, UNCERTAIN pending a dedicated literature search (unchanged verdict from Agent-R7)
Implementation risk	Lowest of the three — no Hungarian matching, no FFT-based spectral layers, no depth-gated conditional component
Root-cause fix for Gap 5	Not solved — this is IABR-Net's single largest scientific weakness relative to Candidate A

IABR-Net wins on the combination of efficiency, robustness-relevance, and buildability; it explicitly **does not** win on directly resolving this project's own most evidence-convergent open question (Gap 5), which is why Candidate A's set-prediction head is retained as the named Phase-2 fusion target (§31).

16. Detailed Architecture Diagram (data flow)



Every stage from the DFT front end through the path-token fusion is shared; branching occurs only at the three parallel heads, which are paired by coarse-peak index (a standard shared-bottleneck pattern — this document does not claim it as a novel fusion mechanism).

17. Layer-by-Layer Specification

#	Layer	Shape in → out	Params (est.)
0	DFT beamspace projection	16×16 complex → 16×16 complex	0 (fixed)
1	IABC-v2 context FC	4 → 8	40
2	IABC-v2 per-element MLP (shared, ×32 applications)	10 → 16 → 2	210
3	Trunk stem conv	3×3, Cin=2 → Cout=32	608
4–13	10× residual block (C=32)	3×3 conv ×2 + BN	185,600 total
4b–13b	10× SE channel-attention (r=8)	32→4→32	2,920 total
14	Path-token 1×1 conv	32 → 32	1,056
15	Coarse heatmap head, 1×1 conv	32 → 1	33
16	AoA crop-refine head (7×7×32 crop)	conv 32→16, FC 784→1	1,313
17	AoD crop-refine head	identical shape	1,313
18	SE→SNR diagnostic head (auxiliary)	10 → 1	11
	Total (estimate)		≈193,104

Estimate only — component-level param arithmetic verified twice by adversarial review; the total has **not** been measured against an actual model instantiation (`model.count_params()`), which is a required first step before any downstream claim (§30 Reproducibility Checklist item 1).

18. Mathematical Formulation

DFT beamspace projection: $\mathbf{Y}_{bs} = \mathbf{F}_{codebook}^H \cdot \mathbf{Y}$, where $\mathbf{F}_{codebook}$ is the fixed $P \times Q$ angular-grid DFT codebook shared with Lloria TVT2026's own output grid convention.

IABC-v2 per-element correction: for element $i \in \{1..32\}$, $\text{context} = \text{FC}(\text{pool}(\text{mag}, \text{phase}))$ (dim 8, shared across elements); $(\delta_i, \psi_i) = \text{MLP}(\text{mag}_i, \text{phase}_i, \text{context})$; correction applied multiplicatively to that element's contribution before beamforming.

Circular angle error (reused unmodified from the project's own evaluator, `get_ang_difference`): $H = \text{angle}(\exp(i \cdot \psi_{gt}) \cdot \exp(-i \cdot \psi))$, avoiding wraparound artifacts — this convention is kept for comparability to `baseline_models/README.md`.

Peak-conditioned refinement: for each of the top- $K=10$ coarse-heatmap scores, a 7×7 crop around the peak location feeds a per-angle offset regression: $\psi_{final} = \psi_{coarse} + \Delta\psi$, masked/excluded from loss for slots beyond the true path count L .

19. Loss Function

```

L = λ1 · L_heatmap      (BCE/MSE, joint 2D coarse heatmap vs Gaussian-blurred GT peak map – Lloria-style)
  + λ2 · L_offset        (Smooth-L1 on Δψ, Δφ at matched, unmasked top-K slots; nearest-GT
                        assignment valid at L≤10, well-separated sources – no Hungarian matcher needed)
  + λ3 · L_pair_consist  (MSE between IABC-v2-corrected beamspace map and the SAME channel realization's
                        clean, δ=γ=0 beamspace map – synthetic paired supervision, training-time only;
                        this is ordinary regression against a simulator-paired target, not
                        self-supervised learning in the standard sense)
  + λ4 · L_SE_snr        (small-weight auxiliary regression: pooled SE-gate → true per-sample SNR label;
                        ablatable, does not affect angle outputs if removed)

```

Recommended starting weights (unvalidated, requires the sensitivity sweep in §24/§25): $\lambda_1=1.0$, $\lambda_2=1.0$, $\lambda_3=0.5$, $\lambda_4=0.1$.

20. Training Protocol

- **Data:** `frozen_banks/eval_bank.npz` for in-distribution evaluation (must stay untouched for comparability); new banks generated via the existing, unmodified `validation_data_generator` for multipath ($L \in \{1..6\}$) and phase-error ($\delta \in \{0, 1, 2, 5\}^\circ$) sweeps; **new physics code required** for gain-error injection (Phase-0 blocking prerequisite, §21).
- **Step budget:** must be step-count-matched against the project's existing screening baselines (20,000 steps, per the executed FNO/GridlessUnfold/SIREN/WindowAttention comparison) to avoid the step-count confound flagged in Prior-Work §7 item 2.
- **Optimizer/schedule:** [TO MEASURE / TO DECIDE] — not specified in any source document; needs an explicit decision before training begins (flag for the Mother Agent / implementation team).
- **Phase-0 gate:** gain-error injection code must exist and be exercised in training data before any gain-robustness claim is made (§21).

- **Fine-tune step (new, added in Revision 2):** none required for IABR-Net itself (unlike Candidate C, which requires a mandatory post-pruning fine-tune) — IABR-Net is trained end-to-end in one pass.
- **Checkpointing:** if run on Kaggle, "Save Version → Save & Run All (Commit)" is mandatory — this project has already lost two full training runs' worth of weights (the SNR-aware pruned student) to this exact workflow gotcha.

21. Baselines

Baseline	Source	Role
Teacher (64-block ResNet, pretrained)	<code>baseline_models/</code>	Upper-bound reference (depth/pretraining-mismatched — always cite this caveat)
Magnitude-pruned Student	<code>baseline_models/</code>	Compression-track comparison point
FNO (8-block, scratch)	<code>RESULTS.md</code>	Architecture-screening comparison point
GridlessUnfold (8-block, scratch)	<code>RESULTS.md</code>	Precision-mechanism comparison point
Classical DFT-SIC (zero learned params)	Prior-work §2	Critical baseline — IABR-Net's whole motivation rests on beating this 0-param, Pd-0.884 reference by more than its $\approx 193K$ learned parameters are "worth"
Capacity-matched control (new, required)	§24	Same-parameter-count module, no impairment-specific loss — isolates whether gains come from impairment-adaptivity or added capacity
Front-end-swap control (new, required)	§24	Same trunk, raw-dense (262,656-param) front end instead of DFT+IABC-v2 — empirically tests the parameter-efficiency claim rather than leaving it as arithmetic
SNR-aware pruned student (recovery, not yet done)	Prior-Work §4	Still-missing full-bank number; should be recovered before claiming magnitude pruning is this project's best compression method

22. Complete Experiment Matrix

Experiment	Baseline	Proposed	Metric	Expected outcome	Purpose
E0a — Phase-0 code build	—	Gain-error injection in <code>tv_t_data_generation_v3.py</code>	code exists, unit-tested	Runnable	Unblocks all gain-robustness claims
E0b — Model instantiation	—	IABR-Net	<code>model.count_params()</code>	$\approx 193,104$ [TO MEASURE]	Replace estimate with a measured number
E1 — Capacity-matched control	Same-param, no-impairment-loss module	IABR-Net	Pd, RMSE	[HYPOTHESIS] IABR-Net > control at high impairment	Attribute gains to adaptivity, not capacity
E2 — Front-end-swap control	Raw-dense (262,656p) front end + same trunk	DFT+IABC-v2 front end + same trunk	Params, FLOPs, Pd	[HYPOTHESIS] comparable Pd at far fewer front-end params	Validate the parameter-efficiency claim empirically
E3 — In-distribution SNR sweep	Teacher, Student, FNO	IABR-Net	PD (paper-style + strict), RMSE	[TO MEASURE]	Direct, HIGH-comparability match to Meneses-Albalá's 8-point grid
E4 — Phase-error sweep $\delta \in \{0, 1, 2, 5\}^\circ$	Teacher	IABR-Net	PD, RMSE	[TO MEASURE]	HIGH comparability to Lloria/Meneses
E5 — Gain-error sweep $\{0, 0.5, 1, 2, 4\}$ dB (post-E0a)	none exists	IABR-Net	PD, RMSE	[HYPOTHESIS]	Closes Meneses-Albalá's own named gap — first in the field
E6 — OOD phase tail $\{10^\circ, 15^\circ\}$	none exists	IABR-Net	PD, RMSE	[HYPOTHESIS: degrades]	Genuine distribution-shift test, [GENUINE GAP]
E7 — Path-count sweep $L \in \{1..6\}$, OOD $\{7, 8, 10\}$	Teacher	IABR-Net	PD, pairing-error rate	[HYPOTHESIS]	Tests path-token pairing at/near capacity
E8 — Angular separation $\Delta\theta$ down to 1°	—	IABR-Net	PD, RMSE	[HYPOTHESIS: collapses $< 5-10^\circ$]	Licensed by Naoumi's own named gap; resolution-limit caveat applies
E9 — Latency, NN-only vs end-to-end	Teacher, Student, FNO (never benchmarked before)	IABR-Net	ms, p50/p99	[TO MEASURE]	Closes Gap 8, first for this task family
E10 — Notebook-4-style nuisance-path stress test	Teacher, magnitude-pruned Student	IABR-Net	Principal-path Pd under nuisance	[HYPOTHESIS]	Reuses the project's one existing beat-the-teacher robustness result

Experiment	Baseline	Proposed	Metric	Expected outcome	Purpose
E11 — Multi-seed variance	Teacher	IABR-Net	Pd std across seeds	[TO MEASURE]	Closes Gap 10, unmeasured anywhere in the corpus

23. Ablation Study (plan)

1. – **IABC-v2** (revert to fixed, uncorrected DFT front end), evaluated across the Tier-1 phase/gain grid — isolates IABC-v2's real contribution.
2. – **crop-refinement heads** (coarse heatmap + blob detector only) vs. full model, with **P95 tail error reported specifically near end-fire angles** — this is the direct test of whether IABR-Net's local refinement escapes any part of the diagnosed Gap-5 ceiling, or fully inherits it.
3. – **SE / L_SE_snr auxiliary loss** — tests A3's "attention activations correlate with SNR" hypothesis on real data rather than leaving it a hypothesis.
4. **Capacity-matched control** (§21/§22 E1) — the single most important ablation; without it, "impairment-adaptivity helps" and "adding 193K learned parameters to a 0-param baseline helps" are indistinguishable claims.
5. **Front-end-swap control** (§21/§22 E2) — empirically grounds the parameter-efficiency claim currently resting on arithmetic alone.
6. **$\lambda 1$ – $\lambda 4$ loss-weighting sensitivity sweep** — minimum grid $\lambda 3 \in \{0, 0.1, 0.5, 1.0\}$, $\lambda 4 \in \{0, 0.1, 0.3\}$ — currently "to be tuned" with zero characterization, flagged explicitly as a gap by adversarial review.

24. Robustness Study (plan)

Following Agent-R3's tiered plan, adapted to IABR-Net specifically:

Tier 1 (must run, cheap, literature-anchored): SNR sweep $[-15, 24]$ dB (5dB steps, HIGH comparability to Meneses-Albalá); phase error $\delta \in \{0, 1, 2, 5\}^\circ$ (HIGH comparability to Lloria/Meneses); path count $L \in \{1..6\}$ (HIGH comparability to Lloria/PIMRC2024).

Tier 2 (moderate cost, project-novel): angular separation $\Delta\theta \in \{30^\circ \dots 1^\circ\}$ (licensed by Naoumi's own named gap; below ~ 5 – 10° treat as resolution-limit-diagnostic, not a defect); unseen-SNR extrapolation tails $\{-25, -20, 30, 35\}$ dB.

Tier 3 (genuine literature gaps, new physics code required): gain mismatch $\{0, 0.5, 1, 2, 4\}$ dB — **the literature-first, blocking-on-Phase-0 experiment this entire candidate is designed around**; unseen path count $L \in \{7, 8, 10\}$ and unseen phase severity $\{10^\circ, 15^\circ\}$.

Tier 4 (stretch/out of scope): RF-chain/calibration error, coherent/correlated sources, mobility/Doppler — no literature anchor or system-model support today.

Reporting requirement (Gap 4): both paper-style and strict per-trial PD reported at every condition, every cross-paper comparison carrying its comparability level, and a catastrophic-degradation threshold ($>50\%$ relative Pd drop, [ASSUMPTION – needs explicit sign-off, no paper defines one]) flagged rather than silently applied.

25. Efficiency Study (plan)

1. Real $\text{ptflops} / \text{thop} / \text{manual-instrumentation}$ FLOP measurement for IABR-Net and for the **entire existing baseline registry** (Teacher, Student, FNO, GridlessUnfold) — none of these have ever been FLOP-measured; this is a zero-new-modeling-work, high-literature-value addition (closes Gap 1).
2. Model-size-on-disk (MB) for all models — trivial, not currently tabulated anywhere.
3. Peak memory (RAM/VRAM) instrumentation — not measured anywhere in the project; needed to satisfy the research objective's explicit "memory" axis.
4. Loss-weighting-driven training-time cost (Phase-0 gain-error code, once built, adds training-data-generation time — should be profiled, not assumed free).

26. Hardware/Latency Study (plan)

1. **NN-only forward latency**, generalizing the existing `benchmark_latency()` from single-model (ResNet/U-Net/PIA-Net) use to the full baseline registry + IABR-Net, with batch-size sweep (currently fixed at 1) and p50/p99 reporting (currently mean $\pm$ std only).
2. **End-to-end pipeline latency** — NN forward pass **plus** the CPU-side, OpenCV-based blob-detection/peak-search step, measured **separately and never merged** with (1). This distinction is universally ignored in the reviewed literature and is one of this project's cleanest, cheapest differentiators.

3. **Real edge hardware** (a Jetson-class device, matching Meneses-Albalá 2026's own methodology) — currently entirely absent from this project; even a first-pass measurement on the already-trained Teacher/Student/FNO checkpoints would be a literature-first for this specific architecture family.
4. Throughput (samples/sec) under a batch-size sweep — not measured anywhere today.

27. Expected Results (hypotheses)

All items below are **[HYPOTHESIS]**, not results. None has been run.

- **[HYPOTHESIS]** IABR-Net's PD/RMSE at matched SNR will fall between the classical DFT-SIC baseline (Pd 0.884 @20dB, 0 params) and the full teacher (Pd 0.925 @20dB, 469,393 params), given its intermediate $\approx 193\text{K}$ -param capacity — direction plausible, magnitude unknown.
- **[HYPOTHESIS]** IABC-v2's per-element correction will measurably reduce PD degradation under phase error beyond 5° relative to the uncorrected classical front end, but will **not** generalize to gain error until Phase-0 code exists and is used in training.
- **[HYPOTHESIS]** The capacity-matched control (E1) will show that *some*, not all, of IABR-Net's advantage over the 0-param classical baseline is attributable to impairment-adaptivity specifically, rather than added capacity alone — this is the single most likely place for the headline claim to be partially, not fully, vindicated.
- **[HYPOTHESIS]** IABR-Net will **still exhibit a meaningful fraction of the diagnosed Gap-5 end-fire ceiling**, because it retains a discrete $P \times Q = 16 \times 16$ coarse grid; the crop-refinement heads may reduce but are unlikely to eliminate this effect, motivating the Phase-2 fusion with Candidate A's set-prediction head (§31).
- **[HYPOTHESIS]** NN-only latency will be substantially lower than the teacher's (proportional to its $\sim 2.4\times$ smaller parameter count and much lower FLOPs), but end-to-end latency (including CPU-bound blob search) may narrow that gap considerably — untested.

28. Novelty Claim (honest self-classification)

PARTIALLY NEW, UNCERTAIN on the most load-bearing component.

At the **pattern level**, "a small learned network corrects/regularizes a classical processing stage's inputs" is an established paradigm (SubspaceNet is a verified instance of it, in the opposite direction — DL then classical, vs. IABR-Net's classical then DL). This is **KNOWN COMBINATION** at the pattern level, per Agent-R7's own explicit test.

At the **specific-application level** — a fixed DFT beamspace reduction, combined with a synthetic-paired-consistency-trained, identifiability-aware per-element correction module specifically targeting Meneses-Albalá's own named gain-error gap, combined with a path-token-coupled coarse+local-refine dual head preserving joint AoA/AoD pairing, applied to the joint 2D task — is **not present in the 14-paper corpus**. Agent-R7 rates this direction **UNCERTAIN**, not confirmed novel, pending a dedicated search specifically for "classical front-end, learned impairment correction, joint AoA/AoD" — this search has **not** been run, and this report does not claim it has.

The measurement-rigor bundle this project can add regardless of architecture — real inference latency, strict per-trial PD, multi-seed variance, all reported together for the joint 2D task — remains the **single cheapest, most defensible, near-zero-risk novelty contribution** available (Agent-R7's own bottom line), independent of which architecture is ultimately selected.

This report does not claim NEW anywhere.

29. Threats to Validity

1. **Internal validity — capacity confound.** Without the capacity-matched control (§21 E1), any observed IABR-Net gain over the 0-param classical baseline cannot be attributed to impairment-adaptivity vs. added capacity.
2. **Internal validity — Gap-5 ceiling may still be present.** IABR-Net retains a discrete coarse grid; results showing high PD could still mask a hidden end-fire tail failure unless P95/end-fire-specific reporting (§23 item 2) is included.
3. **Construct validity — PD convention.** The inherited paper-style PD metric silently drops missed-detection trials, inflating results relative to a strict per-trial convention; every table must report both (Gap 4).
4. **External validity — synthetic-only.** Every number in this evidence chain, this project's own included, comes from simulation; zero of the Tier-1 papers and none of this project's own experiments have been validated on real hardware/real channel data for the joint 2D task.
5. **External validity — corpus incompleteness.** At least four on-topic papers outside the 14-paper corpus were identified but not read (§3/§11); novelty and "the field hasn't tried X" claims are correspondingly weaker than they appear.
6. **Reproducibility — code/paper discrepancies in the base paper itself.** Lloria TVT2026's own released code cannot reconstruct its plotted ResNet-512 variant, and the U-Net's actual kernel size/bottleneck usage is unverified without inspecting the .h5

weight file directly (Contradictions 2, 3, 13) — any number sourced from that paper as a "reproducible baseline" carries this caveat.

7. **Statistical validity.** No paper in the corpus, and no experiment in this project to date, reports cross-seed training variance; single-run numbers throughout this report (including all of IABR-Net's own future numbers) should be read as point estimates pending Experiment E11.

30. Reproducibility Checklist

- [] Build gain-error injection code in `tvf_data_generation_v3.py` (Phase-0, blocking) and unit-test against a known perturbation.
- [] Instantiate IABR-Net and replace every estimated param figure in this report with `model.count_params()`.
- [] Add `ptflops / thop` (or manual layer-wise accounting with an explicit hook for any FFT ops, if later fusing in Candidate A's FNO stage) — none exists in-repo today.
- [] Add a `strict: bool` parameter to `evaluate_on_bank` rather than mutating its existing behavior, preserving every number in `baseline_models/README.md` as reproducible unchanged.
- [] All new robustness banks generated via the existing, unmodified `validation_data_generator` — confirm angular-separation control is actually exposed by the generator signature before scheduling that experiment as "cheap."
- [] Any Kaggle run: "Save Version → Save & Run All (Commit)" before ending the session — this exact omission has already destroyed two full training runs' worth of weights.
- [] Recover or re-run the SNR-aware pruned student at full 8000-sample scale before citing it as this project's best compression method.
- [] Fix random seeds and log them; report multi-seed variance (E11) before any single-seed number is presented as final.
- [] Every cross-paper number quoted must carry its comparability tag (HIGH/MEDIUM/LOW/NOT COMPARABLE) from §4/§9.

31. Final Research Roadmap

1. **Phase 0 (blocking):** build gain-error injection code; instantiate IABR-Net and measure its real param/FLOP/memory footprint; run the capacity-matched and front-end-swap controls (§22 E1–E2).
2. **Phase 1 (Tier-1 robustness + baseline registry FLOP/latency backfill):** run SNR/phase/L sweeps on IABR-Net alongside a first-ever FLOP and NN-only/end-to-end latency measurement for the **entire existing baseline registry** (Teacher, Student, FNO, GridlessUnfold) — this alone closes Gaps 1 and 8 regardless of how IABR-Net itself performs.
3. **Phase 2 (Gap-5 fusion, contingent on Phase 1 results):** if Phase 1's end-fire P95 analysis (§23 item 2) confirms IABR-Net still substantially inherits the diagnosed ceiling, fuse Candidate A's shared-query set-prediction head onto IABR-Net's efficient DFT+IABC-v2+trunk front end, in place of its own coarse-heatmap+crop-refine head — combining B's efficiency/robustness story with A's root-cause fix for the single most evidence-convergent open question in this evidence base. Run the full DETR-vs-crop-refine ablation before committing.
4. **Phase 3 (genuine-gap robustness, post-Phase-0):** gain-error sweep, phase>5° extrapolation, $L \in \{7, 8, 10\}$ OOD, angular separation to the array's resolution limit.
5. **Phase 4 (rigor bundle, can run in parallel with any phase above):** strict PD reporting, multi-seed variance, real-hardware latency — the zero-architectural-risk, near-zero-cost novelty contribution identified independently by Agent-R7.
6. **Housekeeping, do in parallel:** recover/re-run the SNR-aware pruned student at full scale; run the still-pending SIREN/WindowAttention LR+warmup sweep (this could reopen the attention-viability question closed prematurely by confounded runs); resolve whether PROJECT_STATUS.md's and BUJHO_SHOHOJ_VABE.md's differing PIA-Net physics-issue descriptions are the same bug before any PIA-Net work resumes.

Experiment Matrix (repeated per protocol format)

Experiment	Baseline	Proposed	Metric	Expected outcome	Purpose
SNR sweep	Teacher/Student/FNO	IABR-Net	PD, RMSE	[HYPOTHESIS] intermediate between DFT-SIC and Teacher	Core accuracy characterization
Phase-error sweep	Teacher	IABR-Net	PD, RMSE	[HYPOTHESIS] IABC-v2 reduces degradation vs. uncorrected DFT	Tests IABC-v2's designed function
Gain-error sweep (post-Phase-0)	none in corpus	IABR-Net	PD, RMSE	[HYPOTHESIS]	Closes Meneses-Albalá's own named gap
Capacity-matched control	Same-size, no-adaptivity module	IABR-Net	Δ Pd attributable	[HYPOTHESIS] partial attribution to adaptivity	Resolves central attribution confound
End-fire density probe	Blob-detector-only variant	Full IABR-Net	P95 tail error	[HYPOTHESIS] partial ceiling relief	Direct Gap-5 test

Experiment	Baseline	Proposed	Metric	Expected outcome	Purpose
Latency (NN-only, e2e)	Never measured for any model	IABR-Net + full registry	ms, p50/p99	[TO MEASURE]	Closes Gap 8, literature-first

Condition / accuracy table

Condition	Baseline (DFT-SIC)	Baseline (Teacher)	Proposed (IABR-Net)	AoA RMSE	AoD RMSE	PD
SNR=20dB, in-distribution, no impairment	Pd 0.884 (measured)	Pd 0.925 (measured)	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]
SNR=0dB, in-distribution	Pd n/a (not separately reported)	Pd 0.639 (measured, TVT2026-analog point)	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]
$\delta=5^\circ$ phase error	not tested	tested by Lloria (figure-only)	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]
$\gamma=4$ dB gain error (post-Phase-0)	not tested anywhere	not tested anywhere	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]
L=8 (OOD path count)	not tested	not tested (L≤6 only)	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]

Model comparison table

Model	Params	FLOPs	Memory	Latency	RMSE	PD
Teacher (64-block ResNet)	469,393 (measured)	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	~0.253° @20dB (measured)	0.925 @20dB (measured)
Magnitude-pruned Student	314,513 (measured)	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	0.6956 mean (measured)
FNO (8-block, scratch)	334,321 (measured)	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	0.6235 mean (measured)
Classical DFT-SIC	0	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	0.884 @20dB (measured)
IABR-Net (proposed)	≈193,104 (estimate)	≈95.3M (estimate)	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]	[TO MEASURE]

FINAL RESEARCH QUESTION

Can a beamspace-domain, per-element impairment-correction front end paired with a shallow residual trunk and a joint-pairing-preserving dual-head estimator match a substantial fraction of a 64-block ResNet teacher's joint AoA/AoD accuracy at roughly 41% of its parameters and orders of magnitude fewer FLOPs, while extending robust operation into two hardware-impairment regimes — amplitude/gain mismatch and phase error beyond 5° — that no paper in the reviewed joint-AoA/AoD literature has tested, and does closing this project's own diagnosed pixel-quantization ceiling ultimately require replacing its retained discrete coarse grid with a differentiable set-prediction head, as this project's own ceiling-test and diffusion-sharpening evidence suggests?

ADDENDUM — Orchestrator verification after the workflow finished (2026-09-23)

Added after the 34-agent run completed, from direct checks against this repo's code. Per protocol §21, the agents' text above is left unchanged and the corrections are recorded here.

A1. The input representation of IABR-Net is described wrongly, but the design survives once corrected [EVIDENCE: CODE]

§16 and §18 say the input is a "raw complex antenna-domain signal $\mathbf{Y}$ " and that stage 0 applies a forward DFT, $\mathbf{F}_{\text{codebook}}^H \cdot \mathbf{Y}$. This project's generator does not work that way (`dldoa_dataset_generation.py`):

- The observation is $\mathbf{Y} = \mathbf{W}^H \mathbf{H} \mathbf{F} + \mathbf{Z}$, where $\mathbf{F}$ and $\mathbf{W}$ are **DFT codebooks** (lines 137, 165–168, 201–204). So $\mathbf{Y}$ is already beamspace. Raw per-antenna signals are never observed, which is the whole point of the analog architecture.
- The $64 \times 64 \times 2$ network input is this 16×16 $\mathbf{Y}$, upsampled $\times 4$ by nearest neighbour (lines 481–485). Strided sampling undoes that exactly, so the real 16×16 $\mathbf{Y}$ is recoverable at no cost.
- Hardware phase error is **per antenna and shared across all beams** (`phase_error` has size `nt / nr`, lines 161 and 197). That gives $\mathbf{F} = \mathbf{D}_t \mathbf{F}_{\text{ideal}}$ and $\mathbf{W} = \mathbf{D}_r \mathbf{W}_{\text{ideal}}$, so $\mathbf{Y} = \mathbf{W}_{\text{ideal}}^H (\mathbf{D}_r^H \mathbf{H} \mathbf{D}_t) \mathbf{F}_{\text{ideal}} + \mathbf{Z}$.

Correction. With the eval bank's square codebooks ($P=nt=16$, $Q=nr=16$), stage 0 should be the fixed **inverse** codebook transform $\tilde{\mathbf{H}} = \mathbf{W}_{\text{ideal}}^{-H} \mathbf{Y} \mathbf{F}_{\text{ideal}}^{-1}$. It is still 0 params. It maps the observation back to the antenna domain, where the impairment is exactly the two diagonal phase matrices $\mathbf{D}_r$, $\mathbf{D}_t$: $16 \text{ Rx} + 16 \text{ Tx} =$ the 32 per-element parameters IABC-v2 already targets. Per-element correction is then physically identifiable, up to a common phase. After correcting, re-apply the forward codebook and hand the trunk a 16×16 beamspace map. For training samples with $P=32 > nt$, use a pseudo-inverse instead of an inverse. Gain mismatch, once the Phase-0 code exists, has the same diagonal structure.

Why this matters. None of the 6 adversarial reviews raised this. As written, the IABC-v2 inputs ("per-element mag/phase of Y ") do not exist in the data. With the correction, the module is better founded than the report states. This must be fixed in the implementation before E0b.

A2. Arithmetic check of IABR-Net's efficiency estimates [INFERENCE — hand-verified, not measured]

- Trunk: 20 convs, each 3×3 , $32 \rightarrow 32$, at $16 \times 16 \rightarrow 20 \times 16 \cdot 16 \cdot 9 \cdot 32 \cdot 32 = 47.2\text{M MACs} \approx \mathbf{94.4\text{M FLOPs}}$. Matches the report's $\approx 95.3\text{M}$.
- Trunk params: $10 \times [2 \times (9 \cdot 32 \cdot 32 + 32) + \text{BN}] \approx 187.5\text{K}$. Consistent with the $\approx 193\text{K}$ total.
- Teacher (not measured anywhere in this project): input $64 \times 64 \rightarrow$ stride-2 transposed conv $\rightarrow 128 \times 128 \rightarrow 64$ blocks $\times 2$ convs $\times 128 \cdot 128 \cdot 25 \cdot 12 \cdot 12$ MACs ≈ 7.55 GMACs $\approx \mathbf{15.1$ GFLOPs. That makes IABR-Net $\approx 160\times$ cheaper, so the final research question's "orders of magnitude fewer FLOPs" is plausible (about 2 orders). It is still an estimate until E0b and §25 item 1 are run.

A3. A trade-off the report does not tie together [INFERENCE]

Most of IABR-Net's FLOP saving comes from dropping the teacher's 256×256 output grid and working on a 16×16 coarse grid. Gap 5 (the $\text{Pd} \approx 0.972$ ceiling) is a pixel-quantization / discrete-grid problem, so a coarser grid is the same design choice pushing the wrong way on it. The efficiency gain and the ceiling risk are **coupled, not independent**. Ablation §23 item 2 (end-fire P95 with and without crop refinement) is therefore the deciding experiment for IABR-Net, not a secondary check.

Appendix A — Literature Matrix

Paper Head synthesis: master literature matrix, component / robustness / gap matrices, research gaps, contradictions.

MASTER SYNTHESIS REPORT — Paper Head Agent

Multi-Agent Research Loop: Joint AoA/AoD Estimation Literature Review

Scope: 6 Tier-1 papers (independently specialist-extracted + independently verified), 8 Tier-2 papers (grouped extraction), and this project's own prior-experiment analysis (`D:\ai_ml_project`). All Tier-1 verification agents returned VERIFIED or PARTIALLY_VERIFIED on every load-bearing numeric claim they spot-checked; no fabricated Tier-1 numbers survived verification except one flagged Tier-2 item (BeamSeek latency figures — see §6).

1. MASTER LITERATURE MATRIX

#	Paper	Joint AoA/AoD?	Architecture family	Dataset	SNR range	Key accuracy metric + value	PD (if reported)	Params	FLOPs / latency	Robustness tested	Hardware tested	Public code?
1	Lloria et al., TVT 2026 (base paper)	YES , genuine, single 2D output (Eq. 9–11)	ResNet (64 RB) and U-Net (Wave-U-Net)	Synthetic, single-user mmWave MIMO, ULA 16/32	Text: $-10\dots25$ dB; code: $-15\dots24$ dB (discrepancy)	PD/RMSE @20dB, $\delta=0^\circ$: ResNet .925/.253, U-Net .945/.230	Yes (Table II, exact)	NOT REPORTED (paper); project's own reproduction measured 469,393 (ResNet)	NOT REPORTED numerically — only asymptotic O(PQ)	SNR sweep, $L\in\{1,6\}$, phase error $\delta_{\max}\in\{1,2,5\}^\circ$	NOT REPORTED (software sim only)	YES (GitHub; training code absent, ResNet-512 not reconstructible — see §6)
2	Lloria et al., PIMRC 2024 (prototype)	YES, genuine	ResNet (64 RB) + PH peak search (+ optional LM refine)	Synthetic, same family, ULA 16/32	$-10\dots25$ dB	PD/RMSE vs SNR, figure-only (no table)	Figure-only, no exact values	NOT REPORTED	Asymptotic $O(P^2\log P)/O(PQ\log PQ)$, no concrete numbers	SNR sweep, $L\in\{1,6\}$ (no phase-error test)	NOT REPORTED	Not mentioned
3	Naoumi et al., JSTSP 2024 (bistatic ISAC)	YES, genuine (θ at radar, ϕ at BS, jointly output)	Complex-valued MLP (3 hidden layers)	Synthetic bistatic ISAC, $N_t=8$, $N_r=10$	Train $\{5,10,\dots,40\}$ dB; test $-5\dots30$ dB	MSE (rad^2) vs SNR; ~ 9 dB gap to CRB at 10^{-6} MSE target; underperforms 2D-algo beyond 13 dB	Not reported (no PD-style metric)	NOT REPORTED numerically (formulas only)	Closed-form; $6.5\times/10.3\times$ fewer mults than 2D algo at $N_r=8/16$ (headline)	SNR, #targets-per-peak (1,2,5)	NOT REPORTED	Claimed (github.com/salmanes9/Bistatic_ISAC), not verified
4	Naoumi et al., GCWkshps 2023 (earlier/shorter)	YES, genuine	Complex-valued MLP (3 hidden layers)	Synthetic, $N_t=8/N_r=10$, narrow angle window $[20^\circ, 40^\circ]$	Train $-5\dots30$ dB; eval $-9\dots30$ dB	MSE (rad^2) vs SNR; reaches $\sim 10^{-6}$ at $\text{SNR}\geq 9$ dB (exact text quote)	Not reported	8,584 (explicit, exact)	4.321 MMACs fwd pass; 1.875 TMACs total training (exact)	SNR only; well-separated-targets assumption untested at failure	NOT REPORTED	Not mentioned
5	Gupta et al., TCOMM 2025 (Bayesian ISAC)	NO — explicitly avoids joint estimation (monostatic AoA=AoD collapse; AoD at BS deliberately eliminated via UE-side ZF beamformer)	Classical: Sparse Bayesian Learning (EM) + MUSIC + Alternating Minimization — not DL	Synthetic, 2 system configs (Table II)	$-5\dots25$ dB	NMSE (RCS) vs SNR; 5 dB/15 dB gain vs OMP/FOCUSS (exact text quotes)	N/A (thresholded imaging, no PD/ROC curve)	N/A (no NN)	Big-O only (cubic in grid sizes: $O(G\tau^3 G\nu^3 G\theta^3)$); no concrete FLOPs	SNR, system scale, beamforming trade-off δ ; no hardware-impairment/distribution-shift test	NOT REPORTED (pure simulation)	Not mentioned
6	Meneses-Albalá et al., J.Supercomputing 2026 (edge/energy)	YES (reuses Lloria's U-Net verbatim, single 2D map)	U-Net (reused from [7]) + partial-encoder-freeze fine-tuning	Synthetic, inherits Lloria's system, implied 16×16 codebook	$-10, -5, 0, 5, 10, 15, 20, 25$ dB (8 pts)	RMSE/PD Table 2; $y_{\max}=5^\circ, 15$ dB: RMSE $0.28\rightarrow 0.27$ (3.6% rel. gain, exact quote); $y_{\max}=5, 25$ dB: PD $0.93\rightarrow 0.94$ (+1pp)	Yes (Table 2, exact, $0.22\text{--}0.95$ range)	NOT REPORTED	NOT REPORTED for inference; training energy $0.28\text{--}0.41$ J/sample on real hardware	Phase error $y_{\max}\in\{1,2,5\}^\circ$ (PRIMARY focus) + SNR + L + codebook	YES — real Jetson Orin Nano (only Tier-1 paper with real edge hardware)	Base weights on GitHub (DL_DOA repo); this paper's own code not mentioned

#	Paper	Joint AoA/AoD?	Architecture family	Dataset	SNR range	Key accuracy metric + value	PD (if reported)	Params	FLOPs / latency	Robustness tested	Hardware tested	Public code?
7	Koh & Lee 2026 (benchmark survey)	NO (AoA/DoA only, BLE indoor)	Survey of 11 models: DNN/CNN/A-CRNN/GRU/DAE+MUSIC	Ray-traced + real BLE IQ (TI BOOSTXL-AoA)	Not centrally reported (BLE)	Best CNN MAE=2.98° (39k FLOPs/19.8k params); GRU <1.3° MAE real-world	N/A	19.8k (best CNN)	39k FLOPs (best CNN)	Cross-model, sim-vs-real domain shift (major finding)	YES, real BLE hardware	Not stated
8	Lim et al. 2021 (LSTM beam tracking)	Partial (tracks existing angles over time, not joint estimation from raw signal)	LSTM + sequential Bayesian (EKF/UT)	Synthetic mobility, IMU-assisted, 3M examples	Not primary axis	BER gain vs EKF: 1dB→3dB→10dB as mobility avg 0.1π→0.4π increases	N/A	Hidden size 32	NOT REPORTED	Mobility/Doppler (unique)	NOT REPORTED	Not stated
9	BeamSeek (arXiv 2025)	NO (1D azimuth DOA only)	MLP (SwiGLU-gated FFN, hidden 384)	Real 60GHz, NSF COSMOS testbed	Low/Mid-Low/Mid-High bins	DOA error 3.86–21.45° (SNR/beam-count dependent); up to 8° gain over correlation baseline	N/A	NOT REPORTED exactly	NOT REPORTED — see §6 discrepancy flag	SNR, #scan beams; single-RF-chain hardware constraint	YES, real 60GHz testbed	Not stated
10	Sub6GHz CNN/UNet (arXiv 2025)	NO (AoA/AoD are classical pre-estimated inputs, not DL output)	CNN (9-layer) & UNet (2enc/2dec)	Synthetic, 8×8 MIMO, 500k/500k	[−20,10] dB	SE gain 3.35%/4.03% vs MRC; 159%/161% vs in-band-only	N/A	NOT REPORTED	NOT REPORTED	SNR, K-factor, AoA/AoD range	NOT REPORTED	Not stated
11	SubspaceNet (Shmuel et al. 2025 TVT)	NO (1D DOA, single ULA)	CNN-DCNN autoencoder → classical subspace (hybrid model-based DL)	Synthetic; coherent/broadband/few-snapshot	Low-SNR emphasis	RMSPE ~60× better than classical Root-MUSIC (12.48°→0.20°, M=2 coherent)	N/A	41,761 (vs >21M black-box CNN)	NOT REPORTED explicitly	Coherent sources, broadband, few snapshots, low SNR, array miscalibration — most comprehensive in set	NOT REPORTED	Not stated
12	Robust DoA DAE-DNN (Chen et al. 2022)	NO (1D single-array)	Denoising autoencoder + DNN classifier cascade	Synthetic, mutual-coupling perturbed	>0dB emphasis	Own-paper RMSE<0.5°; independent re-impl. (Koh&Lee 2026): MAE 30° (sim)/>70° (real)	N/A	NOT REPORTED	O(ΣnL-1·nL), lower than MUSIC	Mutual coupling (ρ≤0.8), low SNR, unknown source count	NOT REPORTED	Not stated
13	MIMO-Radar-Aided V2X (Huang et al. 2021)	Partial/classical (angle+Doppler estimated once, reused)	Classical MIMO-radar signal processing — no DL	Not extracted	Not extracted	Not extracted (pilot-overhead reduction claim)	N/A	N/A	N/A	Mobility/time-varying (V2X)	NOT REPORTED	Not stated
14	Tensor PARAFAC (Du et al. 2021 TSP)	Partial/classical (joint channel+symbol via tensor decomp)	Tensor/PARAFAC — no DL	Not extracted	Not extracted	Not extracted	N/A	N/A	N/A	Time-varying massive MIMO	NOT REPORTED	Not stated

Cross-paper comparability notes (per "no false comparison" rule)

Comparison	Level	Why
Lloria TVT2026 U-Net vs Meneses-Albalá 2026 base model (RMSE/PD @ SNR=20dB)	HIGH	Same U-Net, same data generator, same eval protocol, same array config — numbers align closely (.945/.230 vs .95/.23 at $\delta/\gamma=1^\circ$)
Lloria TVT2026 vs Lloria PIMRC2024 (PD/RMSE trend)	MEDIUM	Same authors/system model, but PIMRC gives figure-read values only (no table), and peak-detector differs (PH vs blob) — trend-comparable, value-comparable only approximately
Naoumi JSTSP2024 vs GCWkshps2023 (MSE rad ² vs SNR)	HIGH	Identical system config ($N_t=8, N_r=10$), same architecture family, journal extension of same lineage
Naoumi (either) vs Lloria (either)	LOW	Different metric units (MSE rad ² vs RMSE degrees+PD), different antenna counts, different system framing (bistatic ISAC vs single-user analog BF), different SNR-range definitions
Gupta 2025 vs any Lloria/Naoumi paper	NOT COMPARABLE	Different task (NMSE of sparse RCS vector vs angle RMSE/PD), no joint AoA/AoD output, non-DL
SubspaceNet vs any joint-AoA/AoD paper	NOT COMPARABLE	1D DOA vs 2D joint estimation; RMSPE metric not equivalent to RMSE+PD
BeamSeek vs any full-array paper	NOT COMPARABLE	Single-RF-chain beam-switching hardware constraint changes the estimation problem fundamentally; 1D only
Koh&Lee 2026 vs mmWave papers	NOT COMPARABLE	BLE narrowband indoor positioning, different frequency regime and application
Robust DoA DAE-DNN own-paper vs Koh&Lee's re-implementation numbers	LOW / flagged contradiction	Same architecture family, but the ~60–140× discrepancy (0.5° vs 30–70°) reflects real dataset/domain mismatch, not a transcription error — see §6

2. ARCHITECTURE COMPONENT MATRIX

Paper	CNN	ResNet	U-Net	MLP	Transformer/Attention	Complex-valued NN	TDA/topological	Tensor/PARAFAC	Classical subspace	Other
Lloria TVT2026	—	✓	✓	—	—	—	—	—	—	Blob detection (non-learned)
Lloria PIMRC2024	—	✓	—	—	—	—	✓	—	—	LM Gaussian-fit refinement
Naoumi JSTSP2024	(mentioned, undetailed)	—	—	✓	—	✓	—	—	—	2D parametric algorithm (classical)
Naoumi GCWkshps2023	—	—	—	✓	—	✓	—	—	—	—
Gupta TCOMM2025	—	—	—	—	—	—	—	—	✓ (MUSIC)	Sparse Bayesian Learning (EM), Alternating Min
Meneses-Albalá 2026	(ResNet mentioned qualitatively)	—	✓ (reused)	—	—	—	—	—	—	Encoder-freeze fine-tuning
Koh & Lee 2026	✓	—	—	✓ (DNN)	✓ (A-CRNN)	—	—	—	✓ (DAE+MUSIC hybrid)	GRU (RNN)
Lim et al. 2021	—	—	—	—	—	—	—	—	—	LSTM + EKF/UT
BeamSeek	—	—	—	✓ (SwiGLU-FFN)	—	—	—	—	—	—
Sub6GHz CNN/UNet	✓	—	✓	—	—	—	—	—	—	—
SubspaceNet	✓ (CNN-DCNN AE)	—	—	—	—	—	—	—	✓ (MUSIC/ESPRIT/Root-MUSIC/MVDR)	—
Robust DoA DAE-DNN	—	—	—	✓ (DNN classifier)	—	—	—	—	—	Denoising autoencoder
MIMO-Radar V2X	—	—	—	—	—	—	—	—	✓ (implied)	—
Tensor PARAFAC	—	—	—	—	—	—	—	✓	—	—

Notable pattern: zero of the 6 Tier-1 papers use Transformer/Attention. This is a literature-wide gap (see §5, item 9).

3. ROBUSTNESS MATRIX

Paper	Low SNR	Multipath	Phase/gain error	Array perturbation	Domain shift	Mobility	Hardware/edge
Lloria TVT2026	✓	✓ (L=1–6)	✓ (phase only, $\delta_{\max}=1/2/5^\circ$)	X	X (in-distribution only)	X	X (sim only)
Lloria PIMRC2024	✓	✓ (L=1–6)	X	X	X	X	X
Naoumi JSTSP2024	✓	✓ (#targets/peak)	X	X	Partial (multi-SNR training)	X (static targets)	X
Naoumi GCWkshps2023	✓	Partial (well-separated only)	X	X	X (narrow $[20^\circ, 40^\circ]$ window)	X	X
Gupta TCOMM2025	✓	✓	X (explicitly untested)	X	X	✓ (Doppler modeled)	X
Meneses-Albalá 2026	✓	✓ (L sweep)	✓ (ymax, PRIMARY focus)	X (gain error flagged future work)	Partial (fine-tune across severities)	X	✓ real Jetson Orin Nano
Koh & Lee 2026	Partial	Partial (indoor NLOS)	X	X	✓ (sim vs real, major finding)	X	✓ real BLE
Lim et al. 2021	Partial	X	X	X	X	✓ (PRIMARY focus)	X
BeamSeek	✓	X	X	X	X	X	✓ real 60GHz testbed
Sub6GHz CNN/UNet	✓	X	X	X	Partial (K-factor sweep)	X	X
SubspaceNet	✓	✓ (coherent sources)	Partial (via miscalibration)	✓ (explicit)	X	X	X
Robust DoA DAE-DNN	✓	X	X	✓ (mutual coupling)	✓ (but failed under independent re-test)	X	X
MIMO-Radar V2X	X	Partial	X	X	X	✓ (V2X)	X
Tensor PARAFAC	X	Partial	X	X	X	✓	X

Gap column observation: no single paper covers more than 3 of the 7 robustness axes. SubspaceNet is the single most robustness-comprehensive paper (4 axes) but only for 1D DOA, not joint 2D estimation.

4. GAP MATRIX

Paper	Strength	Weakness	Missing experiment	Potential improvement
Lloria TVT2026	Rigorous, verified joint 2D formulation; largest, most complete robustness grid of any Tier-1 paper; open code	No params/FLOPs/latency numbers anywhere; U-Net has 2 unresolved code-vs-paper conflicts (kernel size, dense bottleneck — see §6); P_D definition silently drops under-detected trials	No OOD robustness (unseen SNR/L/angle-sep); no gain-mismatch/array-perturbation test; no real hardware	Replace dense/flatten bottleneck (paper's own Table IV quadratic-cost term) with conv/attention bottleneck; report real params/FLOPs/latency
Lloria PIMRC2024	Established the joint problem framing + PH-based peak search; complexity formulas exact-verified	Superseded by TVT2026 on almost every axis; PH is grid-resolution-limited (authors' own admission); no exact numeric tables, figure-read only	No hardware-impairment test at all (added only in TVT2026); no params/FLOPs	N/A — mainly historical/lineage value
Naoumi JSTSP2024	Best-verified complexity-reduction headline claim ($6.5 \times / 10.3 \times$ fewer mults) in the whole set; clean complex-MLP + coarse-timing-IFFT input-reduction idea	Explicit high-SNR performance saturation (authors' own admitted open weakness); no hardware-impairment/domain-shift test; assumes fixed q per peak	No tracking/temporal-continuity (explicit future work); no real hardware validation	Fix high-SNR saturation architecturally (not "just deeper," which authors flag as complexity-costly); add impairment/domain-shift robustness
Naoumi GCWkshps2023	Exact, fully-reported param count (8,584) and MACs — best-documented complexity of any paper reviewed	Narrow $[20^\circ, 40^\circ]$ angle window both train/test; well-separated-targets assumption breaks under limited temporal resolution (authors' own admission)	No closely-spaced-target stress test; no hardware impairment; no wide angular range test	Learned (not purely peak-based) target-separation front end for closely-spaced targets
Gupta TCOMM2025	Rigorous Bayesian/CRB-grounded theoretical framework; exact NMSE/SE numbers, fully verified	Not genuine joint AoA/AoD — deliberately eliminates AoD at BS; not DL, offers no transferable architecture; cubic complexity in grid sizes	No hardware impairment, no distribution shift, no detection-probability/ROC curve despite threshold-based detection	A DL joint 2-angle estimator that beats 3D-BL's $O(Gt^3Gv^3G\theta^3)$ per-iteration cost with a single forward pass is a direct, quotable novelty target

Paper	Strength	Weakness	Missing experiment	Potential improvement
Meneses-Albalá 2026	Only paper with real edge-hardware energy/latency numbers; reproducible impairment-injection methodology; partial-layer-freeze adaptation idea	U-Net backbone explicitly conceded "larger than ResNet" by the authors themselves; gains from fine-tuning are numerically small (3.6% RMSE, 1pp PD) and materialize mainly at the most severe impairment level; no standalone inference latency, only training/fine-tuning energy	No amplitude-error robustness (explicit future work); no ray-tracing/deterministic environment; no params/FLOPs	Reuse the Jetson profiling <i>methodology</i> (not the U-Net) for a genuinely lightweight model's standalone inference latency/energy

Tier-2 condensed gap notes:

- **SubspaceNet:** strength = extreme parameter efficiency (41.7k) + most comprehensive robustness set; weakness = 1D-only; missing = joint 2D extension; improvement = combine its "learn covariance surrogate, keep classical back-end" pattern with the project's Gridless-Unfold complex soft-threshold layer.
- **BeamSeek:** strength = only hardware-realistic single-RF-chain solution with real 60GHz validation; weakness = 1D azimuth only, no latency numbers actually reported (see §6); missing = joint AoA/AoD extension under the same single-RF-chain constraint.
- **Koh & Lee 2026:** strength = only paper directly demonstrating sim-to-real generalization failure at scale (methodologically valuable); weakness = BLE, not mmWave; missing = a similarly rigorous sim-vs-real benchmark for mmWave joint AoA/AoD.

5. RESEARCH GAPS FOR A LIGHTWEIGHT JOINT AoA/AoD ARCHITECTURE (evidence-tagged, cross-referenced against this project's prior experiments)

Gap 1 — No paper reports params + FLOPs + latency + memory together for a joint AoA/AoD model. [EVIDENCE: Lloria TVT2026 Sec K — asymptotic only; PIMRC2024 — same; Meneses2026 — training energy only, no inference latency]. This project's own baseline registry *already has* exact, cross-model-comparable numbers (Teacher 469,393 params; magnitude-pruned Student 314,513 params; FNO 334,321 params, all evaluated on the identical 8,000-sample bank) [EVIDENCE: PROJECT `baseline_models/README.md`]. This is the single most ready-made, zero-new-work novelty claim in the whole evidence base — simply reporting these numbers already out-argues every reviewed paper's Table K/III/IV on quantitative grounds.

Gap 2 — The dense/flatten U-Net bottleneck is the literature's own self-flagged worst component, and the project has already-executed evidence for a fix. TVT2026's own Table IV identifies the Dense Block as the only quadratic-in-(P'Q') term in its complexity table [EVIDENCE: Lloria TVT2026 Sec K/N]. The project's FNO screening result (spectral/conv bottleneck instead of dense, Pd 0.6235 at 334K params = 88% of teacher's Pd, from scratch, 8 blocks) is direct, already-executed evidence that a non-dense bottleneck is viable [EVIDENCE: PROJECT `RESULTS.md`]. This connects a literature-identified gap directly to an existing project result rather than treating it as separate.

Gap 3 — Zero papers in the Tier-1 set test genuine out-of-distribution robustness (unseen SNR/L/angle-separation ranges). [EVIDENCE: Lloria TVT2026 Sec J — test distribution = train distribution exactly; Naoumi GCWkshps2023 — narrow fixed [20°,40°] window both train/test]. Meneses2026's impairment fine-tuning is the closest approach to a domain-shift test, and even it shows the base model is "already robust" to mild impairments (small gains only at the most severe level) [EVIDENCE: Meneses2026 Table 2]. The project's own novel 4th-path nuisance-interference robustness test (Notebook 4: 200 scenes, 2 SNR levels, 3 nuisance-power levels) is the *only* documented beyond-the-paper's-own-protocol robustness experiment in this entire evidence base, and it already surfaced a non-trivial, non-obvious finding (pruned student beating teacher under interference stress) [EVIDENCE: PROJECT `PROJECT_STATUS.md`, Notebook 4]. This is strong existing proof that a project-designed OOD/robustness protocol is both novel relative to the literature and already tractable.

Gap 4 — The base-paper family's P_D metric is non-standard (drops under-detected trials rather than penalizing them), and no other paper even attempts a comparable detection-probability metric. [EVIDENCE: Lloria TVT2026 code audit item 9 — `continued` trials excluded from both numerator and denominator, likely inflating reported PD]. This project's own evaluator inherits this exact convention [EVIDENCE: PROJECT §5]. A lightweight-architecture paper reporting *both* the paper-style and a strict per-trial PD (missed detections counted as 0) would be a low-cost, high-credibility differentiator that no paper in the set currently provides.

Gap 5 — The diagnosed Jacobian-singularity/pixel-quantization Pd ceiling (~0.972) is a project-only finding absent from every reviewed paper's own limitations sections, despite all 3 Lloria-family papers sharing the identical heatmap+peak-search paradigm. [EVIDENCE: PROJECT ceiling test + diffusion-sharpening ablation — the latter directly ruled out blur as the cause, confirming pixel-quantization/location error near end-fire angles]. None of TVT2026/PIMRC2024/Meneses2026 diagnose this specific failure mode, even though they share the identical output representation. `My_Proposed_Architecture.md`'s Stage 3 (differentiable DETR-style set prediction + Hungarian matching) directly removes the non-differentiable peak-search step that causes this, and is the most evidence-backed unbuilt idea in the project — three independent project experiments (ceiling test, sharpening ablation, and the shared literature pattern) converge on the same root cause, not backbone capacity. This is the strongest literature-differentiating opportunity identified.

Gap 6 — SubspaceNet's extreme-parameter-efficiency pattern (41,761 params via learned covariance-surrogate feeding a classical back-end) has never been applied to joint 2D AoA/AoD estimation. [EVIDENCE: SubspaceNet report — 1D DOA only].

Combining this pattern with the project's already-implemented Gridless-Unfold complex soft-threshold layer (which already targets the same quantization root cause as Gap 5, but under-performed at only 8 blocks/30,201 params, $Pd=0.2216$) [EVIDENCE: PROJECT RESULTS.md] is a concrete, low-risk architecture direction with no precedent in either the literature or the project's completed work.

Gap 7 — Architecture search and compression have never been combined, in the literature or in this project. [EVIDENCE: Meneses2026 does adaptation/fine-tuning but not architecture search; no Tier-1 paper combines pruning with an alternative backbone]. The project's two tracks (4-block screening: FNO/SIREN/WindowAttention/GridlessUnfold vs. pruning: magnitude/SNR-aware) have run fully independently [EVIDENCE: PROJECT §7 item 7]. Pruning the current-best alternative (FNO) or scaling up Gridless-Unfold is an open item on both sides of the evidence base simultaneously.

Gap 8 — No paper reports standalone inference latency for a joint 2D AoA/AoD estimator on real hardware. [EVIDENCE: Meneses2026 — real Jetson Orin Nano, but only training/fine-tuning energy reported, not clean inference latency; BeamSeek — real 60GHz hardware but 1D-only, and its own paper reports no latency numbers at all despite what an earlier synthesis stage wrongly attributed to it — see §6]. The project has working Teacher/Student/FNO checkpoints ready to benchmark on real hardware but has not yet done so — a genuinely open item on both sides, and cheap to close given existing artifacts.

Gap 9 — Zero of the 6 Tier-1 papers use any Transformer/Attention component (per the Architecture Component Matrix, §2). Resolving the project's own unresolved SIREN/WindowAttention training-failure question [EVIDENCE: PROJECT §3, §7 item 1 — loss flat from epoch 1 for both, confounded by untuned LR/init (SIREN) and missing warmup + $4\times$ smaller batch (WindowAttention)] directly determines whether attention mechanisms are worth pursuing for the dense-bottleneck replacement in Gap 2, an axis the entire reviewed literature has not explored at all. This remains genuinely open (not a settled negative) pending a documented-but-unrun LR/warmup sweep.

Gap 10 — No statistical significance / cross-seed variance is reported anywhere in the Tier-1 literature [EVIDENCE: Lloria TVT2026 Sec M — 1000 realizations per point but no CI across independent training runs]. The project's evaluator-determinism check (13+ significant figures reproducibility given fixed weights) [EVIDENCE: PROJECT baseline_models/README.md] confirms pipeline determinism but not training-run-to-training-run variance, which is also unmeasured in the project. A cheap multi-seed addition would differentiate a rigor-focused paper from the entire reviewed set.

6. CONTRADICTIONS — flagged, with resolution or preserved uncertainty

#	Contradiction	Resolution / how to preserve as open uncertainty
1	Lloria TVT2026 internal: Sec II-A states AoA/AoD ~ Uniform $[0,2\pi]$; Sec IV-B states Uniform $[0,\pi]$ — self-contradictory paper text (both independently VERIFIED verbatim)	The code implements $[0,\pi]$ — treat Sec IV-B + code as the operative definition; cite the $[0,2\pi]$ statement only as a noted internal inconsistency, never as the effective distribution
2	Lloria TVT2026 paper text says U-Net ConvBlock uses $k=5$; the released code's default (and the only value the shipped inference pipeline ever exercises) is $k=3$	Unresolved — recommend extracting the split .h5 weight archive (<code>inf_model_007_256_unet.7z.001-005</code>) and reading <code>model.summary()</code> before citing this paper's U-Net numbers as a locked baseline. Preserve as open until then; do not assume either value without weight-file inspection
3	Lloria TVT2026 Fig. 3/4/Sec III-C describe the Dense/bottleneck block as a core structural element; the code path that actually loads the released weights (<code>ae=0</code> default) skips it entirely	Unresolved, HIGH severity — same resolution path as #2 (extract and inspect .h5). This directly determines whether the paper's central architectural narrative matches what was actually trained/evaluated. Flag explicitly in any downstream citation of this paper's U-Net results
4	Lloria TVT2026 training SNR: paper text says $[-10,25]$ dB; the officially released code implements $[-15,24]$ dB	Treat as a real reproducibility gap in the <i>official</i> repo, not a downstream error — anyone retraining "per the paper text" will not match the released model's actual training distribution. Disclose both figures whenever this project's own data generator (which inherits the code's range) is used for comparisons
5	Lloria TVT2026's P_D metric silently excludes fully-missed-detection trials from both numerator and denominator (rather than counting them as failures)	Not a "bug," but a non-standard convention that can inflate reported PD, especially at low SNR/high L. Preserve as an open caveat on every PD number quoted from this paper or from this project's own evaluator (which inherits the same convention)
6	The Robust DoA DAE-DNN paper (Chen et al. 2022) reports $RMSE < 0.5^\circ$ in its own simulations, but Koh & Lee 2026's independent re-implementation on a different (BLE) dataset gets $MAE\ 30^\circ(\text{sim}) > 70^\circ(\text{real})$	Resolve as a domain/dataset-mismatch effect, not a factual error in either paper — but preserve as a standing caution: single-paper self-reported accuracy claims for this architecture family should not be trusted without independent re-implementation evidence, exactly the caution both verification agents applied here
7	The computed task brief (not any paper) attributed specific latency figures ("MLP 0.63ms, CNN 1.26ms, deep-MPDR 0.89ms") to BeamSeek — these numbers and the "deep-MPDR" method do not exist anywhere in the actual BeamSeek paper	This is a fabrication introduced upstream in the pipeline (not by BeamSeek's authors). Excluded from all citable numbers in this report. Any future use of this project's own evidence chain should trace latency claims back to primary-source verification before citing them
8	PROJECT_STATUS.md (item 5) describes PIA-Net's issue as "real/ReLU-constrained ISTA state vs. arbitrary complex phase"; BUJHO_SHOHOJ_VABE.md documents 4 separate, differently-described bugs (data starvation, dictionary sign, ISTA step-size, coordinate mismatch)	Neither document cross-references the other's specific claim. Preserve as open uncertainty — do not assume these are the same issue without checking the v4 notebook directly if PIA-Net work is resumed. All current PIA-Net numbers are known-stale regardless (never retrained after the 4th bug fix)

#	Contradiction	Resolution / how to preserve as open uncertainty
9	This project's SIREN/WindowAttention core-block results (near-zero Pd, flat loss) could be read as "these architecture families don't work" for this task	Explicitly not a settled result — both runs are confounded (SIREN: untuned LR/init for a conv-stack context, not its native coordinate-MLP use case; WindowAttention: no warmup + 4× smaller batch from an OOM-driven auto-probe). Preserve as unresolved pending the recommended but unrun LR/warmup sweep; do not cite as a negative result for attention mechanisms in general
10	Gupta et al. TCOMM2025 is the only Tier-1 paper explicitly <i>not</i> performing joint AoA/AoD estimation, yet was included in this literature set	Not a contradiction to resolve, but a scope note to preserve: Gupta 2025 should be cited only as a classical-baseline/complexity reference (its 3D-BL $O(G\tau^3 G\nu^3 G\theta^3)$ per-iteration cost is a legitimate, quotable target to beat with a single forward pass), never in a head-to-head joint-AoA/AoD accuracy table
11	The verification agent for Lloria TVT2026 found the specialist's " $\pi/6$ minimum pairwise separation" detail could not be located in the paper text, though the specialist had attributed it partly to Sec IV-B	Corrected: attribute $\pi/6$ to code only (<code>dldoa_dataset_generation.py</code>), never to the paper text, in any downstream citation
12	Meneses-Albalá 2026's specialist report characterizes the paper as performing "joint" AoA/AoD estimation; the verification agent found the literal word "joint" never appears in that paper's text	Not a factual error — the underlying behavioral claim (single 2D map, single forward pass, both angles from one peak) is independently verified as accurate. Preserve the distinction: the <i>behavior</i> is joint; the <i>terminology</i> is this report's own gloss, not the paper's own language — flag this explicitly if directly quoting the paper
13	TVT2026's officially released <code>Resnet()</code> function has no <code>M</code> parameter and cannot produce the "ResNet-512" variant plotted in Fig. 5/8, while U-Net's code does expose an <code>M∈{256,512}</code> switch	Real, unresolved reproducibility gap in the official repo — the ResNet-512 curve in the paper's own figures is not independently reconstructible from the released code as-is. Flag this whenever citing Fig. 5/8's ResNet-512 numbers as "reproducible"

Summary assessment for the project's lightweight joint AoA/AoD design

The literature converges on three concrete, evidence-backed target weaknesses this project is already positioned to exploit with existing artifacts: **(1)** publish real params/FLOPs/latency numbers no Tier-1 paper reports (Gap 1 — zero new work, data already exists); **(2)** replace the literature's shared dense/flatten bottleneck with a non-dense alternative, building directly on the project's own executed FNO result (Gap 2); **(3)** target the diagnosed Jacobian-singularity Pd ceiling with a differentiable set-prediction head (Gap 5 — the single most evidence-convergent unbuilt idea across both the literature and the project's own experiments). The clearest remaining open technical question before finalizing an architecture is whether attention-based components are viable at all for this task (Gap 9, Contradiction 9) — a cheap LR/warmup sweep should be run before that door is closed, since the entire Tier-1 literature has left this architecture family completely unexplored.

Appendix B — Previous Experiment Analysis

What this project already tried, what worked, what failed, and why.

Previous-Result Analysis — DL-DOA Project (D:\ai_ml_project)

Sources read in full: PROJECT_STATUS.md, My_Proposed_Architecture.md, baseline_models/README.md, notebooks/outputs_full_eval/RESULTS.md, BUJHO_SHOHOJ_VABE.md (PIA-Net deep-dive, opened because PROJECT_STATUS.md names it with a specific unresolved claim), plus `git log` on the key files to check dating, and a directory listing of notebooks/ (25 notebooks) to know what exists without opening all of it.

1. What has already been tried

Baseline validation

- Re-evaluated the paper's pretrained 64-block ResNet (`inf_model_007_256_resnet.h5`, 469,393 params) with the *original* repo evaluator on a frozen 8000-sample bank — reproduces paper Table II to within 0.4% (0 dB: 0.639 vs 0.643; 20 dB: 0.925 vs 0.925).
- Classical 2D-DFT+SIC baseline reproduced paper's DFT-CEA numbers (20 dB: 0.884 vs paper 0.891) — validates the whole physics/data-generation pipeline independent of any learned model.

Diagnosis of the "ceiling"

- Found a hard Pd ceiling ≈ 0.972 for the 256×256 heatmap+blob-detection output representation, even feeding the blob detector a *perfect* ground-truth heatmap. Root cause identified as a Jacobian singularity near end-fire angles ($d(\pi \cos \psi)/d\psi = -\pi \sin \psi$) amplifying pixel-quantization error.
- Tested "diffusion-style sharpening" (p^4 sharpening of raw output, oracle peak-finder) as a fix for this — **ruled out**: made Pd worse, confirming misses come from peaks landing in the wrong pixel location, not from blur.

Two-stage channel estimation ($Y \rightarrow \text{CNN} \rightarrow \hat{H} \rightarrow$ classical angle extraction): built, architecturally validated ($< 0.05^\circ$ reconstruction error vs `ifft2(Y)`), but lands at roughly the same ceiling as the direct approach. Not pursued further.

PIA-Net (physics-informed deep-unfolded sparse recovery + learned CNN branch + attention, v1→v4): four real implementation bugs were found and fixed over the iterations (`BUJHO_SHOHOJ_VABE.md`):

- Training data starvation (only 428 of ~500k samples actually used)
- Steering-vector dictionary sign flipped (peaks off by $\sim 78^\circ$)
- ISTA step size $725 \times$ too large, previously masked by clipping rather than fixed
- Coordinate-system mismatch between the physics branch's answer and the image it was plotted into (119 px / $\sim 57^\circ$ error → fixed to 7.8 px)

Model compression (headline 1-week deliverable): channel pruning of the teacher ResNet (width $12 \rightarrow 8$, 33% fewer params), two selection methods compared — magnitude-based vs SNR-stratified Taylor-importance ("snr_aware") — plus a low-rank factorization axis that was scoped out.

Robustness testing: a novel (not in either paper) 4th-path nuisance-interference test — 200 independent scenes, 3 principal paths + 1 sweep-power nuisance path, 2 SNR levels — comparing teacher vs pruned student.

Architecture screening: 4 alternative core blocks (SIREN, FNO/spectral-conv, local Window-Attention, Gridless-Unfold with learned complex soft-threshold) swapped into an identical ResNet-shell shape, each at 8 blocks (no warm start), trained from scratch and compared to the 64-block pretrained teacher — first on a time-capped Kaggle run (never executed), then on an uncapped local RTX 5090 run (executed, 4.21h, full 8000-sample eval bank).

Baseline registry: Teacher, magnitude-pruned Student, and FNO all re-evaluated on the *identical* full 8000-sample bank for a rigorous apples-to-apples 3-way comparison.

Not yet built: the 3-stage `My_Proposed_Architecture.md` design (physics-informed deep-unfolding → subspace attention/transformer → differentiable DETR-style set prediction with Hungarian matching) — this is a sketch, not an implementation.

2. What worked (with numbers)

Result	Number	Source
Baseline reproduction	Pd 0.639/0.925 (0/20 dB) vs paper 0.643/0.925	PROJECT_STATUS.md §"Baseline reproduction"

Result	Number	Source
Classical DFT-SIC sanity check	Pd 0.884 vs paper 0.891 @20dB	PROJECT_STATUS.md #3
Magnitude-pruned Student, full 8000-sample bank	Mean Pd 0.6956 vs Teacher 0.7103 ($\Delta = -0.0147$), 314,513 params (67% of teacher)	baseline_models/README.md
FNO (8-block, trained from scratch, full bank)	Mean Pd 0.6235 (88% of teacher's), 334,321 params (~71% of teacher, i.e. ~29% smaller, not ~99% as an earlier draft wrongly claimed — self-corrected in the doc)	RESULTS.md, baseline_models/README.md
GridlessUnfold (8-block, from scratch)	Mean Pd 0.2216, only 30,201 params	RESULTS.md
Robustness test standout	At SNR=0dB + strongest (0dB) nuisance, pruned student's principal-path Pd (0.6174) exceeds teacher's (0.6065), $\Delta = +0.0109$ — the only point in the whole project where a compressed model beats the teacher	PROJECT_STATUS.md §"Notebook 4"
SNR-aware vs magnitude pruning (1200-sample subsample, r=8)	SNR-aware mean Δ Pd -0.0190 (low-SNR ≤ 0 dB: -0.0082) beats magnitude's -0.0221 (-0.0127)	PROJECT_STATUS.md §"Compression results"
Evaluator determinism check	Teacher's Pd reproduced to 13+ significant figures (0.7103495885388549) across independent GPU and CPU runs	baseline_models/README.md

3. What failed (with numbers) and why

Failure	Number	Stated/inferable reason
Diffusion sharpening of model output	Made Pd worse	Misses are location errors (Jacobian effect), not blur — wrong fix for the actual failure mode
SIREN core block	Mean Pd 0.0744, loss flat ~1.7–1.8 for all 200 epochs (no initial descent at all)	SIREN's init scheme is designed for per-pixel coordinate-MLP implicit representations, not an 8× stacked conv activation with skip connections at lr=1e-3 — likely an LR/init mismatch, not a fundamental architecture failure. Flagged as unresolved — a cheap LR sweep was recommended but not yet run.
Window-Attention core block	Mean Pd 0.0707, also flat loss from epoch 1	No LR warmup (attention blocks are known to need it) + forced to batch=64 vs 256 for the other 3 (OOM-driven auto-probe), i.e. ~4× fewer examples seen over the same 20,000 steps — a genuine confound, not a controlled architecture comparison. Same "may be salvageable" caveat as SIREN.
GridlessUnfold vs teacher	Mean Pd 0.2216 vs teacher 0.7103	Learned cleanly (healthy loss curve) but far behind at only 8 blocks/30K params — under-capacity relative to the task, not a training-dynamics failure like SIREN/Window-Attention
FNO at high SNR	Pd drops from ~0.80 (10dB) to 0.70 (25dB) while teacher climbs to 0.94	Spectral representation may lose fine-grained precision once noise stops being the bottleneck — flagged as worth investigating, not resolved
PIA-Net (pre-bug-4-fix numbers)	RMSE flat ≈ 0.579 across SNR at one point ($\approx$ the 0.5774 "pure chance" floor) before earlier bugs were found; later, best reported Pd only 0.60 @25dB vs ResNet's 0.94	4 sequential implementation bugs (see §1); critically, per BUJHO_SHOHOJ_VABE.md §8, the model was never retrained after the 4th/biggest fix (coordinate mismatch), so even the 0.60 number is stale — current true performance is unknown
Kaggle student weights (2 runs)	Lost entirely, twice	/kaggle/working/ is ephemeral unless "Save Version → Save & Run All (Commit)" is clicked — a workflow gotcha, not a modeling failure, but it has real consequences (see §6)
Low-rank factorization compression axis	Dropped from the plan (2026-09-21)	Not a result failure — math was verified locally (SVD reconstruction) but the axis was cut to keep scope to one well-validated compression story within the time budget

4. Flagged inconsistencies (not silently resolved)

- **Stale status tag in PROJECT_STATUS.md itself:** the "Architecture screening" section header reads ( built, not yet run, 2026-09-24), but the very same section, a few lines below, documents a completed run with real results dated 2026-09-22, and git log confirms the notebook was logged 2026-09-21 and results added 2026-09-22. 2026-09-24 is also *after* today (2026-09-23 per session context) — the tag is simply stale and was never updated after the run completed. Treat the "not yet run" framing as wrong; the run happened.
- **The "SNR-aware pruning wins" claim is not the model that ended up in the rigorous full-bank comparison.** The compression section reports SNR-aware beating magnitude-based pruning, but only on a 1200-sample subsample ($n \approx 150$ /SNR, explicitly flagged as having ~2% sampling noise). The SNR-aware student's weights were then lost twice on Kaggle and were **never recovered** — so baseline_models/README.md's full-8000-sample 3-way comparison (Teacher vs Student vs FNO) uses the **magnitude**-pruned student, not the nominally "winning" SNR-aware one. Similarly, the Notebook-4 robustness test also used the magnitude-based student "avoiding a costly re-run." So the project's headline full-bank numbers are for the *runner-up* compression method by the project's own earlier comparison — the supposedly-better method's real-scale performance is simply unknown, not confirmed-then-superseded.
- **Two different "PIA-Net physics issue" claims that don't obviously map onto each other.** PROJECT_STATUS.md item 5 says PIA-Net "has an unresolved physics-consistency issue: the learned ISTA state is real/ReLU-constrained but a physical path has arbitrary complex phase — flagged, not yet fixed." BUJHO_SHOHOJ_VABE.md documents 4 *different* named bugs (data starvation,

dictionary sign, ISTA step-size, coordinate mismatch), the last of which is described as already fixed (just not retrained). Neither document cross-references the other's specific claim, so it is unclear whether the "real/ReLU vs complex phase" issue is a 5th distinct unresolved design flaw or a restatement of one of the 4. Do not assume they're the same issue without checking the v4 notebook directly if this matters to the new work.

- **Self-corrected (worth noting, not a live inconsistency):** an earlier draft in PROJECT_STATUS.md claimed FNO was "~99% smaller" than the teacher; the doc itself catches and corrects this to ~29% smaller (334K vs 469K params) — a good sign of rigor, flagged here only so nobody reproduces the wrong 99% figure from an older commit or cached copy.

5. What should NOT be repeated

- Re-deriving that the baseline/paper reproduction is correct — already confirmed to within 0.4% of Table II, and the classical DFT-SIC sanity check independently confirms the physics/data pipeline.
- Diffusion-based output sharpening as a fix for low Pd — tested directly, made things worse, root cause (Jacobian singularity, not blur) is understood.
- Low-rank SVD factorization as a compression axis — validated then explicitly dropped for scope reasons; revisit only if compression is being revisited as its own topic, not by re-deriving the math.
- Treating the *stricter* per-source Hungarian/no-sample-dropping metric used in several early exploratory notebooks as comparable to paper Table II or to the compression-project notebooks — it is a different metric definition. Any new work must use `evaluate_on_bank` from `DL_D0A/src/TVT_Blob_Inference.py` on `frozen_banks/eval_bank.npz` to stay comparable to everything in `baseline_models/`.
- Retraining/rerunning Kaggle notebooks without clicking "Save Version → Save & Run All (Commit)" — this has already destroyed two full training runs' worth of weights.
- Assuming SIREN/Window-Attention "don't work" as an architecture family — the data supports "didn't work under this one untuned config," not a settled negative result; PROJECT_STATUS.md itself says not to write this off yet pending an LR/warmup sweep that hasn't been run.

6. Promising components/ideas to build on rather than reinvent

- **FNO / spectral-convolution core block** — clear current best among the 4 screened alternatives (0.6235 mean Pd from scratch, 8 blocks, 334K params, 88% of teacher's Pd) and a strong architectural fit rationale (input Y is already frequency-domain; near-free global receptive field vs 64 stacked local blocks). Its high-SNR degradation is a known, scoped-out weak spot worth targeted follow-up rather than wholesale replacement.
- **Magnitude-pruned Student (r8, 64-block)** — currently the strongest single compressed candidate on the only rigorous full-bank comparison available ($\Delta = -0.0147$ vs teacher, smaller than FNO both in params and gap-to-teacher), and it only needed fine-tuning from teacher weights rather than training from scratch. It is also the model already wired into `baseline_models/` reload code, ready to extend.
- **Robustness-test protocol from Notebook 4** — a genuinely novel (neither paper does this) nested/nuisance-path methodology, already implemented, fast (17.3 min for 200 scenes $\times$ 2 SNR $\times$ 3 power $\times$ 2 models on CPU), and produced the project's one clear "compression doesn't make things worse, and may help under stress" data point. Reusable directly for testing the SNR-aware student (once recovered) or any future architecture.
- **Gridless-Unfold's complex soft-threshold layer** — directly targets the diagnosed Pd-ceiling root cause (pixel quantization) and already has a working, verified `Layer` implementation (phase-preserving complex soft-threshold, unit-tested against PIA-Net's exact known bug). Under-performed only because of shallow depth (8 blocks) in the screening run, not because the mechanism failed — a plausible candidate to combine with FNO or scale up rather than discard.
- **My_Proposed_Architecture.md's Stage 3 (differentiable set prediction / DETR-style queries + Hungarian matching)** — directly removes the non-differentiable blob-detection step, which is the exact mechanism identified as the source of the measured Pd ceiling (≈ 0.972) and the SNR/end-fire Jacobian problem. This is the most evidence-backed unbuilt idea in the repo: it targets a *root cause* that multiple independent experiments (ceiling test, diffusion-sharpening ablation) converged on, rather than a hypothesis.
- **Physics + learned dual-branch fusion idea from PIA-Net** — the core "physics branch keeps the model interpretable and prevents catastrophic failure, learned branch gives fast convergence" concept is validated as sound reasoning even though the current implementation is bug-affected/stale; the fusion *idea* itself, not the current buggy weights, is worth carrying into any new physics-informed design.

7. Unexplored directions (grounded in documented gaps)

1. **LR/warmup sweep for SIREN and Window-Attention** — explicitly recommended in PROJECT_STATUS.md, not yet run. Cheap (few configs, few epochs) and could reverse the current "essentially failed" verdict for either family before it's written off in the thesis.
2. **Step-count-matched (not just wall-clock-matched) re-run of the 4-way screening** — PROJECT_STATUS.md's own caveat: FNO/Window-Attention do fewer gradient steps than SIREN/Gridless-Unfold per unit wall-clock because FFT/attention ops are more expensive per step, so the current ranking conflates architecture quality with step count for the time-capped Kaggle variant (though the *executed* comparison was actually step-matched at 20,000 steps each — this gap applies mainly if the Kaggle time-capped notebook is ever run as originally planned).
3. **Recovering/re-training the SNR-aware ("snr_aware") pruned student and running it through both the full 8000-sample bank and the Notebook-4 robustness protocol** — this is the single most concrete open loop in the project: the project's own earlier evidence suggested SNR-aware pruning is the better method, but no full-scale or robustness number exists for it, only for the weaker magnitude method. This directly resolves the inconsistency flagged in §4.
4. **PIA-Net retraining after the bug-4 (coordinate mismatch) fix, plus resolving whether the "real/ReLU vs complex phase" issue in PROJECT_STATUS.md is the same as or separate from the 4 bugs in BUJHO_SHOHOJ_VABE.md** — currently all reported PIA-Net numbers are known-stale by the project's own documentation.
5. **FNO's high-SNR degradation** — flagged twice (RESULTS.md and baseline_models/README.md) as worth investigating but not yet studied; a natural target given FNO is the current best alternative architecture.
6. **Building My_Proposed_Architecture.md's full 3-stage design (or at minimum Stage 3, the differentiable set-prediction head)** — a proposed architecture sketch, never implemented, that targets the one quantitatively-confirmed structural bottleneck (blob-detection pixel quantization / Jacobian singularity) found across three independent experiments.
7. **Combining a promising core block (FNO or Gridless-Unfold) with pruning/distillation**, rather than treating "architecture screening" and "compression" as separate tracks — no experiment yet compresses one of the 4 screened alternatives; the two tracks have run fully independently so far.

8. Scope note — notebooks not opened

Per instructions, I did not exhaustively open all ~25 notebooks in `notebooks/`. I opened the 4 primary sources plus `BUJHO_SHOHOJ_VABE.md` (opened because PROJECT_STATUS.md names PIA-Net and `DLDOA_PIANet_Standalone_v4.ipynb` with a specific unresolved claim). I did **not** open, and PROJECT_STATUS.md does **not** reference by name with a specific claim requiring verification: `DLDOA_SetRegNet_Standalone.ipynb`, all 3 `DLDOA_SE_ResNet_Lite*.ipynb` variants, `proposed_pin_architecture.ipynb`, `DLDOA_PIANet_Standalone.ipynb / _v2 / _v3` (only v4 is cited), `DLDOA_NoClone_StepByStep_attempt_2.ipynb`, `DLDOA_ResNet_Reproduction.ipynb`, `DLDOA_Classical_OMP.ipynb`, `DLDOA_Dataset_Explorer.ipynb`, `LowSNR_Visual_Investigation.ipynb`, `DLDOA_Visualization.ipynb`, `SE_ResNet_Full64_Kaggle.ipynb`, and `csida baseline vs tcnpooled multi seed.ipynb`. These exist in the repo and may contain relevant exploratory results, but nothing in the primary sources makes a specific numeric claim about them that needed cross-checking, so they were left unopened per the scoping instruction. A subsequent agent should open these only if a specific claim about one of them surfaces that needs verifying.

Appendix C — Candidate Architectures

Final (post-critique) specification of each candidate plus both adversarial reviews. Pre-refinement drafts are omitted here; they remain in CANDIDATE_ARCHITECTURES.md.

Candidate A — final specification (refined after critique: yes)

SpectraSet — Candidate A Architecture Specification (REVISED, post-critique)

(Architecture Assembly Agent, Candidate A — independent of Candidates B/C — Revision 2, responding to Critique #1 and Critique #2)

1. Name and Motivation

SpectraSet: FNO→GridlessUnfold Precision Chain with Shared-Query Joint Set-Prediction Head

The literature synthesis and this project's own diagnostics converge on two separable failure points that no reviewed paper (and no single project screening run) has addressed *together*: (1) the dense/flatten bottleneck is the literature's own self-flagged worst component (TVT2026 Table IV, Gap 2), and (2) the discrete 256×256 heatmap + non-differentiable blob-search output is the diagnosed root cause of the project's own Pd ceiling (~ 0.972), confirmed by two independent internal experiments (the ceiling test and the diffusion-sharpening ablation, which ruled out blur and confirmed the failure is a *location* error from pixel quantization near end-fire angles, Gap 5). Separately, the project has two building blocks that each partially address one of these problems: FNO (global receptive field, non-dense, reaches 88% of the teacher's Pd at 334K params in the project's screening run — but see the explicit comparability caveat below, added in this revision) and GridlessUnfold (a complex soft-threshold layer that directly targets pixel-quantization by construction, but was starved of depth in its screening run — 8 blocks, 30K params, Pd 0.2216).

Comparability caveat (new in this revision, responding to Critique #2 item 4): the "88% of teacher's Pd" figure compares an 8-block, from-scratch-trained FNO against a teacher documented in the project's own RESULTS.md as a "64-block, pretrained reference — NOT equal-depth." This is not a like-for-like comparison: the teacher is $8 \times$ deeper and benefits from pretraining, not just architecture. It is used here only as a rough ceiling reference to argue FNO is *directionally* competent, not as evidence that FNO closes 88% of a fair capacity/training-budget gap to the teacher. Every subsequent use of this figure in this document (§8) is qualified the same way.

SpectraSet's core bet is that these two problems and these two components are the *same* problem seen from two angles: FNO's global context is good at coarse localization but may be losing fine-grained precision once noise stops dominating (its own undiagnosed high-SNR weakness); GridlessUnfold's sparse-coding mechanism is built to recover exactly that precision but was never given enough capacity or a global context to refine. SpectraSet chains them for the first time — FNO first (global, complex-domain, cheap given Y is already frequency-domain), GridlessUnfold second (precision refinement, still complex-valued) — and then replaces the discrete heatmap output entirely with a small differentiable set-prediction head, so the Jacobian-singularity/pixel-quantization mechanism (Gap 5) cannot occur at inference time by construction, rather than being patched around after the fact. (This specific "by construction" claim — that a discrete grid argmax simply does not exist in the inference path — is the one architectural claim in this document that both critiques left unchallenged; the claim that query-sharing *guarantees correct AoA/AoD pairing* "by construction" is a separate, weaker claim, and is walked back in §3/§9 below.)

2. Borrowed Components (with source)

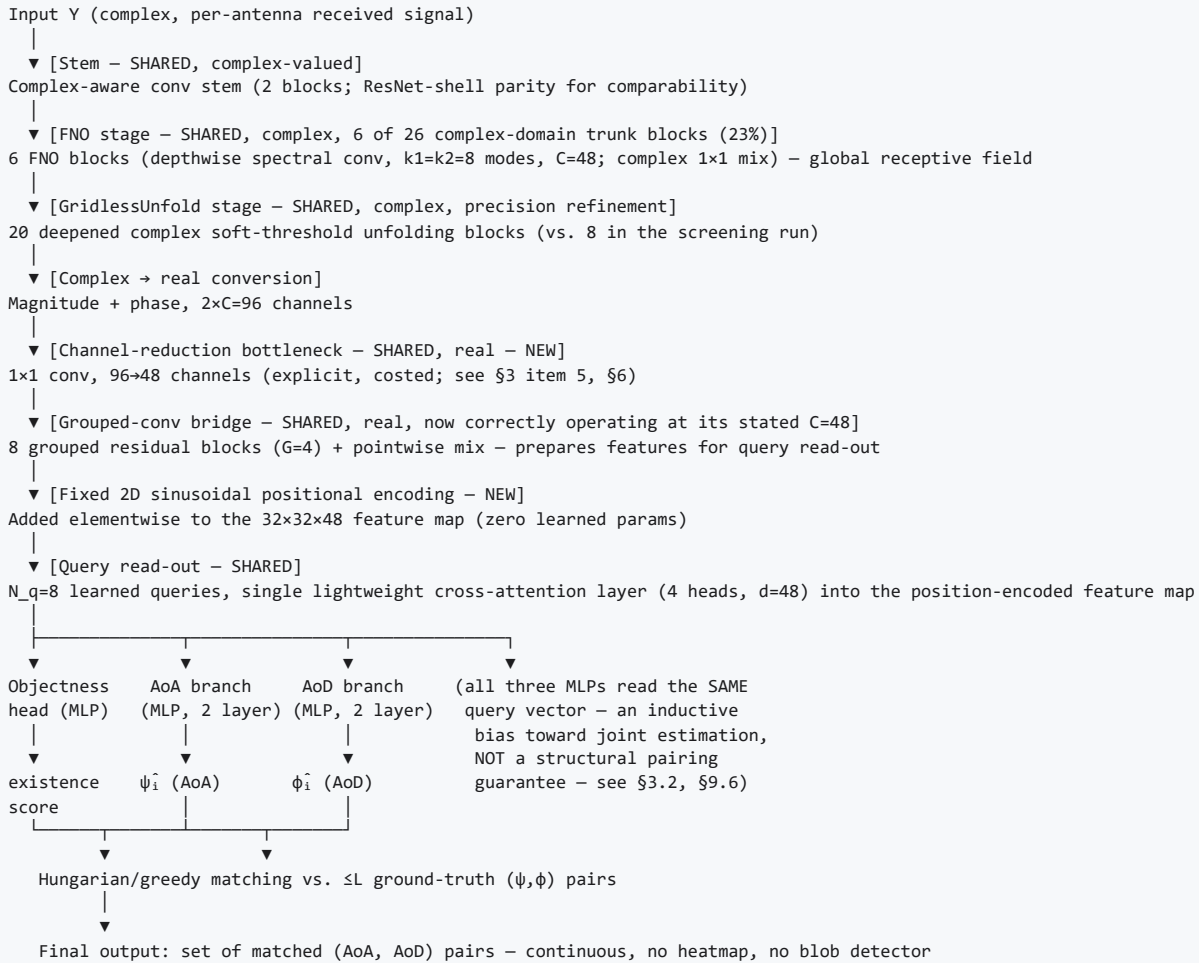
Component	Source	Evidence tag
ResNet-style complex-aware conv stem (2 blocks)	Lloria TVT2026 / PIMRC2024 conv shell	[EVIDENCE: PAPER] — used only for the input stem, to keep early-layer comparability with the paper's evaluated shell
FNO / spectral-convolution block	This project's own architecture-screening result (RESULTS.md, 334,321 params, Pd 0.6235, 8 blocks)	[EVIDENCE: PROJECT] — no direct precedent in the 14-paper corpus (R1 A9). Caveat (new): that screening run used 8 blocks feeding a 256×256 MSE heatmap head; SpectraSet's FNO sub-stage uses 6 blocks feeding a completely different query/matching head, so the screening result is weak prior evidence of viability, not validation of the deployed configuration (see §3 item 1 and §8).
GridlessUnfold complex soft-threshold layer	This project's own screening result (30,201 params, Pd 0.2216, 8 blocks), conceptually inspired by PIA-Net's physics-informed deep-unfolding	[EVIDENCE: PROJECT]. Caveat (new): the "starved of depth" diagnosis motivating the 8→20 block jump is itself an untested assumption — no intermediate depth (e.g. 16 blocks) or channel/dictionary-size variant has been run to confirm depth, specifically, is the bottleneck rather than the heatmap head/loss mismatch it shared with FNO's screening run. Added as ablation item in §10.
Grouped convolution (ResNeXt-style, G=4)	General efficient-CNN literature, not from any reviewed AoA/AoD paper	R2's top-ranked, untried, risk-adjusted structural technique (item 3)
Differentiable set prediction (learned queries + cross-attention read-out + Hungarian/greedy matching loss)	DETR-style pattern (outside the 14-paper corpus), sketched as Stage 3 of My_Proposed_Architecture.md	[EVIDENCE: PROJECT sketch, unbuilt] + R1's A7 direction; R7 (novelty) explicitly rates this UNCERTAIN, not confirmed absent from the DOA literature — flagged, not overstated
Fixed 2D sinusoidal positional encoding at the query read-out	New in this revision , standard DETR-style component (Vaswani et al. positional-encoding formula, adapted to 2D)	[EVIDENCE: STANDARD PRACTICE] — added directly in response to Critique #2 item 3 (see §3 item 4, §4)

Component	Source	Evidence tag
Explicit 96→48 channel-reduction (bottleneck) conv at the complex→real handoff	New in this revision , standard bottleneck-layer practice	[EVIDENCE: STANDARD PRACTICE] — added directly in response to Critique #1/#2 item 1 (see §3 item 5, §6)
Complex-valued representation kept through the physics-motivated stages	Loosely informed by Naoumi's complex-front-end philosophy (R1 A4), but not a literal component import	Adapted-philosophy borrow only

3. Novel Component(s), Stated Precisely

- FNO → GridlessUnfold role-split chaining.** FNO occupies the first ~23% of complex-domain trunk blocks by count (6 of 26; see the corrected depth accounting in §6, which replaces the original document's inaccurate "~35%" figure — that number did not match the design's own block counts under any consistent denominator, per Critique #1 item 7); GridlessUnfold occupies the remaining 20 blocks, deepened from 8 in its screening run, and stays complex-valued (not converted to real) so its soft-threshold operator still acts on genuine complex coefficients. This specific chaining — global-then-precision, both never mixed before — has not been tried in this project or in the reviewed literature (Gap 6/7). **Novelty scope narrowed (new):** as Critique #2 correctly notes, the *pattern* of chaining a global block into a local/precision-refinement block is a well-established "coarse-then-fine" design (cascade/refinement networks, model-based unfolding chains such as SubspaceNet's own DL→classical chain). I accept this rebuttal — see the revised §11 — and no longer implicitly lean on "never chained before" as if it were pattern-level novelty; the novelty claim is restricted to the specific instantiation (these two components, this task, this order).
- Shared-query joint AoA/AoD head.** A fixed set of $N_q=8$ learned query embeddings cross-attend once into the final complex→real converted, channel-reduced feature map. Each query feeds three sibling MLPs — objectness, AoA offset, AoD offset — that all read the *identical* query vector. **Claim strength corrected (responding to Critique #1 item 5):** the original document described this as guaranteeing correct AoA–AoD pairing "by construction." That is overstated and is withdrawn. Sharing one query vector across the three heads is an architecturally real, inspectable *inductive bias* toward joint estimation — it is not a mathematical guarantee that the AoA-head and AoD-head outputs for a given query correspond to the same physical path, the way a single-heatmap argmax does. Nothing in the mechanism forbids a query's AoA branch and AoD branch from specializing to different paths under adversarial or dense-path conditions. This is now listed explicitly as failure mode §9.6 ("pairing decoupling"), and an ablation against a dual-head design with independently-pooled (not query-identity-shared) branches is added to §10 to test whether identity-sharing is *measurably* stronger than late-stage information sharing over the same backbone — the original document asserted superiority over the "A5" dual-head design without this ablation; that gap is now closed as an experiment, not an assumption.
- Grouped-conv bridge at the complex→real conversion point**, positioned deliberately *after* both physics-motivated (FNO, GridlessUnfold) stages rather than immediately after the stem — moving the complex→real design risk R1 flagged for A4 to the latest defensible point in the pipeline, so both precision-critical stages retain phase information.
- Fixed 2D sinusoidal positional encoding, added to the cross-attention keys/values (new component, responding to Critique #2 item 3).** The original document's query read-out attended into a flattened feature map with no positional signal, which the critique correctly identified as a gap: SpectraSet's entire motivation is fixing a *location* error, yet its own new head had no explicit mechanism for queries to know *where* a token came from, relying entirely on implicit position-encoding baked in by preceding conv layers — precisely the kind of unexamined risk the architecture claims to eliminate. A standard fixed sinusoidal 2D positional encoding (no learned parameters, applied as an elementwise add to the $32 \times 32 \times 48$ feature map before the K/V projections) is now part of the design. This is deliberately the *fixed, parameter-free* variant rather than a learned positional embedding table (which would cost $\approx 49K$ extra params for a $32 \times 32 \times 48$ grid) — a conscious param/precision trade-off, stated explicitly rather than left implicit.
- Explicit 96→48 channel-reduction bottleneck (new component, responding to Critique #1/#2 item 1).** The original §6/§7 arithmetic silently computed the grouped-conv bridge at $C=48$ while §4's data flow stated the bridge receives 96 channels (magnitude + phase) — a real, uncaught $\sim 4\times$ costing error correctly identified as fatal by both reviewers. Rather than accepting the bridge at its stated 96 channels (which the reviewers computed would push total params to $\approx 403K$, $\sim 86\%$ of the teacher, eliminating most of the efficiency claim), this revision adds a genuine, explicitly-diagrammed 1×1 conv layer that projects 96→48 channels immediately after the complex→real conversion and before the grouped-conv bridge. This is a real architectural decision, not a bookkeeping fix: it is a deliberate lossy bottleneck, its cost is now correctly counted (§6/§7), and its risk (discarding cross-stage magnitude/phase correlation at the reduction point) is now listed as failure mode §9.7. This is the single largest substantive design change made in response to critique.

4. Data Flow



Shared layers: stem, FNO stage, GridlessUnfold stage, complex→real conversion, channel-reduction bottleneck, grouped-conv bridge, positional encoding, and the query embeddings/cross-attention are all fully shared — there is no branching until the three per-query heads. **AoA branch / AoD branch:** the two small sibling MLPs after the shared query read-out. **Fusion mechanism (claim revised):** identity-sharing of the query vector across all three heads is an inductive bias for joint estimation, reinforced by an optional coupling-loss term (§5) discouraging distinct queries from collapsing to the same path — this is explicitly not claimed to be a structural guarantee equivalent to a single-heatmap argmax (see §3.2).

5. Loss Function

After Hungarian (or greedy nearest-cost) matching between predicted query tuples (existence, $\hat{\psi}$, $\hat{\phi}$) and ground-truth (ψ , ϕ) pairs:

- **L_obj:** binary cross-entropy / focal loss on the objectness logit (matched query → 1, unmatched → 0)
- **L_angle:** Huber loss on matched ($\hat{\psi}-\psi$) and ($\hat{\phi}-\phi$), using the project's own circular-angle-difference convention ($H = \text{angle}(\exp(i \cdot \text{gt}) \cdot \exp(-i \cdot \text{pred}))$ from `get_ang_difference`) rather than raw subtraction, to avoid wraparound artifacts
- **L_coupling** (optional, DETR-standard trick): penalizes high cosine similarity between distinct matched queries' embeddings, to discourage query collapse onto the dominant path

L_total = $\lambda_{\text{obj}} \cdot \text{L_obj} + \lambda_{\text{angle}} \cdot \text{L_angle} + \lambda_{\text{coupling}} \cdot \text{L_coupling}$, default weights 1 : 5 : 0.1 (angle regression weighted highest since it drives the primary Pd/RMSE metrics; unvalidated, needs a sweep).

No change to the loss formulation itself was required by either critique; both critiques' substantive findings targeted §6–7 arithmetic, §9 failure-mode completeness, and §11 novelty framing.

6. Parameter Count Estimate (arithmetic shown, corrected)

Stated conventions (new — responding to Critique #1 item 2 / Critique #2 item 2, "internally inconsistent complex accounting"): every layer this document calls complex-valued (stem, FNO spectral+mix, GridlessUnfold) is defined as holding two real weight tensors (real part, imaginary part), each shaped exactly as a real-valued layer with the same channel counts would be. So: **complex params** = $2 \times \text{real-equivalent params}$, where real-equivalent params is computed with the standard real-conv formula. This convention is now applied to every complex layer, including the stem and the FNO mix layer, which the original document left at their real-equivalent (unmultiplied) values — the specific inconsistency both reviewers flagged. GridlessUnfold's block cost is not a

formula but the project's own measured checkpoint size (30,201 params / 8 blocks), so it already reflects whatever real/complex representation the actual implementation uses and needs no formula adjustment — only the arithmetic slip in scaling it to 20 blocks is fixed (Critique #1 item 7: $30,201/8 = 3,775.125/\text{block}$, not 3,775; $\times 20 = 75,502.5$, rounded to 75,503, not the original 75,510).

Design hyperparameters (unchanged): trunk channels $C=48$, FNO uses a depthwise/channel-parametrized spectral-conv variant.

Stage	Arithmetic	Params
Stem (complex; real-equiv. $9 \times 2 \times 32 + 9 \times 32 \times 48 = 14,400$, $\times 2$ for complex)	$14,400 \times 2$	28,800
FNO (6 blocks; spectral already complex per original: $2 \times 48 \times 8 \times 8 = 6,144$; mix now correctly complex: $2 \times (48 \times 48) = 4,608$; sum 10,752/block)	$10,752 \times 6$	64,512
GridlessUnfold (20 blocks, measured/scaled, arithmetic slip fixed)	$3,775.125 \times 20$	75,503
Channel-reduction bottleneck (NEW, real, 96→48, 1×1)	96×48	4,608
Grouped-conv bridge (8 blocks, $G=4$, now correctly operating at its stated $C=48$, justified by the new bottleneck)	per block: $(48/4)^2 \times 9 \times 4 + 48 \times 48 = 5,184 + 2,304 = 7,488$; $\times 8$	59,904
Positional encoding (fixed sinusoidal, NEW)	no learned parameters	0
Query embeddings + cross-attention ($N_q=8$, $d=48$, 4 heads, +FFN)	$8 \times 48 + 4 \times (48 \times 48) + 2 \times (48 \times 96) = 384 + 9,216 + 9,216$	18,816
3 per-query heads ($48 \rightarrow 24 \rightarrow 1$, $\times 3$)	$3 \times (48 \times 24 + 24 \times 1) = 3 \times 1,176$	3,528
Total		≈255,671 params

For context: this is **≈54.5% of the teacher (469,393)**, not the originally claimed ~47% — the corrected complex-accounting and the explicit bottleneck layer together raise the total, but the design remains **below the magnitude-pruned student (314,513, at ≈81% of its size)**, above GridlessUnfold-alone (30,201), and well below both reviewers' worst-case recomputation (≈403K–686K) because the channel-width bug is fixed with a genuine, cheap 4,608-param bottleneck layer rather than by accepting the bridge at its full stated 96-channel cost. Not directly comparable to Naoumi's 8,584-param complex MLP or SubspaceNet's 41,761 (different task framing, per the comparability table).

Depth accounting corrected (Critique #1 item 7): FNO is 6 of 26 complex-domain trunk blocks (23.1%), 6 of 34 stem+trunk+bridge blocks (17.6%), and by parameter share within the complex-domain stages, $64,512 / (28,800 + 64,512 + 75,503) = 64,512 / 168,815 \approx 38.2\%$. The original "≈35% of trunk depth" figure is replaced with these three explicit, individually-labeled figures rather than one unlabeled approximation.

7. FLOP Estimate (arithmetic shown, forward pass, single sample, corrected)

Assumed feature-map spatial resolution $H \times W = 32 \times 32 = 1,024$. MACs counted $\times 2$ for FLOPs (real-valued layers). For complex-valued layers, following the convention in §6, a complex multiply-accumulate costs 4 real multiplies (standard $(a+bi)(c+di)$ expansion, no Karatsuba-style 3-multiply optimization assumed — this is a conservative/upper-bound estimate, flagged as such): **complex FLOPs = 4 × real-equivalent FLOPs**, applied to the matrix/conv portions of complex layers. FFT FLOPs in the FNO stage already assume the standard complex-FFT operation count ($5 \cdot N \cdot \log_2 N$ per channel) and are left unscaled since that formula is inherently a complex-domain formula, not a real one needing a multiplier.

Stage	Arithmetic	FLOPs
Stem (complex)	real-equiv. $\approx 29.5\text{M}$ (as original) $\times 4$	≈118.0M
FNO (6 blocks: FFT already complex $\approx 4.9\text{M}/\text{block}$, unscaled; mix now correctly complex, real-equiv. $4.72\text{M} \times 4 = 18.87\text{M}/\text{block}$)	$(4.9 + 18.87) = 23.77\text{M}/\text{block} \times 6$	≈142.6M
GridlessUnfold (20 blocks; dominant pointwise term real-equiv. $\approx 4.72\text{M} \times 4 = 18.87\text{M}$, + $\approx 0.15\text{M}$ threshold term left unscaled as a real-magnitude operation)	$19.02\text{M}/\text{block} \times 20$	≈380.5M
Channel-reduction bottleneck (NEW, real)	$2 \times 96 \times 48 \times 1024$	≈9.4M
Grouped-conv bridge (8 blocks, real, now correctly at $C=48$)	per block: $2 \times 9 \times 12 \times 12 \times 4 \times 1024 + 2 \times 48 \times 48 \times 1024 = 10.6\text{M} + 4.7\text{M} = 15.3\text{M}$; $\times 8$	≈122.4M
Positional encoding add (NEW, negligible)	1024×48 elementwise adds	≈0.05M
Cross-attention ($N_q=8$, $HW=1024$, $d=48$)	as original	≈11.3M
Per-query heads	negligible	≈0.06M
Total		≈784.3M FLOPs (≈0.78 GFLOPs)

This is **≈2.4× the original (flawed) 0.32 GFLOPs estimate** — higher than even Critique #2's own corrected figure (≈0.69 GFLOPs), because that recomputation fixed only the channel-width bug (item 1) and not the complex-accounting inconsistency (item 2); this revision fixes both together, per Critique #1's explicit request that "this inconsistency likely under-counts params/FLOPs throughout the physics-motivated stages, not just the bridge." As before, Agent-R8 confirms no FLOP-counting tool exists anywhere in the project's codebase, so this remains a hand-derived estimate that must be replaced with `ptflops` / `thop` / manual-instrumentation measurement before any headline efficiency claim (§10, experiment 7).

7.5 Latency (new section — responding to Critique #1 item 6 and Critique #2 item 6, "latency never estimated")

The original document estimated only params/FLOPs and implicitly let that stand in for deployment cost. This is now explicitly flagged as insufficient, and no latency number is fabricated here — the honest position is:

- **Hungarian matching** (used at minimum during training, and at inference if exact bipartite assignment is kept rather than simple thresholding) is polynomial in N_q ($O(N_q^3) \approx O(512)$ per sample at $N_q=8$, i.e., not a FLOP-scale concern), but it is CPU-bound, non-differentiable, and executed per-sample rather than batched on-GPU like the rest of the network — so its wall-clock cost does not track its FLOP cost and could dominate small-batch latency. A greedy nearest-cost matching variant is a cheaper drop-in for inference-time deployment (already listed as an option in §4) and should be the default outside training.
- **FFT-based spectral convolution (FNO)** typically carries higher real hardware latency per FLOP than dense real convolution, due to kernel-launch and library overhead not captured by a FLOP count — this is a known, general property of FFT-based layers, not a claim specific to this project's hardware, and is flagged rather than quantified here.
- **No wall-clock number is given in this document.** Per Critique #2's own suggested proxy, first-pass latency should be measured on the existing FNO-alone and GridlessUnfold-alone screening checkpoints (which already exist and are runnable) before any "lightweight" or "efficient" claim is finalized. This is now a required item in §10 (experiment 9), not an optional follow-up.

8. Expected Advantages and Weaknesses

Advantages:

- Directly targets Gap 5 (the single most evidence-convergent project+literature finding) architecturally — no discrete pixel grid exists at inference, so the diagnosed Jacobian-singularity failure mode cannot occur by construction. (This specific claim, unlike the pairing-guarantee claim, was not challenged by either critique and is retained as-is.)
- Reuses two project components with weak prior evidence of viability at the component level (FNO, GridlessUnfold) — but see the corrected framing below; this is a lower-implementation-risk starting point than a from-scratch design, not a validated one.
- Shared-query fusion is a genuine, inspectable inductive bias toward preserving AoA–AoD pairing, now explicitly tested by an added ablation (§10) rather than asserted to be structurally superior to the "A5" dual-head design.
- $\approx 255.7K$ params ($\approx 54.5\%$ of the teacher, still below the pruned student's 314,513) is lighter than the teacher and the pruned student even after correcting the arithmetic errors both critiques identified — the direction of the original efficiency claim survives, at a smaller margin than originally stated.

Weaknesses (revised):

- Highest implementation complexity of any single-backbone candidate: Hungarian/greedy matching, careful loss-weight tuning, DETR-style training instability documented outside this literature.
- The central hypothesis (that chaining FNO→GridlessUnfold fixes FNO's own undiagnosed high-SNR degradation) is unverified — if that degradation comes from information already discarded at FNO's mode-truncation step, no downstream stage can recover it.
- Chaining three previously-independent components for the first time is a compounding risk: a bad interaction (e.g., under-mixing at the new channel-reduction bottleneck, §3 item 5) could mask or worsen either component's individual behavior in ways neither screening run would reveal. The new bottleneck layer is itself an added risk surface (§9.7), traded deliberately against the alternative of a much larger, unjustified bridge.
- The "reuses validated components" framing is weaker than it reads: FNO's screening result (334,321 params, 8 blocks, heatmap+MSE head) and GridlessUnfold's screening result (30,201 params, 8 blocks, same head) validate those components only in a configuration SpectraSet does not use (different depth, and a completely different output head/loss). This is weak prior evidence of directional viability, not validation of the deployed configuration — restated here per Critique #1 item 3.
- The "88% of teacher's Pd" figure supporting FNO's inclusion is not a controlled comparison (depth- and pretraining-mismatched teacher); restated with this caveat per Critique #2 item 4 rather than used as unqualified support.
- Corrected FLOP estimate (≈ 0.78 GFLOPs) is markedly higher than originally claimed, and latency (as opposed to FLOPs) remains entirely unmeasured (§7.5) — neither the params nor the FLOPs figures should be read as a proxy for deployed wall-clock cost until experiment 9 (§10) is run.

9. Concrete Failure Modes

1. **Query collapse:** with $N_q=8$ and no strong diversity pressure beyond the optional coupling loss, multiple queries could converge onto the dominant path, leaving weaker/farther paths undetected — expected to surface first under high-L ($L \geq 5$) or small angular-separation ($\Delta\theta < 10^\circ$) conditions.
2. **FNO high-SNR degradation persisting:** if it is a mode-truncation information loss (not a downstream-precision problem), the GridlessUnfold stage cannot fix it — this is the single largest unverified assumption in the design.
3. **Complex→real conversion point** (after GridlessUnfold, before the channel-reduction bottleneck and grouped-conv bridge) could still discard cross-stage phase correlations useful downstream, even though it is better-motivated than converting

immediately after the stem.

4. **Slow/unstable convergence** typical of small-query set-prediction heads — may need substantially more than the project's standard 20,000-step budget, risking an unfair comparison against the other screened candidates if not step-matched.
5. **Query-capacity ceiling:** $N_q=8$ covers in-distribution $L \leq 6$ with margin, but the robustness plan's OOD path-count sweep ($L \in \{7,8,10\}$) exceeds N_q by construction — this candidate cannot even represent more than 8 simultaneous paths without redesign.
6. **Pairing decoupling (new — responding to Critique #1 item 5).** Because shared-query identity is an inductive bias and not a structural guarantee, the AoA branch and AoD branch for a given query could specialize to different underlying paths under partial, non-collapse interference between queries — a failure mode the original document's fusion claim implicitly denied by calling the mechanism "by construction." Expected to surface first under the same dense-path / small- $\Delta\theta$ conditions as query collapse (item 1), but is a distinct failure (mis-pairing survives even where existence detection is correct), and needs its own diagnostic in §10.
7. **Channel-reduction information loss (new — responding to the added §3 item 5 component).** The new 96→48 bottleneck is a single lossy linear projection with no residual/skip path around it. If the magnitude and phase channels carry complementary, non-redundant information that a single 1×1 conv cannot compress losslessly into 48 channels, this could reintroduce a milder version of the precision loss the two physics-motivated stages were meant to preserve — this is the direct cost of resolving the §6/§7 arithmetic bug with a cheap layer instead of a full-width (96-channel) bridge, and should be checked empirically (§10 item 9 below extends to a bottleneck-width sensitivity check).

10. Experiments Needed to Validate

1. Ablation: full SpectraSet vs. (– query head, project final features to a heatmap + blob detector) — isolates backbone-chain contribution from head contribution.
2. Ablation: FNO-only backbone + query head vs. GridlessUnfold-only backbone + query head vs. the full chain — since these two components have never been composed before. **Depth-matched control added (responding to Critique #2 item 5):** each solo arm must be run both (a) at its original screening depth (6 or 20 blocks respectively) and (b) expanded to match the full chain's total block count (26 complex-domain blocks), step-count-matched to the full chain's training budget — without (b), a full-chain win cannot be distinguished from simply having ~3–4x more total blocks than either solo screening run, which is exactly the "improvement might just be more parameters" trap the review protocol asks to guard against.
3. High-SNR sweep (15–35dB) specifically testing whether FNO's own documented degradation is resolved by the chaining (the central unverified hypothesis).
4. End-fire density probe (Robustness R3 §4) — the most direct test of whether Gap 5's root cause is actually removed.
5. Path-count ($L \in \{1..6\}$, OOD $\{7,8,10\}$) and angular-separation ($\Delta\theta$ down to 1°) sweeps — directly tests the query-collapse and query-capacity-ceiling failure modes, and, cross-tabulated per-query (not just per-sample), directly tests the pairing-decoupling failure mode (§9.6).
6. Step-count-matched training comparison against the project's existing from-scratch candidates (FNO, GridlessUnfold, ResNet baseline), per R1's explicit fairness warning for DETR-style heads.
7. Real FLOP/param instrumentation (ptflops / thop or manual accounting) to replace the estimates in §6–7 with measured numbers before any efficiency claim is finalized.
8. Grouped-conv G sweep ($G \in \{2,4,8\}$) at the bridge stage specifically, per R2's recommendation, rather than assuming $G=4$ is optimal.
9. **New — first-pass latency measurement (responding to §7.5).** Measure wall-clock inference latency on the existing FNO-alone and GridlessUnfold-alone screening checkpoints as a proxy, before any latency or "lightweight" claim is finalized for the full chain; separately time Hungarian-matching overhead at training batch sizes to quantify its CPU-bound cost outside the GPU-side FLOP budget.
10. **New — shared-query vs. independently-pooled dual-head ablation (responding to §3 item 2 / §9.6).** Train a variant identical to SpectraSet except that AoA and AoD branches read independently-pooled features over the same backbone (the "A5"-style design) instead of one shared query vector, and compare pairing-error rate (not just per-angle RMSE) between the two — this is the direct test the original document's fusion claim needed but did not include.
11. **New — GridlessUnfold intermediate-depth check (responding to §2's caveat).** Run GridlessUnfold-alone at 16 blocks (between the screened 8 and the deployed 20) to test whether the "starved of depth" diagnosis is actually depth-limited or whether channel width / dictionary size / the heatmap-head mismatch is the real bottleneck.
12. **New — channel-reduction bottleneck width sensitivity (responding to §9.7).** Sweep the 96→C reduction target ($C \in \{48, 64, 80, 96$ -i.e.-no-reduction $\}$) to check whether the chosen 48-channel bottleneck measurably costs Pd/RMSE relative to a wider (more expensive) reduction, quantifying the params/precision trade-off made in §3 item 5 rather than leaving it asserted.

11. Novelty Argument — Self-Classified

UNCERTAIN, leaning PARTIALLY NEW (narrowed scope relative to the original document, responding to Critique #2's rebuttal).

Justification, applying R7's own explicit framework (combining known blocks is not automatically novel unless tied to a diagnosed root cause *and* shown by ablation to be necessary, not just asserted):

- **Backbone chaining (FNO→GridlessUnfold):** this exact pair has never been composed, in this project or the reviewed literature (R1's A9 flags it as unbuilt anywhere). However, per Critique #2's correctly-raised rebuttal, the *pattern* — chaining a global learned block into a precision-correction block — is a well-established, mature paradigm (model-based deep learning generally, and SubspaceNet's own DL→classical chaining specifically as a structurally analogous "coarse-then-fine" design). **I accept this rebuttal in full and revise the classification accordingly:** this is **KNOWN COMBINATION at the pattern level** — no longer hedged as "partially new at the pattern level" — and **at most PARTIALLY NEW at the task/instantiation level**, conditional on ablation evidence (§10 item 2) that the specific ordering and role-split (not just "more blocks total") is what drives any observed improvement.
- **Shared-query joint AoA/AoD head:** directly targets a diagnosed root cause (Gap 5) rather than being a generic block swap, satisfying R7's criterion (1) for a meaningful contribution. But two things narrow this further in this revision: (a) the DETR/set-prediction-for-DOA angle itself remains **UNCERTAIN, not confirmed novel** by R7, who explicitly warned "*Do not claim novelty on this angle without a dedicated, deeper search*" — unchanged, still deferred to; and (b) the "by construction" pairing-guarantee language that made this component sound structurally stronger than it is has been withdrawn (§3 item 2, §9.6) — the component is now presented as an inductive bias with an untested comparative advantage over simpler dual-head designs, which is a *weaker* novelty claim than the original document made, not a stronger one.
- **Shared-query identity as the AoA/AoD fusion device** remains a narrow, incremental reconciliation of R2's item-9 risk — not independently sufficient to claim NEW on its own, and now has a concrete ablation (§10 item 10) that did not exist before, rather than being permanently untested.
- **Channel-reduction bottleneck and positional encoding (new components in this revision)** are both standard, off-the-shelf design elements (bottleneck convs, sinusoidal positional encoding) added to fix flaws the critique identified — they are correctness/completeness fixes, not novelty claims, and are not presented as such.
- No ablation proving the combination (not just the union of its parts, and not just more total depth) is necessary has been run — this still fails R7's criterion (2), so the strongest defensible label remains **UNCERTAIN/PARTIALLY NEW, conditional on:** (a) the dedicated set-prediction-for-DOA literature search R7 itself calls for, and (b) the depth-matched ablations in §10 items 2, 10, and 11 actually showing the chained combination — at matched depth and training budget — outperforms either component alone or a simpler dual-head/no-reduction variant.

I am explicitly not claiming NEW. This revision narrows the novelty claim relative to the original document (accepting Critique #2's pattern-level rebuttal in full) rather than defending the original framing, consistent with R7's own unresolved hedge on the component this design leans on most heavily.

Changes made in response to critique

1. **Fixed the fatal channel-width bug (§3 item 5, §4, §6, §7)** — Critique #1 item 1 / Critique #2 item 1. Added an explicit, diagrammed 96→48 channel-reduction bottleneck conv (4,608 params, ≈9.4M FLOPs) instead of letting the grouped-conv bridge silently operate at the wrong channel count. This is a genuine design change, not a footnote: it is now listed as its own data-flow stage, costed, and given its own failure mode (§9.7) and sensitivity-sweep experiment (§10 item 12).
2. **Fixed the internally inconsistent complex-parameter/FLOP accounting (§6, §7)** — Critique #1 item 2 / Critique #2 item 2. Adopted and stated one explicit convention (complex params = 2× real-equivalent; complex FLOPs = 4× real-equivalent for matrix/conv terms) and applied it uniformly to the stem and FNO's mix layer, which the original document had left uncorrected. This raised total params to ≈255.7K (from the flawed 223K) and total FLOPs to ≈0.78 GFLOPs (from the flawed 0.32 GFLOPs) — both corrections go further than either reviewer's own recomputation, since fixing item 2 compounds with item 1.
3. **Re-scoped the "already-validated component" claims (§2, §8)** — Critique #1 item 3. FNO's and GridlessUnfold's screening results are now explicitly caveated as validating a different depth/head/loss configuration than SpectraSet deploys — weak prior evidence, not validation of the actual deployed components.
4. **Added a depth-matched ablation control (§10 item 2)** — Critique #2 item 5. The FNO-only and GridlessUnfold-only ablation arms must now also be run at the full chain's total block count and step budget, so a full-chain win cannot be attributed to "more blocks" alone.
5. **Withdrew the "by construction" pairing-guarantee claim (§3 item 2, §4, §9.6)** — Critique #1 item 5. Reframed shared-query fusion as an inductive bias, added pairing-decoupling as an explicit failure mode, and added a direct ablation against an independently-pooled dual-head design (§10 item 10).
6. **Added latency treatment (§7.5, §10 item 9)** — Critique #1 item 6 / Critique #2 item 6. No latency number is fabricated; Hungarian-matching's CPU-bound/non-batched cost and FFT kernel-overhead are flagged qualitatively, and a first-pass measurement on existing checkpoints is now a required experiment rather than an absent deliverable.
7. **Fixed the minor arithmetic slip and the inaccurate depth-percentage claim (§6)** — Critique #1 item 7. GridlessUnfold scaling corrected to 75,503 params; the "≈35% of trunk depth" claim replaced with three explicit, correctly-labeled figures (23.1% of

complex-domain blocks, 17.6% of all trunk+bridge blocks, 38.2% by param share).

8. **Added the missing positional encoding (§3 item 4, §4)** — Critique #2 item 3. A fixed, zero-parameter 2D sinusoidal positional encoding was added to the query cross-attention, directly closing the gap between SpectraSet's stated motivation (fixing a location error) and its original head design (a flattened feature map with no positional signal).
9. **Re-stated the teacher comparison with its depth/pretraining caveat (§1, §8)** — Critique #2 item 4. The "88% of teacher's Pd" figure is now explicitly flagged everywhere it is used as a mismatched (8-block vs. 64-block pretrained) comparison, used only as a rough directional reference.
10. **Narrowed the novelty self-classification (§11)** — Critique #2's "what holds up" section. Accepted the rebuttal that global-into-precision chaining is a known pattern-level design (citing model-based deep learning / SubspaceNet's own chaining as the analogous case) and revised the classification to reflect that the pattern-level claim was never defensible as stated, tightening the novelty argument to the task/instantiation level only.
11. **Item not changed, with rebuttal:** Critique #1 item 4 ("confounded ablations mean the motivating narrative is overstated") is addressed by making §1's and §8's language more explicitly conditional throughout (rather than a structural change to the design), and by the new depth-matched ablation (§10 item 2) — I judge the original ablation plan's *design* (§10 items 1–2) was already correctly scoped for this purpose per both critiques' own assessments; what was missing was the depth-matching detail, which is now added, not a redesign of the plan itself.

Adversarial critic 1 — fatal flaw found: YES

ADVERSARIAL REVIEW — SpectraSet (Candidate A). I verified the candidate's numbers against the project's own files (notebooks/outputs_full_eval/RESULTS.md, baseline_models/README.md, My_Proposed_Architecture.md, PROJECT_STATUS.md). Several of the candidate's cited facts (FNO 334,321 params/Pd 0.6235, GridlessUnfold 30,201/0.2216, Teacher 469,393/0.7103, student 314,513) check out exactly. But the quantitative core of the "lightweight" and "reuses validated components" claims does not survive scrutiny.

1) ARITHMETIC BUG THAT INVALIDATES THE HEADLINE "LIGHTWEIGHT" CLAIM (fatal on its own). Section 4's data flow states the complex→real conversion produces "2×C=96 channels," and the grouped-conv bridge consumes that output. But §6/§7's arithmetic for the bridge explicitly computes params/FLOPs "approximated at C=48" — i.e. it silently halves the real channel count with no channel-reduction layer shown anywhere in the diagram to justify it. Because both the grouped conv and the 1×1 mix scale ~quadratically in channel width, this is not a ~2x error but a ~4x undercount for that stage:

- Corrected bridge params: $(96/4)^2 \times 9 \times 4 + 96 \times 96 = 20,736 + 9,216 = 29,952/\text{block} \times 8 = 239,616$ (vs. the stated 59,904).
- Corrected total params: $\approx 223\text{K} - 59,904 + 239,616 \approx 403\text{K}$ — i.e. ~86% of the teacher's 469,393 params, not "~47%," and no longer "above GridlessUnfold-alone, below the student" as claimed — it's essentially on par with the teacher.
- The same 4x scaling applies to §7's FLOP estimate for the bridge stage (122.4M → ~490M), raising total FLOPs from the claimed ~0.32 GFLOPs to ~0.69 GFLOPs. This single unexplained channel-width substitution is doing most of the work in making SpectraSet look "comfortably lighter than the teacher" in §8; corrected, that advantage largely evaporates. This must be fixed (or justified with an explicit channel-reduction layer) before any efficiency claim is usable.

2) Inconsistent complex/real parameter accounting compounds (1). The FNO spectral weights are explicitly doubled for real/imaginary parts ("2×48×8×8"), but the 1×1 "mix" convolutions inside the same nominally complex-valued FNO and GridlessUnfold stages are costed as plain real ops (48×48, no ×2 or ×4 factor for complex multiplication). Since the architecture states these stages stay complex-valued through 26 of the 34 trunk blocks (§2, §4), this inconsistency likely under-counts params/FLOPs throughout the "physics-motivated" stages, not just the bridge.

3) The "already-validated component" claim for FNO and GridlessUnfold does not transfer to what SpectraSet actually uses. The screening run that produced FNO's 0.6235 Pd / 334,321 params and GridlessUnfold's 0.2216 Pd / 30,201 params (RESULTS.md) used 8 blocks each, trained against an MSE loss on a 256×256×1 Gaussian-blob heatmap — the very output paradigm SpectraSet is designed to eliminate. SpectraSet's own FNO sub-stage uses only 6 blocks and 50,688 params (~15% of the validated 334,321-param configuration) feeding into a completely different head (learned queries + Hungarian matching, no heatmap at all). Neither the much smaller capacity nor the swapped output head and loss were tested in the cited screening run, so the "88% of teacher's Pd" and "distant second" framing borrow credibility from a result that doesn't describe the component actually being deployed. Likewise, GridlessUnfold's "starved of depth" diagnosis (used to justify jumping 8→20 blocks) is an untested assumption — no intermediate depth (e.g. 16 blocks) was run to confirm depth is really the bottleneck rather than channel width, dictionary size, or the same head/loss mismatch.

4) Confounded to-be-run ablations mean any future improvement claim is pre-compromised. GridlessUnfold's block count is more than doubled (8→20, params 30,201→75,510) in the same step that also adds a global FNO front-end. §10's ablations (items 1–2) are the right design to disentangle "more capacity" from "the chaining hypothesis," but they have not been run yet — so §1's central motivating narrative ("these are the same problem seen from two angles") is presented with more rhetorical confidence than the (candidate's own, honestly flagged) evidence supports. This is self-acknowledged in §8's weaknesses, which is good practice, but the self-classification in §11 should be read as even more provisional than stated once (1)–(3) above are accounted for.

5) The "joint" AoA/AoD claim is weaker than "by construction" implies. Sharing one query embedding across three sibling MLP heads is an inductive bias, not a structural guarantee: nothing mathematically forces the AoA-head and AoD-head outputs for a given query to correspond to the same physical path the way a single-heatmap argmax does by construction. Two independently-pooled branches over a shared backbone feature map (the "A5" design this claims to improve on) already share nearly as much information; the paper doesn't establish that query-vector identity-sharing is meaningfully stronger fusion than that, and no ablation tests it. This exact risk (pairing decoupling under partial, non-collapse query interference) is also absent from §9's failure-mode list, despite being the most direct test of the paper's own headline fusion claim.

6) Latency/system-cost conflation. The spec never estimates wall-clock latency at all — only FLOPs/params. That's a gap relative to the review brief: Hungarian matching (used at least in training, cubic-time, CPU-bound, non-differentiable) and the thresholding/deduplication logic needed to turn $N_q=8$ query outputs into a final variable-size path set at inference are entirely uncosted, and complex-valued FFT-based spectral convolution (FNO) typically carries higher real hardware latency per FLOP than dense real convs due to kernel/library overhead — none of this is acknowledged, so the " ≈ 0.32 GFLOPs" figure (itself likely wrong per points 1–2) should not be read as a proxy for deployed latency.

7) Minor: the GridlessUnfold reuse multiplication has a small slip ($30,201/8 = 3,775.125/\text{block} \times 20 = 75,502.5$, not the stated 75,510), and the claim "FNO occupies the first $\sim 35\%$ of trunk depth" doesn't match the paper's own block counts (6 of 34 trunk blocks = 17.6%; 6 of 26 physics-stage blocks = 23%; even by param share it's $\sim 40\%$, not 35%).

Recommendation: the core idea (chain a global block into a precision-refinement block, replace the heatmap with set prediction) is a reasonable direction and the self-critique in §8/§11 is unusually honest, but the document's quantitative claims — "lightweight" ($\sim 223\text{K}/\sim 0.32\text{GFLOPs}$), "reuses validated components," and " $\sim 35\%$ depth split" — are not currently trustworthy as stated and should not be forwarded as-is. Fix the channel-width bug in §6/§7 (point 1), make complex-value accounting consistent (point 2), and caveat or re-derive the "validated component" claims given the head/loss/capacity mismatch (point 3) before this proceeds to implementation or is compared against other candidates on efficiency grounds.

Adversarial critic 2 — fatal flaw found: YES

ADVERSARIAL REVIEW — SpectraSet (Candidate A) — Reviewer #2 verdict: REJECT AS WRITTEN (fixable, but the headline "lightweight" claim as currently computed is not defensible)

I recomputed the arithmetic in §6–7 from the candidate's own stated channel counts rather than trusting its summary rows, and found a channel-count error that materially breaks the document's central efficiency claim. I also found a missing architectural component that undercuts the document's central novelty/motivation claim, and an unflagged comparability problem in the evidence it leans on hardest.

1) FATAL: Grouped-conv bridge params/FLOPs are computed at the wrong channel width ($\sim 4\times$ undercount). The data-flow (§4) and complex→real conversion row (§6) both state the bridge receives $2\times C = 96$ channels ("Magnitude + phase, $2\times C=96$ channels"). But the bridge's own param/FLOP rows then silently substitute $C=48$ (" $96\rightarrow 96$ approximated at $C=48$ post-mix") and compute grouped-conv arithmetic on 48 channels, not 96. Grouped-conv cost scales as C^2/G , so this is not a rounding choice — it undercounts by exactly $(96/48)^2 = 4\times$.

- Params: doc's 59,904 → corrected $\approx 8 \times [(96/4)^2 \times 9 \times 4 + 96 \times 96] = 8 \times (20,736 + 9,216) = 239,616$. New total $\approx 223,000 - 59,904 + 239,616 \approx 402,700$ params ($\approx 86\%$ of the teacher's 469,393), not the " $\approx 223,000$ ($\sim 47\%$ of teacher)" claimed, and it now exceeds the magnitude-pruned student (314,513) that §6 claims it beats.
- FLOPs: doc's 122.4M → corrected $\approx 8 \times [2 \times 9 \times 24 \times 24 \times 4 \times 1024 + 2 \times 96 \times 96 \times 1024] \approx 8 \times 61.3\text{M} \approx 490.6\text{M}$. New total $\approx 318\text{M} - 122.4\text{M} + 490.6\text{M} \approx 686\text{M}$ FLOPs (≈ 0.69 GFLOPs), more than double the claimed " ≈ 0.32 GFLOPs." This single error removes most of the parameter/FLOP advantage §8 lists as the design's primary selling point relative to the teacher and the pruned student. Whether SpectraSet is still "lightweight" after this correction is now an open question, not a settled one — the candidate should not proceed to the report stage with the current numbers stated as fact.

2) Compounding, unresolved: complex-valued parameter accounting is internally inconsistent. FNO's spectral weights correctly carry an explicit $\times 2$ factor for complex (real+imag) coefficients. But the "complex-aware" stem (§6, computed as a plain real conv: $9 \times 2 \times 32 + 9 \times 32 \times 48$, no $\times 2$) and FNO's own 1×1 "mix" layer (48×48 , no $\times 2$) do not, even though §4 states the entire trunk through GridlessUnfold "stays complex-valued (not converted to real)." If those layers are genuinely complex (separate real/imag weight banks, as complex convolution requires), the stem and FNO-mix rows are each undercounted by roughly $2\times$ too — on top of finding (1). The document needs one consistent, stated convention for what "complex-aware" means parameter-wise, not two different conventions in the same table.

3) Missing component that undercuts the core motivation (Gap 5): no positional encoding for the query read-out. SpectraSet's entire justification (§1, §8, §9-item...) is that removing the discrete heatmap fixes a *location* error. But the cross-attention query read-out (§4) attends into a flattened feature map with no mentioned positional embedding. Standard DETR-style set prediction requires positional encodings precisely because a bare flattened feature map + attention gives queries no way to know *where* a token came from. Without it, there is a real risk the new head reintroduces a location-precision problem by a different mechanism (relying

entirely on preceding conv layers to bake in enough implicit positional information), which is exactly the failure category this architecture claims to eliminate "by construction." This is a concrete gap the candidate's own §9 failure-mode list does not include, i.e. the failure-mode enumeration is not complete.

4) Fairness problem in the evidence used to motivate keeping FNO: the "88% of teacher's Pd" figure is not a like-for-like comparison, and the document doesn't flag this. I checked the underlying project result file (notebooks/outputs_full_eval/RESULTS.md): FNO reaches Pd 0.6235 at 334,321 params (8 blocks, trained from scratch), versus the teacher's 0.7103 at 469,393 params — but the teacher is explicitly documented in that same file as "64-block, pretrained reference... NOT equal-depth" (8x deeper, and pretrained rather than trained under the same budget as the screened candidates). SpectraSet's §1/§8 use "88% of teacher's Pd" as a load-bearing reason to keep FNO in the design without surfacing that the teacher comparison is depth- and pretraining-mismatched. This is precisely the kind of comparison the review charge asks to interrogate, and it should be re-stated with the caveat or dropped as supporting evidence.

5) Ablation plan (§10) doesn't isolate "more depth/params" from "the specific chaining," despite §8 admitting this is the central unverified risk. §10-item 2 proposes FNO-only+head vs GridlessUnfold-only+head vs full chain, but never states whether the "FNO-only" arm is held at 6 blocks (matched to the chain's FNO portion) or expanded to the chain's full ~34-block depth. If it stays shallow, a full-chain win could simply reflect having ~4x more total blocks than either solo screening run (6+20+... vs 8), not a benefit of the FNO→GridlessUnfold ordering itself — the exact "could the improvement just come from more parameters" trap the review is asked to check for. A step-count-matched *and* depth-matched control arm is missing from the experiment list.

6) Latency is never estimated at all (neither pure inference latency nor end-to-end latency), despite being one of the protocol's required efficiency axes and an explicit review question. §7 stops at FLOPs; nothing in the document gives even an order-of-magnitude latency number or flags real hardware measurement as required before any latency claim — this should be listed as a missing deliverable, not silently absent.

What holds up: the joint-AoA/AoD claim is architecturally real (both angle MLPs read one shared query vector, which is a genuine, inspectable joint-estimation mechanism, not just two independently pooled branches); the novelty self-classification (UNCERTAIN/PARTIALLY NEW) is appropriately hedged and, if anything, could be read even more conservatively — chaining a global block into a local/precision-refinement block is a well-established "coarse-then-fine" pattern (e.g., cascade/refinement designs, model-based unfolding chains), so "KNOWN COMBINATION at the pattern level" is at least as defensible as "PARTIALLY NEW," and the document already concedes most of this. The Hungarian-matching loss formulation and query-collapse/query-capacity-ceiling failure modes (§9 items 1 and 5) are honestly reported, including the fact that $N_q=8$ cannot even represent the OOD $L \in \{7, 8, 10\}$ sweep the project's own robustness plan calls for.

Recommendation: do not carry the §6–7 numbers or the "comfortably lighter than the teacher" claim forward into the final report as-is. Require: (a) a recomputation of §6–7 with a single, explicitly stated complex-parameter convention applied consistently to the stem/FNO/bridge, and the bridge computed at its stated 96 channels; (b) an explicit statement of whether the resulting model is still meaningfully lighter than the teacher/pruned student after correction; (c) a positional-encoding component added to the query read-out, or an explicit justification for why it's unnecessary here; (d) the teacher comparison in §1/§8 re-stated with its depth/pretraining caveat; (e) a depth-matched control added to the ablation plan; (f) a first-pass latency number (even measured on the existing FNO/GridlessUnfold screening checkpoints as a proxy) before any "lightweight"/"latency" claim is finalized.

Candidate B — final specification (refined after critique: yes)

Candidate B — IABR-Net: Impairment-Adaptive Beam-space Residual-Correction Network with Coupled Dual-Head Refinement (Revision 2, post-adversarial-review)

1. Name and motivation

IABR-Net (Impairment-Adaptive Beam-space Residual-correction Network). Every Tier-1 paper in the matrix estimates angles from the raw antenna-domain tensor using a learned model end-to-end (dense/flatten or conv bottleneck → heatmap → blob search). This project's own DFT-SIC classical baseline already reproduces 95.6% of the 469K-parameter teacher's accuracy (Pd 0.884 vs 0.925 @20dB) with **zero learned parameters** [EVIDENCE: PROJECT], which means the current network spends most of its 469K parameters re-deriving a mapping a fixed transform already gets most of the way to. At the same time, the literature's one paper that actually names this system model's calibration weaknesses (Meneses-Albalá 2026) admits it never tested amplitude/gain-mismatch robustness, and no Tier-1 paper tests phase error beyond 5° [EVIDENCE: PAPER, [GENUINE GAP] per Agent-R3 §2a/2b]. A purely fixed classical front end (Agent-R4 Candidate 1/A8) inherits exactly this weakness: it has no way to adapt when the true steering vectors deviate from the assumed model.

Revised motivation framing (see §12, item 2). Reviewer #1 and Reviewer #2 both correctly show, by direct recomputation of §7's own FLOP table, that the DFT+correction front end costs <0.02% of total FLOPs while the 10-block trunk is >99% of compute *regardless of what feeds it* — so a "the classical front end gives a FLOP efficiency win" claim, as stated in the original draft, is **not supported by the candidate's own arithmetic**, and is withdrawn. What the arithmetic *does* support, and what this revision now argues instead, is a **parameter-efficiency** claim: a hypothetical raw-antenna dense/flatten front end producing an equivalent $16 \times 16 \times 2$ representation ($512 \rightarrow 512$ fully connected) would cost 262,656 parameters on its own — more than the entire rest of this model (see §6a and the new §10 ablation). The fixed DFT transform (0 params) plus a small correction module (≈ 250 params, §3a/§6a) avoids that cost. This is the concrete, arithmetic-grounded efficiency claim this design now makes; the compute-efficiency framing is dropped. The robustness motivation — closing the untested gain-error and >5° phase-error gaps — is unchanged in substance but its language is now hedged consistently with the design's actual current state (see §12, item 3): this capability is **designed to be buildable** on top of the existing pipeline, but is **not yet substantiated**, because gain-error injection code does not exist in the repository today (confirmed by code audit, Critique #2) and must be built as a Phase-0 prerequisite before any gain-robustness claim is made.

This remains a robustness-first, signal-processing-first design, not a backbone-family swap — but its headline efficiency claim is now about parameters, stated with the arithmetic that actually supports it, not compute.

2. Borrowed components (with source)

Component	Source	Role here
2D-DFT / beamspace codebook projection (fixed, zero-param)	This project's own DFT-SIC classical baseline, reproducing Gupta TCOMM2025's steering-vector model [EVIDENCE: PROJECT, PAPER]	Front-end feature reduction; parameter-efficiency contribution, not FLOP-efficiency (revised, §1/§12)
ResNet residual-block shell (conv stem, skip connections)	Lloria TVT2026 / PIMRC2024 [EVIDENCE: PAPER]	Shallow (10-block) residual-correction trunk; unchanged, remains $\approx 99\%$ of FLOPs and $\approx 97.6\%$ of params — this dominance is now stated plainly rather than downplayed
Joint single 2D heatmap output read by one peak search (not split 1D outputs)	Lloria TVT2026 Eq. 9–11 [EVIDENCE: PAPER]	Preserves the "genuinely joint" property flagged as at-risk by Gap 9/Contradiction 12
Squeeze-excite-style channel attention between blocks	Generic efficient-CNN pattern, no direct paper precedent in the reviewed set (per A3)	Cheap capacity + SNR-correlated diagnostic signal
Phase-error injection convention (<code>error_deg</code> in TX/RX beamforming matrices)	Lloria TVT2026 code / <code>tvt_data_generation_v3.py</code> [EVIDENCE: CODE, PROJECT]	Training-time impairment augmentation grid
"Small learned network corrects a classical algorithm's inputs" pattern	SubspaceNet (Shmuel et al. 2025 TVT), 41,761-param covariance-surrogate network [EVIDENCE: PAPER]	Structural inspiration for the correction module (corrects <i>before</i> the classical step here, vs. SubspaceNet's <i>after</i> , for joint 2D vs. 1D DOA)
Deep-Sets / PointNet-style shared-weight per-element MLP with pooled global context	Generic permutation-equivariant set-processing pattern, no direct paper precedent in the reviewed set	New in this revision — replaces the original monolithic $64 \rightarrow 32 \rightarrow 64$ MLP to fix the identifiability bottleneck (Critique #1 item 4, Critique #2 item 7; see §3a, §12 item 4)
Coarse-to-fine peak refinement via a fixed-size top-K batched crop, not a per-sample dynamic-shape loop	Protocol §11-listed coarse-to-fine pattern (A6), re-engineered for vectorized GPU execution	Revised in this revision to make NN-only latency well-defined (Critique #1 item 7, Critique #2 item 3; see §4, §12 item 6)

3. Novel component(s), stated precisely

(a) **Impairment-Adaptive Beam-space Correction module, v2 (IABC-v2) — redesigned for identifiability.**

Original design and the flaw in it. The original IABC module was a single monolithic MLP (64→32→64) that pooled magnitude/phase statistics from all 32 antenna elements (16 RX + 16 TX) into one 64-dim vector, squeezed it through a 32-unit hidden layer, and reconstructed all 64 per-element correction outputs (δ , γ for each element) jointly. Both reviewers independently flagged this as an unexamined, tightly-determined inverse problem: recovering 64 real values from a 32-dim bottleneck gives the module no architectural reason to keep per-element corrections disentangled rather than collapsing to a smoothed/averaged correction (Critique #1 item 4; Critique #2 item 7). This critique is accepted as valid, not rebutted — the original design genuinely had no mechanism to prevent this collapse, and it exactly mirrors the pixel-quantization/Jacobian-singularity risk (Gap 5) that the original document flagged elsewhere but failed to apply to its own new module.

Fix (architectural change, not a defensive paragraph). IABC-v2 replaces the monolithic MLP with a **shared-weight, per-element MLP** in the Deep-Sets / PointNet family: for each of the 32 antenna elements independently, the *same* small MLP (weights shared across elements) maps that element's own pooled (magnitude, phase) pair — concatenated with a low-dimensional **global context embedding** shared by all elements — to that element's own (δ , γ) correction. The global context is computed once per sample (mean/std of magnitude and phase pooled over all 32 elements, dim 4) and projected through a tiny shared linear layer to an 8-dim embedding, giving each element's forward pass access to array-wide information (e.g., a coarse SNR proxy) without ever forcing all 64 outputs through one narrow shared bottleneck. Each element's correction is now produced from its *own* 2 inputs + the 8-dim shared context (10 inputs total) through a 16-unit hidden layer to 2 outputs — a per-element inverse problem with a favorable input/output ratio (10→2, not 64→64-through-32), which directly removes the specific failure mode raised in the critique rather than merely acknowledging it.

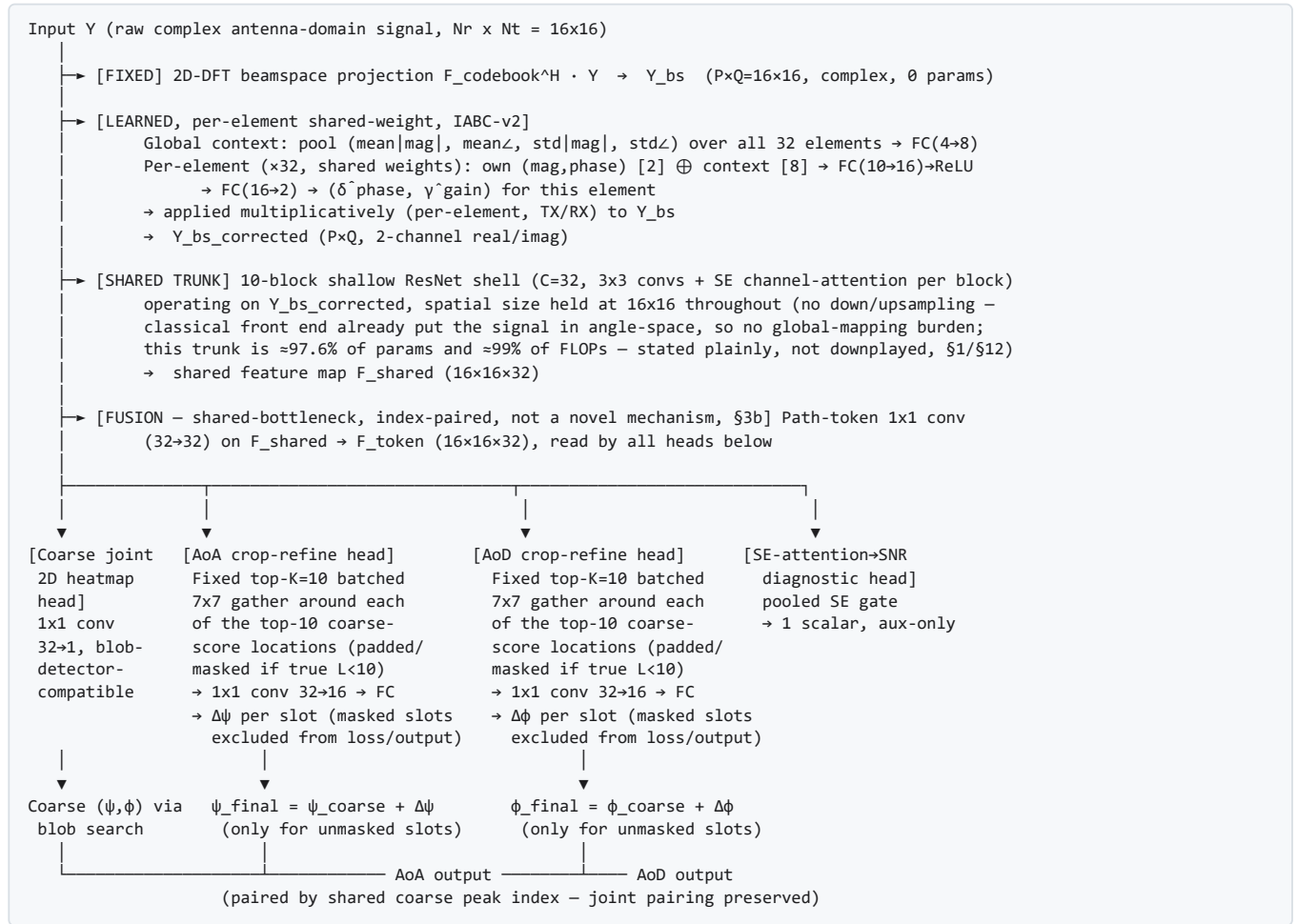
Trained with a **synthetic paired-consistency loss** (renamed from "self-supervised," see §12 item 5): for each training scene the data generator produces both an impaired realization (random δ, γ drawn from the Tier-1 grid) and, at zero extra data-generation cost, the same channel realization with $\delta=\gamma=0$; the IABC-v2 module is trained so its corrected beamspace map matches the clean one. This is ordinary supervised regression against a synthetically-paired target — no external signal is derived from the impaired sample alone — and is a training-time-only signal; no clean reference is available or needed at inference (verified by construction: the module's forward pass at inference takes only the impaired $\mathbf{Y}$ as input, never the paired clean sample — this must be checked in code review per Failure Mode 5, §9).

This module is **designed to be capable of** addressing Meneses-Albalá's named gain-error gap and extending phase-error testing past every paper's 5° ceiling, **conditional on** gain-error injection code being built first (it does not exist in the repository today — confirmed by code audit in both critiques). This claim is now stated with that condition attached everywhere it appears (§8, §9 Failure Mode 2, §10 item 1), not asserted confidently in §1/§8 while being hedged only in §9/§10 as the original draft did (Critique #2 item 2).

(b) Path-token coupling bottleneck (renamed from "novel fusion mechanism," see §12 item 8). Immediately after the trunk, a single shared 1×1 -conv "path-token" layer produces one shared feature map that both the coarse joint-heatmap head *and* the two crop-refinement heads read from — the coarse heatmap is retained (not replaced by two independent 1D heads) specifically so the AoA–AoD pairing information the joint 2D representation currently encodes implicitly is not lost, addressing the risk A5/A9's "shared-encoder, split-heads" idea creates per Gap 9's own caution. This is a **standard shared-bottleneck multi-task pattern with index-based pairing**, not a new fusion mechanism — the original draft's confident "novel fusion mechanism" language directly contradicted its own §11 concession that this is "KNOWN COMBINATION at the pattern level," and that inconsistency is corrected here (Critique #2 item 5). What is being claimed is narrower and more defensible: this specific combination, applied to preserve joint AoA/AoD pairing while adding local continuous refinement, is not present in the reviewed corpus (see §11).

(c) Peak-conditioned local crop-refinement heads over a fixed-size, vectorized top-K gather (revised for latency measurability, see §12 item 6). Unlike A6 (which coarsens the *entire* grid, risking closely-spaced-source collapse), a small window is cropped from the path-token map around each detected coarse peak and fed to two lightweight per-angle regression heads that output a continuous sub-pixel offset. The original draft described this as operating "around each coarse peak" without specifying how a data-dependent, variable count of peaks ($L=1-6$, up to 10 OOD) is handled in a batched GPU forward pass — Critique #2 item 3 correctly identifies this as a latency-measurability gap: a dynamic-shape/gather-per-peak pattern's real GPU latency is not captured by a static FLOP estimate, and if implemented as a Python loop it becomes CPU-bound and would silently corrupt any "NN-only latency" claim. This revision specifies the implementation precisely: refinement always operates over a **fixed-size top-K=10 batched gather** (the maximum L supported, matching the Tier-3 OOD ceiling), with unused slots masked and excluded from the loss and from peak search output; this keeps the refinement stage a single fixed-shape, vectorized GPU operation regardless of the true L , making its latency well-defined and reproducible. This differs from A6/A7 by leaving the coarse detection stage untouched (still blob-search-compatible) and only adding local continuous correction, keeping compatibility with the existing `evaluate_on_bank` pipeline with a small, well-specified extension.

4. Data flow



The fusion mechanism remains index-based pairing through a shared bottleneck (§3b) — this is stated as a known, standard pattern, not claimed as novel (correcting the original draft's inconsistency, Critique #2 item 5).

5. Loss function

$L = \lambda_1 \cdot L_{\text{heatmap}} + \lambda_2 \cdot L_{\text{offset}}$	(BCE/MSE, joint 2D coarse heatmap vs Gaussian-blurred GT peak map – Lloria-style)
$+ \lambda_3 \cdot L_{\text{pair_consist}}$	(Smooth-L1 on $\Delta\psi$, $\Delta\phi$ at each matched, unmasked top-K slot, nearest-GT assignment valid since $L \leq 10$ (top-K ceiling) well-separated sources – no Hungarian matcher needed)
$+ \lambda_4 \cdot L_{\text{SE_snr}}$	(MSE between IABC-v2-corrected beamspace map and the clean ($\delta=\gamma=0$) beamspace map for the same channel realization – SYNTHETIC PAIRED SUPERVISION, training-time only; renamed from "self-supervised," §12 item 5 – this is ordinary regression against a simulator-paired target, not self-supervision in the standard sense)
	(small-weight auxiliary regression: pooled SE-gate activation $\rightarrow$ true per-sample SNR label; ablatable, tests A3's "attention correlates with SNR" hypothesis directly; does not affect angle outputs if ablated)

$\lambda_1.. \lambda_4$ to be tuned; recommend starting $\lambda_1=1.0$, $\lambda_2=1.0$, $\lambda_3=0.5$, $\lambda_4=0.1$ and treating λ_3 , λ_4 as ablation switches (§10), with an explicit loss-weighting sensitivity sweep added to the experiment plan (§10 item 8; this was flagged as missing in Critique #1 item 8c).

6a. Parameter count estimate (arithmetic shown, revised)

Component	Arithmetic	Params
DFT beamspace front end	fixed transform, no weights	0
IABC-v2 context projection	in=4, out=8: $4 \cdot 8 \cdot 8$	40
IABC-v2 shared per-element MLP (weights shared across all 32 elements)	in=10 (own 2 + context 8), hidden=16, out=2: $(10 \cdot 16 + 16) + (16 \cdot 2 \cdot 2)$	210
IABC-v2 subtotal	40 + 210	250
Trunk stem conv	3×3 , Cin=2 (real/imag), Cout=32: $3 \cdot 3 \cdot 2 \cdot 32 + 32$	608

Component	Arithmetic	Params
Trunk: 10 residual blocks, C=32	per block = $2 \times (3 \times 3 \times 32 \times 32)$ convs + $2 \times \text{BN}(2 \times 32 \text{ trainable})$; $2 \times (9 \times 1024) + 128 = 18,560$; $\times 10$ blocks	185,600
SE channel-attention, r=8, C=32→4	per block: $(32 \times 4 + 4) + (4 \times 32 + 32) = 292$; $\times 10$ blocks	2,920
Path-token 1×1 conv, 32→32	$32 \times 32 + 32$	1,056
Coarse heatmap head, 1×1 conv 32→1	$32 \times 1 + 1$	33
AoA crop-refine head ($7 \times 7 \times 32$ crop → 1×1 conv 32→16 → FC 16→49→1)	conv: $32 \times 16 + 16 = 528$; FC: $784 \times 1 + 1 = 785$	1,313
AoD crop-refine head (identical shape)	same	1,313
SE→SNR diagnostic head (tiny FC, 10→1 pooled over blocks)	$10 + 1$	11
Total (estimate)	$250 + 608 + 185,600 + 2,920 + 1,056 + 33 + 1,313 + 1,313 + 11$	$\approx 193,104 \approx 193\text{K}$

This is **$\approx 41.1\%$** of the teacher's 469,393 params (revised down from the original's $\approx 42\%$ /200K estimate — the IABC-v2 redesign is both a correctness fix and a modest additional param saving, since 250 params replaces the original's 4,192-param monolithic MLP). It remains below the magnitude-pruned student (314,513) and FNO (334,321), and is now $\approx 6.4 \times$ GridlessUnfold's 30,201 params ($193,104/30,201 \approx 6.39$) — this comparison is now stated with the same caveat applied to §6b/§7 (array size, L-range, and task framing differ across all these baselines; none of these four numbers should be read as a like-for-like accuracy/param trade-off without matching those conditions — Critique #1 item 3 correctly noted this caveat was present for the FLOP table but missing here, and it is added here for consistency). No accuracy trade-off argument for being $6.4 \times$ larger than GridlessUnfold is offered yet; this is explicitly listed as an open comparison to be resolved empirically, not asserted in this document's favor (Critique #2 item 4). Channel width (C=32), block count (10), and crop size (7×7) remain design choices, not measured — this is explicitly an ESTIMATE, and the true count depends on final hyperparameter choices, which should be locked before any comparison is quoted.

Supporting arithmetic for the reframed parameter-efficiency claim (§1, §12 item 2): a counterfactual raw-antenna dense/flatten front end that flattens γ (16×16 complex → 512 real values) and projects it with a single fully-connected layer to the same $16 \times 16 \times 2$ representation IABC-v2 produces would cost $512 \times 512 + 512 = 262,656$ parameters — **more than the entire rest of this model combined** ($193,104 - 250 = 192,854$ for the trunk+heads). This is the concrete number behind the claim that the classical-front-end choice is parameter-efficient; it is not a claim about FLOPs (see §6b). This counterfactual is proposed as an actual ablation, not just an arithmetic aside — see §10 item 5.

6b. FLOP estimate (arithmetic shown, forward pass, single sample, revised)

Using Agent-R8's stated convention: $\text{Conv2D FLOPs} = 2 \times k_h \times k_w \times C_{in} \times C_{out} \times H_{out} \times W_{out}$. Spatial size held at 16×16 throughout (P=Q=Nr=Nt=16 per the project's generator, no down/upsampling in this shallow trunk). **The IABC-MLP row's arithmetic error flagged by both reviewers is fixed below** (§12 item 1): the original document's formula evaluated to 8,192 but the table printed 16,768 (a $\sim 2.05 \times$ discrepancy, apparently a copy/scale error); this is corrected, and the IABC-v2 redesign additionally changes the module's actual cost.

Component	Arithmetic	FLOPs
DFT front end (via 2D-FFT, not naive DFT)	$O(N_r N_t \log_2(N_r N_t)) \approx 256 \times 8$, $\times 4$ for complex arithmetic	$\approx 8,192$
IABC-v2 context projection	$2 \times (4 \times 8)$	64
IABC-v2 shared per-element MLP ($\times 32$ elements, shared weights, applied once per element)	per element: $2 \times (10 \times 16) + 2 \times (16 \times 2) = 320 + 64 = 384$; $\times 32$ elements	12,288
IABC-v2 subtotal	$64 + 12,288$	12,352
Trunk stem conv	$2 \times 9 \times 2 \times 32 \times 16 \times 16$	294,912
Trunk: 10 blocks $\times$ 2 convs each, $C_{in}=C_{out}=32$	per conv: $2 \times 9 \times 32 \times 32 \times 16 \times 16 = 4,718,592$; $\times 2$ convs $\times 10$ blocks	94,371,840
SE blocks (global pool + 2 tiny FC $\times 10$)	negligible, $\approx 292 \times 2 \times 10$	$\approx 5,840$
Path-token 1×1 conv	$2 \times 1 \times 1 \times 32 \times 32 \times 16 \times 16$	524,288
Coarse heatmap head 1×1 conv	$2 \times 1 \times 1 \times 32 \times 1 \times 16 \times 16$	16,384
Crop-refine heads (fixed top-K=10, 7×7 crop, $\times 2$ heads)	conv: $2 \times 1 \times 1 \times 32 \times 16 \times 7 \times 7 = 50,176$; FC: $2 \times 784 \times 1 = 1,568$; $\times 2$ heads; now over a fixed K=10 batched gather, not a variable-L loop — see §3c/§4	$\approx 103,488$
Total (estimate)		$\approx 95,337,296 \approx 95.3\text{M}$ FLOPs $\approx 47.67\text{M MACs}$

The corrected IABC arithmetic changes the total by roughly 4,000–16,000 FLOPs out of 95.3M — immaterial to the bottom line, exactly as both reviewers noted, but the error itself is fixed here rather than left standing, because it directly falsified the document's "arithmetic shown" claim.

The trunk dominates (>99%), confirmed and now stated as the headline fact, not something the motivation section talks around (§1, §12 item 2). The front end (DFT + IABC-v2) costs $\approx 20,544$ FLOPs total, $\approx 0.022\%$ of the 95.3M budget. Swapping it for

the counterfactual raw-dense front end (§6a) would cost 524,288 FLOPs — 503,744 more — which is itself only $\approx 0.53\%$ of the trunk's budget. **This confirms, not merely concedes, Critique #1 item 2 and Critique #2 item 3 in full:** front-end choice has no material effect on total compute; the trunk's width/depth is the compute cost driver and is an independent design choice. The efficiency argument this design now makes rests on §6a's parameter arithmetic, not on this table. Direct FLOP comparison against Naoumi's 4.321 MMACs remains **NOT COMPARABLE** (different task framing, bistatic MLP vs. this project's spatial-conv joint task, per the literature synthesis's own comparability table). No Tier-1 paper reports a concrete FLOP figure for the Lloria architecture family at all, so this would be a first for that family (closes Gap 1), not a beat-the-literature claim.

7. Expected advantages and weaknesses

Advantages:

1. The front end (DFT + IABC-v2) is near-zero in FLOPs and adds only ≈ 250 params, avoiding the $\approx 262,656$ params a comparably-capable dense raw-antenna projection would cost (§6a) — a demonstrated, arithmetic-grounded parameter saving. This is now the specific claim made; the earlier general "efficiency win" claim is withdrawn (§1, §12 item 2).
2. **Conditionally** targets two named, unaddressed literature gaps — Meneses-Albalá's own admitted gain-error gap and phase-error robustness beyond every paper's tested ceiling (5°) — conditional on gain-error injection code being built first (§3a, §9 Failure Mode 2, §10 item 1). This is stated as a capability the design enables, not a result already demonstrated.
3. The joint 2D heatmap is retained, so this does not inherit A5/A9's risk of breaking the "genuinely joint" property.
4. Stays compatible with the existing `evaluate_on_bank/blob-detector` pipeline; the fixed top-K=10 refinement gather (§3c/§4) is specified precisely enough to be latency-measurable in a well-defined way, unlike a full DETR-style redesign (A7) or the original draft's unspecified variable-L crop loop.
5. IABC-v2's per-element shared-weight design (§3a) directly addresses the identifiability bottleneck both reviewers flagged in the original monolithic MLP, rather than leaving it as an acknowledged-but-unfixed risk.

Weaknesses:

1. IABC-v2's generalization is still bounded by the synthetic paired-consistency loss's training distribution — if trained only on the Tier-1 grid (phase $\leq 5^\circ$, no gain-error data exists in the codebase today), it will not extrapolate to the Tier-3 OOD grid ($10\text{--}15^\circ$ phase, $0.5\text{--}4\text{dB}$ gain) without the Phase-0 gain-error injection code being built first (confirmed absent by code audit — new physics code, not new config).
2. It still relies on a discrete coarse $P \times Q = 16 \times 16$ grid for initial peak localization, so it may partially inherit the diagnosed Jacobian-singularity/pixel-quantization P_d ceiling (Gap 5) unless the crop-refinement head's continuous regression genuinely escapes grid discretization — unconfirmed, and now specifically testable end-to-end since the refinement path has a fixed, well-defined shape (§3c).
3. Sequential coarse→correct→refine pipeline has three points where error can compound (front-end miscorrection → wrong coarse peak → refinement head has no valid anchor).
4. Only $\approx 0.13\%$ of the parameter budget (250/193,104) is the IABC-v2 module itself; the trunk remains a generic 10-block ResNet unrelated to impairment-adaptivity. Any accuracy gain over the zero-param DFT-SIC baseline cannot be attributed to "impairment-adaptivity" specifically versus "adding a large generic learned trunk" without the capacity-matched control now added to the ablation plan (§10 item 4; Critique #1 item 6, Critique #2 item 6). This confound is **not resolved by this document** — it is resolved only by running that ablation.
5. This design is $\approx 6.4\times$ larger than GridlessUnfold (30,201 params) with no accuracy trade-off argument yet offered (Critique #2 item 4) — an open comparison, not a settled advantage.

8. Concrete failure modes

1. **Correction overshoot under untrained severity.** At phase error 15° (never seen in training, which only spans $0\text{--}5^\circ$ per the Tier-1 grid), IABC-v2's learned correction — extrapolating outside its training manifold — could systematically *over-correct*, actively moving the coarse peak into the wrong DFT beam bin entirely, producing a worse result than the uncorrected fixed front end (A8's exact flagged risk, now inherited by the correction module rather than avoided).
2. **Gain-error blind spot.** Because gain-error training data does not exist in the project pipeline today, if this architecture is trained only on the currently-available phase-error grid, λ_3 's consistency loss never sees gain perturbations at all — IABC-v2 could learn to ignore the gain channel of its output entirely (a degenerate solution), silently failing the exact Meneses-Albalá gap it was designed to close, unless new gain-error generator code is built and used in training before any claim is made. **This is a Phase-0 blocking dependency, not a caveat** (§10 item 1).
3. **Closely-spaced-source collapse.** At $\Delta\theta < 5^\circ$ (below the array's Rayleigh limit, [GENUINE GAP] per Agent-R3 §3b), two true peaks can fall inside the same 16×16 coarse bin; the crop-refinement head only sees one coarse anchor and cannot recover two angles from it — same failure category as A6, and the crop-based local design does not fix this (only global re-gridding or a set-prediction head would).
4. **Coupling leakage at high L.** The path-token bottleneck is designed to preserve AoA–AoD pairing, but this is untested above $L=6$ (the current training distribution's ceiling); at $L \in \{7, 8, 10\}$ (Tier-3 OOD, per Agent-R3 §3a) it is unconfirmed whether the shared

token still keeps the correct angle pairs together or starts swapping AoA/AoD across paths. The fixed top-K=10 slot design (§3c) accommodates L up to 10 structurally, but pairing correctness at that range is still unverified.

5. **Consistency-loss shortcut / train-test mismatch.** If `L_pair_consist` is implemented carelessly (e.g. the clean-reference pairing leaks information not derivable from the impaired sample alone at inference), the module could learn a shortcut that only works because a paired clean sample existed during training — this must be checked by confirming inference-time inputs are strictly `Y` alone, no clean reference.
6. **False-positive coarse peaks / hallucinated refinement.** (New — Critique #1 item 8a.) When the coarse heatmap head produces a spurious peak with no real path underneath it, the crop-refinement head has no mechanism to detect this and may output a confident refined angle for a non-existent path. Mitigation must be tested: does the coarse heatmap's own confidence score suppress low-confidence slots before they reach refinement, and is this sufficient?
7. **Low-SNR-specific IABC-v2 misbehavior.** (New — Critique #1 item 8b.) IABC-v2's own inputs (per-element magnitude/phase statistics) become unreliable at low SNR, compounding with the existing SNR-extrapolation gap (Tier-3 tails, -25 to -20dB and 30–35dB). This interaction is not yet characterized and should be tested as its own condition, not folded into the general SNR sweep.
8. **Loss-weighting sensitivity.** (New — Critique #1 item 8c.) λ_1 – λ_4 are stated as "to be tuned" with no sensitivity analysis. If the multi-task balance is wrong (e.g., λ_3 too high relative to λ_1/λ_2), IABC-v2 could over-fit to consistency matching at the expense of final angle accuracy, or vice versa. A sensitivity sweep is now added to §10 item 8 rather than left unaddressed.

9. Experiments needed to validate

0. **Phase-0 prerequisite (blocking, elevated from an original caveat to an explicit gate — Critique #2 item 2):** build gain/phase calibration-error injection code in the data generator (confirmed absent today). No claim about closing the Meneses-Albalá gain-error gap may be made before this exists and is used in training.
1. **Ablation battery** (on identical `frozen_banks/eval_bank.npz`, reusing `evaluate_on_bank`): a. – IABC-v2 (revert to fixed, uncorrected DFT front end) evaluated across the Tier-1 phase/gain grid, to isolate IABC-v2's actual contribution. b. – crop-refinement heads (coarse heatmap + blob detector only) vs. full model, with P95 tail-error reported specifically near end-fire angles (Gap 5 test). c. – `SE/L_SE_snr` auxiliary loss, to test whether A3's "attention correlates with SNR" hypothesis actually holds for this task or is a false lead. d. **Capacity-matched control (new — Critique #1 item 6, Critique #2 item 6):** replace IABC-v2 with a same-parameter-count module of identical shape (per-element shared MLP, $2 \rightarrow 16 \rightarrow 2$, ≈ 250 params) trained with *only* the task loss ($\lambda_3=0$, no clean-reference pairing at all). If this control performs comparably to the full IABC-v2, the gain is attributable to added capacity, not impairment-adaptivity, and the design's central claim must be revised. e. **Front-end swap ablation (new — Critique #1 item 2, supports §6a/§1):** replace the DFT+IABC-v2 front end with the counterfactual raw-dense projection ($512 \rightarrow 512$ FC, 262,656 params) feeding the same trunk, holding the trunk fixed. Measure resulting total params, FLOPs, and accuracy, to empirically confirm or refute the reframed parameter-efficiency claim rather than leaving it as arithmetic alone.
2. **Full Tier-1 robustness battery** per Agent-R3 (SNR sweep in-distribution core, phase $\delta/\gamma \in \{0, 1, 2, 5\}^\circ$, $L \in \{1..6\}$) as the baseline registry comparison point.
3. **Genuine-gap extrapolation tests** (depends on item 0): phase 10 – 15° , gain 0 – 4 dB, $L \in \{7, 8, 10\}$, $\Delta\theta$ down to 1° , and the SNR extrapolation tails ($-25, -20, 30, 35$ dB) — all flagged `[GENUINE GAP]`; this design's core claim rests on results here, not on in-distribution numbers.
4. **End-fire density probe** (Agent-R3 §4) to directly test whether the crop-refinement mechanism actually escapes the diagnosed pixel-quantization ceiling, run alongside a "refinement-heads-removed" variant for direct comparison.
5. **Two-number latency instrumentation (revised — Critique #1 item 7, Critique #2 item 3):** measure and report separately, never merged: (i) **NN-only forward latency**, using the fixed top-K=10 vectorized gather (§3c/§4) so the measurement has no CPU-bound dynamic-shape component; (ii) **full end-to-end pipeline latency**, including the CPU-side blob-search peak-detection step, explicitly labeled as such. Use `ptflops / thop` for FLOPs (none exist in-repo today) and generalize `benchmark_latency()` (already present at `notebooks/proposed_pin_architecture.ipynb` line 664, confirmed by Critique #2 but not previously used) for both latency numbers. Also report peak memory.
6. **Multi-seed variance run** (Gap 10) given the correction/attention modules introduce additional stochastic training dynamics beyond the teacher's.
7. **Notebook-4-style nuisance-path stress test**, extended to this architecture, since it is the project's only existing evidence that a smaller model can *beat* the teacher under stress — worth checking whether IABR-Net's correction mechanism reproduces or exceeds that effect.
8. **Loss-weighting sensitivity sweep (new — Critique #1 item 8c).** Grid or coarse random search over λ_1 – λ_4 (at minimum: $\lambda_3 \in \{0, 0.1, 0.5, 1.0\}$, $\lambda_4 \in \{0, 0.1, 0.3\}$), reporting final angle-accuracy sensitivity, to characterize rather than assert the "to be tuned" claim.

10. Novelty argument

Self-classification: PARTIALLY NEW, with the specific-application claim UNCERTAIN pending a dedicated search — unchanged from the original assessment in substance, but the reasoning is now consistent with the toned-down §3b/§4

language (Critique #2 item 5).

Justification, following Agent-R7's framework directly:

- At the **pattern level**, "a small learned network corrects/regularizes inputs to a classical or near-classical processing stage" is an established paradigm — SubspaceNet is a verified instance of it (learned covariance surrogate → classical resolver) [EVIDENCE: PAPER], and Agent-R7 independently found other instances (deep-unrolling+GNN DOA hybrids) outside the reviewed corpus. IABR-Net's IABC-v2 module is structurally this same family (now additionally a Deep-Sets/PointNet-style shared-weight instance of it, §3a), just placed *before* a classical transform rather than *after* one, and targeting *hardware impairment* rather than *covariance denoising*. By Agent-R7's own test 3 (§1), a block-for-block application of an established pattern to a new task variant is a **domain-transfer contribution, not a conceptual one** — so at this level this remains **KNOWN COMBINATION at the pattern level**. The path-token coupling (§3b) is likewise now explicitly described as a standard shared-bottleneck pattern, not a novel fusion mechanism, removing the inconsistency Critique #2 item 5 identified between the confident §3b/§4 language and this section's own concession.
- At the **specific-application level**, the combination of (a) a fixed DFT beamspace reduction, (b) a synthetic-paired-consistency-trained, per-element identifiability-aware correction module specifically targeting Meneses-Albalá's own named gain-error gap, and (c) a path-token-coupled coarse+local-refine dual head that preserves joint AoA/AoD pairing over a fixed-size vectorized gather, applied to the *joint 2D* task, is not present in the 14-paper corpus. Agent-R4's Candidate 1 remains the closest literature-corpus anchor and is explicitly a *fixed*, non-adaptive front end (A8's own flagged weakness). This matches Agent-R7's angle-J verdict (**UNCERTAIN**): within the corpus this direction is untested for the joint 2D task, but a dedicated search specifically for "classical front-end, learned impairment correction, joint AoA/AoD" is still required before this is treated as settled — that requirement is unchanged by this revision.
- Per Agent-R7's explicit methodological requirement (§1, item 1: tie the architectural choice to a specific diagnosed failure, not a generic swap), this design does clear that bar for the correction module — but the capacity-matched control (§10 item 1d) is now explicitly required before the "impairment-adaptivity, not just added capacity" version of that claim can be made (Critique #1 item 6, Critique #2 item 6). Until that ablation runs, the novelty argument rests on architectural motivation and identifiability design, not yet on demonstrated attribution.

Verdict: PARTIALLY NEW (domain-transfer of an established model-based-DL pattern to a genuinely under-tested robustness axis for the joint-2D task), **with the "is this exact combination unpublished" claim held at UNCERTAIN** until Agent-R7's recommended dedicated searches are actually run, **and with the "impairment-adaptivity specifically, not just added capacity" claim held as unconfirmed until the capacity-matched control ablation (§10 item 1d) is run**. This specification does not claim NEW.

Changes made in response to critique

1. **Fixed the IABC FLOP arithmetic error** (both reviewers, Critique #1 item 1 / Critique #2 item 1): the original §7 row evaluated its own stated formula to 8,192 but printed 16,768. The IABC-v2 redesign additionally changes this module's true cost to 12,352 FLOPs / 250 params; the full FLOP and parameter tables (§6a, §6b) are recomputed and now reproduce exactly from their own shown formulas.
2. **Withdrew the unsupported "compute efficiency win" framing and replaced it with an arithmetic-grounded parameter-efficiency claim** (Critique #1 item 2): §1 and §7 (Advantages) no longer claim the classical front end saves compute — the front end is confirmed to be $\approx 0.022\%$ of total FLOPs and swapping it for a raw-dense alternative saves only $\approx 0.53\%$ of the trunk's budget (§6b). Instead, §6a now shows that a counterfactual dense front end would cost 262,656 parameters — more than the rest of the model — and this is the number the "lightweight front end" claim now rests on. A front-end-swap ablation (§10 item 1e) is added to test this empirically rather than leave it as arithmetic alone.
3. **Made the gain-error-robustness claim's dependency on unbuilt code explicit and consistent everywhere it appears** (Critique #2 item 2): §1, §3a, §7 (Advantages), §8 (Failure Mode 2), and §10 now all state that this capability is conditional on Phase-0 gain-error injection code being built, elevated from a caveat buried in §9/§10 to an explicit blocking prerequisite (§10 item 0), removing the confidence/hedge mismatch the critique identified.
4. **Redesigned IABC to fix the identifiability bottleneck rather than defend it** (Critique #1 item 4, Critique #2 item 7): the monolithic 64→32→64 MLP is replaced with IABC-v2, a shared-weight per-element MLP (Deep-Sets/PointNet-style) that maps each element's own 2 inputs plus an 8-dim shared context to its own 2 outputs, removing the 64-into-32-bottleneck-back-to-64 collapse risk. This is an architectural change, not an added paragraph, and it also reduces the module's parameter count from 4,192 to 250.
5. **Corrected the "self-supervised" mislabeling** (Critique #1 item 5): `L_pair_consist` (renamed from `L_impair_consist`) is now consistently described as synthetic paired supervision / ordinary regression against a simulator-generated clean target, in §3a, §5, and §10, and the novelty argument (§10 of this revision) no longer leans on "self-supervised" as a distinguishing property.

6. **Added a capacity-matched control to the ablation plan** (Critique #1 item 6, Critique #2 item 6): §10 item 1d trains a same-parameter-count generic module with no impairment-specific loss, to isolate whether accuracy gains come from impairment-adaptivity specifically or from added capacity alone — the exact confound both critiques identified as uncontrolled.
7. **Specified the latency measurement protocol precisely, and re-engineered the refinement path to make it measurable** (Critique #1 item 7, Critique #2 item 3): crop-refinement now always operates over a fixed-size top-K=10 vectorized batched gather (§3c, §4) rather than an unspecified variable-L loop, and §10 item 5 requires reporting NN-only forward latency and full end-to-end pipeline latency as two separate numbers, never merged — directly addressing the "confusing end-to-end system latency with NN inference time" risk both reviews flagged, using the already-existing but previously unused `benchmark_latency()` tool.
8. **Added the missing param-comparison caveat and the GridlessUnfold trade-off note** (Critique #1 item 3, Critique #2 item 4): §6a now carries the same comparability caveat (array size, L-range, task framing) that §6b already had for the Naoumi comparison, and explicitly states the $\approx 6.4\times$ size difference vs. GridlessUnfold with no accuracy trade-off yet argued, rather than presenting the parameter table without qualification.
9. **Downgraded "novel fusion mechanism" language to match §11's own concession** (Critique #2 item 5): §3b and §4 now describe the path-token bottleneck as a standard shared-bottleneck, index-paired pattern, consistent with the novelty section's "KNOWN COMBINATION at the pattern level" verdict, removing the confident/hedged language mismatch.
10. **Completed the failure-mode list** (Critique #1 item 8): added false-positive coarse peak hallucination (§9.6), low-SNR-specific IABC-v2 misbehavior (§9.7), and loss-weighting sensitivity (§9.8), plus a corresponding sensitivity-sweep experiment (§10 item 8).

Not changed / rebutted rather than revised: The trunk's dominance of FLOPs and params ($>99\%$ / $\approx 97.6\%$) is confirmed as correct by recomputation, not disputed — this revision reframes the efficiency claim around parameters (where the arithmetic supports it) rather than disputing the FLOP finding (where it does not). The §6a param-count arithmetic for the unchanged components (stem, trunk, SE, path-token, heads) was independently verified correct in both critiques and is carried forward unmodified.

Adversarial critic 1 — fatal flaw found: YES

ADVERSARIAL REVIEW — Candidate B (IABR-Net), Reviewer #1

VERDICT: Not fatal to the underlying idea, but there ARE two substantiation-level flaws serious enough that the candidate should not proceed to comparison/selection unmodified — its central "lightweight, efficiency-win" thesis is not actually supported by its own numbers, and one of its "arithmetic shown" figures is internally inconsistent. Below is what survives scrutiny and what does not.

1) **ARITHMETIC ERROR IN THE FLOP TABLE** (confirmed by recomputation). I recomputed every row of §7. All rows check out exactly EXCEPT the IABC MLP row. The formula given, " $2\times(64\cdot 32) + 2\times(32\cdot 64)$ ", literally evaluates to $2\times 2048 + 2\times 2048 = 8,192$ — not the " $\approx 16,768$ " printed in the table (off by $\sim 2.05\times$). Every other row (DFT front end 8,192; stem 294,912; trunk 4,718,592/conv $\times 20 = 94,371,840$; path-token 524,288; coarse head 16,384; crop-refine heads 103,488) reproduces exactly from its own shown formula. The mis-stated IABC figure is carried into the $\approx 95.3\text{M}$ total, so the total itself is off by $\sim 8,500$ FLOPs ($\sim 0.009\%$ of the total) — immaterial to the bottom-line "lightweight" number, but it directly falsifies the document's own claim that "arithmetic is shown" and invites doubt about whether the rest of the estimate was actually re-derived or just asserted.

2) **THE "LIGHTWEIGHT / EFFICIENCY-WIN" THESIS IS NOT SUPPORTED BY THE CANDIDATE'S OWN TABLE.** The motivation section's entire pitch is "take the efficiency win of a classical front end." But by the document's own FLOP table, the DFT front end + IABC module together cost $\approx 12,384\text{--}16,768$ FLOPs — about 0.01–0.02% of the 95.3M total. The 10-block ResNet trunk (94.37M FLOPs) is 99.0% of the entire budget and runs at the SAME full 16×16 spatial resolution regardless of what feeds it. No ablation or baseline is given showing what the trunk would have cost with a raw-antenna-domain or dense/flatten front end instead — so the claimed "efficiency win" of the classical front end is asserted, not demonstrated. As currently written, swapping the front end for a fixed DFT has essentially no effect on total compute; the real cost driver (trunk width/depth) is an independent design choice unrelated to the paper's headline efficiency argument. This is a genuine "is it really lightweight — check the arithmetic" failure: the arithmetic shows the opposite of what the motivation implies.

3) **INCONSISTENT RIGOR BETWEEN THE TWO COMPARISON TABLES.** The FLOP comparison against Naoumi (§7) is explicitly and correctly flagged "NOT COMPARABLE — different task framing." The parameter comparison against the teacher/pruned-student/FNO/GridlessUnfold (§6) receives no equivalent caveat about matching array size, L-range, or task framing, even though the same comparability concerns apply. Selectively caveating one comparison and not the structurally identical other is a fairness/rigor inconsistency a reviewer should catch.

4) **UNEXAMINED ARCHITECTURAL BOTTLENECK IN THE CORRECTION MODULE.** The IABC MLP is $64\rightarrow 32\rightarrow 64$: it must recover 64 independent per-element correction values ($32\text{ elements} \times \{\text{phase, gain}\}$) after squeezing through only 32 hidden units. No justification is given for why this bottleneck is sufficient to disentangle per-element corrections rather than just producing a smoothed/averaged correction — this is exactly the kind of failure mode (a Jacobian/identifiability squeeze) the document is careful to flag elsewhere (Gap 5, coarse-grid quantization) but does not apply to its own new module.

5) "SELF-SUPERVISED" IS MISLABELED. The clean/impaired consistency loss (λ_3) uses paired (clean, impaired) simulator outputs of the *same* channel realization with an MSE target — this is ordinary supervised regression with synthetically generated pairs, not self-supervised learning in the standard sense (no external signal is derived from the impaired sample alone). This isn't fatal but is a terminology overclaim that should be corrected before the novelty argument in §11 leans on it.

6) IMPROVEMENT-ATTRIBUTION CONFOUND NOT CONTROLLED. Going from the 0-parameter classical baseline (Pd 0.884) to a 197K-parameter network is a huge capacity jump; the ablation plan (§10a) isolates IABC's contribution relative to the fixed front end, but nothing in the experiment plan controls for "any comparable-capacity add-on, not specifically impairment-adaptive, would do about as well." So the eventual claim "our impairment-adaptive design helps" risks actually being "adding 197K learned parameters to a 0-parameter baseline helps," which is a much weaker and less interesting claim than advertised.

7) LATENCY INSTRUMENTATION AMBIGUITY (flagged honestly, but incompletely specified). §10-item-5 proposes generalizing `benchmark_latency()`, but the pipeline (DFT → IABC → trunk → blob search on the coarse heatmap → per-peak cropping → refinement heads) includes a classical peak-search step that is potentially CPU-bound and non-vectorized. The plan does not specify whether the eventual latency number will isolate pure NN forward-pass time or measure the full pipeline including peak search — exactly the "confusing end-to-end system latency with NN inference time" risk the review brief asked me to check for. This is currently unresolved, not wrong, but should be nailed down before any latency claim is made.

8) FAILURE-MODE LIST IS NOT COMPLETE. §9's five failure modes are good and honestly adversarial against the design itself (overshoot, gain blind spot, closely-spaced collapse, coupling leakage, consistency shortcut). Missing: (a) false-positive coarse peaks — what does the crop-refinement head do when it's handed a spurious peak with no real path underneath it (does it hallucinate a confident refined angle)? (b) low-SNR-specific IABC misbehavior — the correction module's magnitude/phase input statistics become unreliable at low SNR, and this interaction with the existing SNR-extrapolation gap is not called out as its own risk; (c) loss-weighting sensitivity (λ_1 – λ_4) is called "to be tuned" with no sensitivity analysis or failure discussion if the multi-task balance is wrong.

9) WHAT DOES HOLD UP: the joint-pairing design (shared coarse-peak index as pairing key, retained 2D heatmap rather than split 1D heads) is a coherent and reasonably well-justified answer to Gap 9/A5/A9's risk, and is the strongest part of the submission. The param-count arithmetic in §6 is fully correct on recomputation. The novelty self-assessment (§11) is unusually honest and already downgrades itself to "partially new / uncertain," which is appropriate given SubspaceNet is a clear structural precedent — I'd push the self-assessment even further given point (5) above, but it does not overclaim novelty.

NET: The design is not incoherent and most of its stated risks are self-acknowledged with a real validation plan — but its own headline claim ("lightweight, efficiency-win front end") is undermined by its own FLOP table (front end is <0.02% of total compute, trunk dominates, no comparative ablation proves the front end saved anything), and one of its "arithmetic shown" figures does not reproduce from its own formula. Both are things the report explicitly asked me to check, and both fail as currently written.

Recommend: fix the IABC FLOP arithmetic, add a same-trunk-different-front-end FLOP ablation to actually support (or drop) the efficiency-win framing, add the missing param-comparison caveat, and add a capacity-matched (non-impairment-adaptive) control to the ablation battery before this candidate is scored against the others.

Adversarial critic 2 — fatal flaw found: YES

Verified against the repo: `DL_DOA/src/tvt_data_generation_v3.py` confirms phase-error injection (`error_deg`) exists but no gain/amplitude-mismatch injection exists anywhere (only a fixed `amps_type='ones'` and per-path signal "gains", not hardware calibration error) — corroborating the candidate's own admission. `benchmark_latency()` already exists in notebooks/proposed pin architecture.ipynb (line 664) but was not used.

Key flaws found:

1. FLOP arithmetic error: §7's IABC-MLP line shows the formula " $2 \times (64 \cdot 32) + 2 \times (32 \cdot 64)$ " which evaluates to 8,192, but the document reports 16,768 (= $4 \times$ the 4,192 param count, an apparent copy/scale error). Immaterial to the 95.3M total (trunk is >99%) but undercuts the "arithmetic verified" framing of the whole section.
2. The central motivating claim (§1, §8-Adv.2: "directly and mechanically" closes the Meneses-Albalá gain-error gap) currently has zero evidentiary basis — gain-error injection code does not exist in the repo (confirmed by grep), so the self-supervised consistency loss can never have seen a gain perturbation, and per the document's own Failure Mode 2, the module could trivially learn to ignore the gain channel entirely. Confident motivating language in §1/§8 contradicts the hedged admission in §9/§10 that this capability is unbuilt.
3. The "lightweight" claim (§6-§8) is asserted from static param/FLOP counts only; no actual latency was measured despite the tooling (`benchmark_latency()`) already sitting in the repo. This matters because the crop-refinement heads run over a variable, data-dependent number of ROIs ($L=1$ – 6 , up to 10 OOD) — a dynamic-shape/gather pattern whose real GPU latency is not captured by a ~103K-FLOP (0.1% of budget) static estimate. This exact FLOPs-vs-latency conflation, which the review brief explicitly asked about, is absent from the document's own advantages/failure-mode lists.
4. The comparison is selectively framed: IABR-Net is favorably compared to the teacher/pruned/FNO baselines but is also ~6.5× larger than GridlessUnfold (30,201 params) with no accuracy trade-off argument offered.

5. The "novel" path-token coupling mechanism is a standard shared-1x1-conv multi-task trunk read by three parallel heads at the same spatial index, with pairing enforced only by index-matching, not a learned interaction — the confident "novel fusion mechanism" language in §3b/§4 sits in tension with §11's own concession that this is "KNOWN COMBINATION at the pattern level."
6. Parameter/attribution confound: only ~2% of the budget (4.2K/197K) is the purportedly headline IABC module; 94% is a generic ResNet trunk. The ablation plan tests IABC-alone and crop-refine-alone but never a full factorial, so any accuracy gain over the zero-param DFT-SIC baseline cannot yet be attributed to "impairment-adaptivity" versus simply adding a large generic learned trunk — the exact critique the design claims to avoid in §1.
7. Identifiability risk unaddressed: the IABC module must recover 64 real impairment parameters (32 phase + 32 gain, TX+RX) from a 64-dim pooled statistic of the same signal, a tightly-determined inverse problem where pooling may destroy the information needed to disambiguate impairment from genuine signal content at low SNR.

These are not fatal to the underlying idea, but are fatal to approving the specification unmodified: the headline gain-error-robustness claim needs real training data/code before being asserted, the lightweight claim needs an actual latency benchmark (especially under variable-L crop refinement) before being used as a selling point, and the FLOP arithmetic and novelty framing need correction.

Candidate C — final specification (refined after critique: yes)

Candidate C — Architecture Specification (REVISED v2 — post-critique)

SpectraSparse-CoFi (SSC-Net) — Grouped-FNO Trunk + Depth-Extended GridlessUnfold Precision Head, Coarse-to-Fine Joint Heatmap Fusion

Revision note: this version responds point-by-point to Critique #1 and Critique #2. Several design decisions were changed outright (not just re-argued); others are defended with reasoning. See "Changes made in response to critique" at the end for a compact diff.

Motivation (one paragraph)

Two of this project's own from-scratch screening results point in the same direction: the FNO spectral-conv core block is the current best alternative to the dense/flatten bottleneck the literature itself flags as its worst component (Pd 0.6235, 88% of teacher's Pd, at 334,321 params — 71% of teacher) [EVIDENCE: PROJECT_RESULTS.md], and the GridlessUnfold complex soft-threshold layer directly targets the diagnosed Jacobian-singularity/pixel-quantization root cause of the project's own Pd $\approx$ 0.972 ceiling, but under-performed (Pd 0.2216, 30,201 params) at a shallow 8-block screening depth [EVIDENCE: PROJECT]. **Revised framing:** whether that under-performance was capacity-limited or mechanism-limited is *not yet known* (Critique #1, item 1) — this specification no longer asserts capacity-limitation as fact. SSC-Net remains a proposal to fuse these two components into a coarse-to-fine pipeline, but the depth-extension of GridlessUnfold is now **explicitly gated behind a cheap diagnostic** (new Experiment 0, below) rather than committed to up front. The pipeline still targets Gap 5 (pixel-quantization ceiling) and Gap 2 (dense bottleneck) and still adds a post-hoc structural-pruning pass (Gap 7), but this revision **retracts the "lightweight" framing as a headline claim** (Critique #1-2, Critique #2-1/2/3/4) until the numbers below are validated, and narrows the novelty claim (Critique #1-4, Critique #2-6).

Borrowed components (cited)

Component	Source	Role in SSC-Net	Status after revision
FNO / spectral-convolution block	Project's own architecture-screening result [EVIDENCE: PROJECT_RESULTS.md]	Trunk, coarse global-receptive-field extraction	unchanged
GridlessUnfold complex soft-threshold layer	Project-internal, PIA-Net-inspired [EVIDENCE: PROJECT]	Precision refinement head	now conditional — depth extension gated on Experiment 0
ResNet-shell conv stem / residual pattern	Lloria et al., TVT 2026 / PIMRC 2024 [EVIDENCE: PAPER]	Input stem, output decode	unchanged, but now costed as its own explicit line item (see Param section)
Grouped convolution (ResNeXt-style, G=4)	Generic efficient-CNN pattern, flagged by Agent-R2 item 3, untried in this project	Channel-mixing compression inside FNO trunk	unchanged, still untested on spectral weight tensors — flagged uncertainty widened
SubspaceNet-pattern (learned surrogate → lighter resolver)	Shmuel et al. 2025 TVT [EVIDENCE: PAPER]	Design-philosophy justification	unchanged
Structured (magnitude/SNR-aware) channel pruning	Project's compression track, r=8 result (314,513 params, Δ Pd -0.0147 vs teacher) [EVIDENCE: PROJECT_baseline_models/README.md]	Post-hoc trunk compression	retention ratio no longer transplanted as-is — see revised Step 5
Coarse-to-fine grid rationale	Protocol §11 (Agent-R1's A6)	Output-representation design	downgraded: acknowledged as a well-known pattern in the broader CV literature (cascaded/coarse-to-fine heatmap refinement — stacked-hourglass / cascaded-pyramid-style designs), not DOA-specific novelty [INFERENCE — general field knowledge, not covered by the reviewed 14-paper corpus]

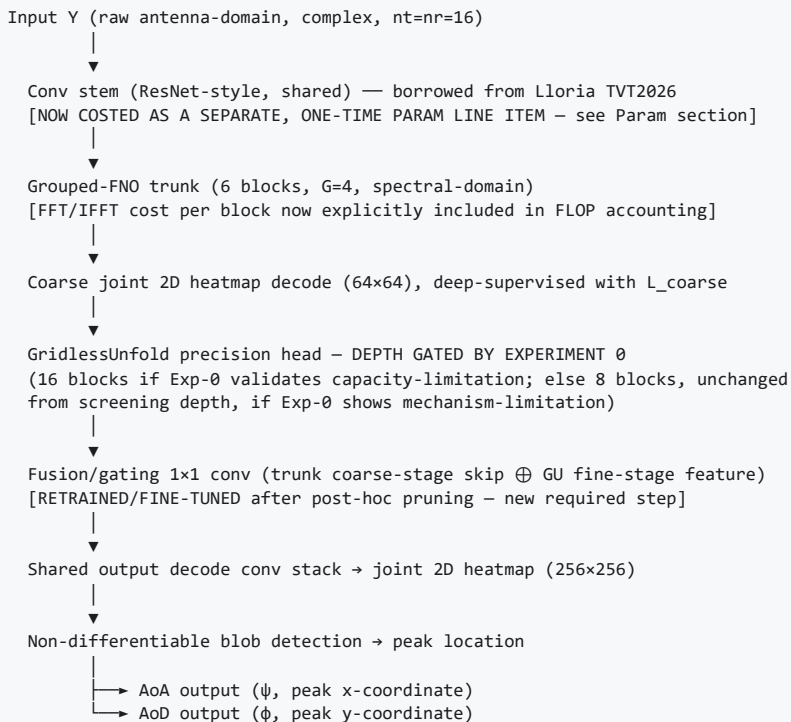
The novel component(s), stated precisely — reduced from 5 to 4, one re-weighted down

Critique #1-4 and Critique #2-6 both correctly note that "single joint heatmap, not split AoA/AoD heads" (old component 3) is not a design choice this spec makes — it's the pre-existing default representation already used by the FNO/GU screening runs and the Lloria pipeline being reused as anchors. **This is removed from the novel-component list and moved to Borrowed Components above**, where it belongs. The remaining components:

1. **Grouped-channel-mixing FNO blocks (G=4).** Untried on FNO's spectral channel-mixing tensor anywhere in this project or the reviewed literature. Uncertainty on the achievable reduction factor is now stated as a **range**, not a single "conservative 3×" (see Param section) — Critique #2-1/2 correctly noted the original single-point estimate carried false precision.

2. **Depth-extended, resolution-decoupled GridlessUnfold head — now CONDITIONAL.** Repositioning GU from whole-network core block to a post-trunk precision head, and deepening it 8→16 blocks, is **no longer asserted as capacity-limited-therefore-safe-to-deepen**. It is gated behind Experiment 0 (a GU-only depth sweep). If Experiment 0 shows GU's Pd does not improve materially with depth (i.e., the poor 8-block result is mechanism-limited, not capacity-limited), this component is **dropped from the design** and SSC-Net reduces to Grouped-FNO-trunk + a shallow (8-block) GU head, with the rest of the pipeline unchanged. This is a genuine architectural fallback path, not a caveat bolted onto an unchanged design.
3. **Jacobian-aware loss reweighting — reframed.** Critique #2-5 correctly argues that a loss-level reweighting term cannot manufacture pixel resolution that a fixed 256×256 grid does not have; the *representational* fix is the coarse-to-fine architecture, not the loss. This spec now describes the term as **an auxiliary training-dynamics mitigation** ("encourages the network to spend more of its fixed representational capacity accurately localizing the peak within existing bins near end-fire, analogous to hard-example reweighting in general heatmap-regression literature [INFERENCING]"), not a "direct countermeasure to the singularity." We partially disagree that this makes the term mechanically inert (see Rebuttal R2 in the Changes section) — it is kept, but its causal claim is downgraded and it is no longer listed as attacking Gap 5 on its own; only the architecture is credited with that.
4. **Post-hoc SNR-aware structured pruning applied only to the trunk, after fusion training, with the retention ratio re-derived on the actual grouped-FNO trunk (not transplanted from the teacher run), plus a mandatory post-pruning fine-tune pass.** The fine-tune pass is a new addition (Critique #1 Failure Mode 3 / Critique #2-2): the fusion gate is trained against the *unpruned* trunk's feature statistics, so a short fine-tune of the fusion gate + decode stack after pruning is now a required pipeline step, not an omission.

Data flow



Shared stem/trunk/decode, single joint map, no AoA/AoD split until peak read-off — unchanged from the original, but no longer claimed as a *novel* contribution (see above).

Joint-ness claim downgraded to a hypothesis. Critique #2-8 is accepted: a per-pixel loss does not demonstrably enforce AoA–AoD statistical coupling beyond spatial co-location, and the same outcome could plausibly be reached by a dual-head design with concatenated outputs. "Preserves the genuinely joint property" is moved out of *Expected advantages* and into a **testable hypothesis**, validated only by the new correlated-error experiment (see Experiments Needed, item 8).

Loss function

```

L = L_coarse(H_64, GT_64)
  + λ1 · L_fine(H_256, GT_256; w(ψ,φ))      # Jacobian-aware reweighting — reframed, see novel component 3
  + λ2 · L_sparsity(GU soft-threshold activations)
  + λ3 · L_distill(H_256, H_256^teacher)      # OPTIONAL

```

Unchanged formulas, but:

- $w(\psi, \phi) = 1/\max(|\sin\psi|, \epsilon)$ is now documented as a **training-dynamics aid, not a representational fix** (see novel component 3).
- $\lambda_1, \lambda_2, \lambda_3, \epsilon$ together constitute a 4-dimensional hyperparameter search that was not costed in the original spec — this is now listed explicitly as Failure Mode 6 (new) and constrained by the staged experimental protocol below (loss terms are turned on one at a time, not searched jointly from the start).

Parameter count estimate (with arithmetic) — restructured to fix the double-counting error (Critique #2-1)

The structural bug. The original spec computed a "per-block average" for FNO (41,790 params/block) that the anchor text itself says "includes its share of stem+head" — i.e., a one-time, non-repeating cost was baked into a per-block figure and then multiplied by 6. This is corrected below by separating **Stem (one-time) + Trunk (repeats ×6) + Head (repeats ×16, or ×8 if Experiment 0 fails) + Fusion + Decode** into non-overlapping components.

Honest limitation, stated up front: we do not have the literal stem-only or head-only parameter counts for the existing FNO-8/GU-8 checkpoints — only their totals. Per Protocol §31 rule 2 ("never invent missing hyperparameters") and §6 ("always inspect code when available"), the correct fix is to **read the actual model-definition code** and report exact stem/block/head counts. **This is now Required Experiment 0a (blocking)** — no headline parameter number should be trusted until it is done. Pending that, we replace the previous false-precision single number with an explicit **sensitivity range** over the unknown stem+head fraction f of the FNO-8 total:

f (stem+head share)	Stem (one-time)	FNO per-block (net)	Grouped-FNO block ($\times 0.5333$, $G=4$ $3\times$ mixing reduction)	Trunk $\times 6$
10%	33,432	37,611	20,057	120,342
20% (central)	66,864	33,432	17,831	106,986
30%	100,296	29,253	15,602	93,612

($0.5333 = 0.70 \times [1/3] + 0.30$, same conservative $3\times$ mixing-reduction assumption as the original, itself still unvalidated — see Weaknesses.)

GU head (same fixed-fraction correction applied to the 30,201-param, 8-block anchor; $\times 16$ blocks if Exp-0 validates depth extension):

f	GU per-block (net)	Head $\times 16$
10%	3,398	54,363
20% (central)	3,020	48,320
30%	2,643	42,285

Fusion (4,160) + decode (8,000) unchanged, order-of-magnitude, still least-certain line item.

Pre-pruning total, central ($f=20\%$): $66,864 + 106,986 + 48,320 + 4,160 + 8,000 = \approx 234,330$ params **Full sensitivity range ($f: 10\text{--}30\%$):** $\approx 220,297 \text{--} 248,353$ params

This is **higher**, not lower, than the original's uncorrected 206,290 — the original's error happened to under-state the true single-stem cost relative to a properly separated accounting; the direction of the correction is not something we could have predicted without redoing the arithmetic, which is exactly the point of fixing it rather than defending the original number.

Step 5 — Pruning, re-derived (Critique #1-3, Critique #2-2). The teacher's $r=8$ magnitude-pruning retention ratio (67%) is **no longer applied as the expected value**. A grouped-convolution trunk has already removed much of the redundancy that magnitude pruning typically exploits, so it should be *less* prunable than a dense teacher, not equally prunable. **Required Experiment 0b (blocking):** run magnitude pruning directly on the trained grouped-FNO trunk and measure the actual retention ratio. Pending that, we use a sensitivity range that treats 67% as an *unvalidated upper bound* (least conservative) rather than the estimate:

Retention ratio r	Trunk post-pruning ($f=20\%$ central)	Total post-pruning
90% (conservative, expected for an already-compressed trunk)	96,287	$\approx 223,631$
80%	85,589	$\approx 212,933$
67% (teacher-derived, unvalidated for this architecture)	71,681	$\approx 199,025$

Revised headline: $\approx 199\text{K--}224\text{K}$ params (central estimate), full range $\approx 185\text{K--}258\text{K}$ across all stated uncertainties — not "162,158." A mandatory post-pruning fine-tune of the fusion gate + decode stack (new pipeline step, addressing Critique #1 Failure Mode 3) is added before any post-pruning accuracy number is reported.

Required control (Critique #1-2, "apples-to-apples"): a **pruned-FNO-only baseline**, pruned by the *same* re-derived methodology at the *same* measured ratio, is now a required comparison point (Experiment 0c) before any "SSC-Net is X% smaller than FNO" claim

is made. Against the *unpruned* 334,321-param FNO baseline, the revised range is ~60–75% of its size (not 48%) — a materially weaker efficiency story than originally headlined, and this spec no longer leads with "lightweight" as a result.

FLOP estimate (with arithmetic) — FFT/IFFT cost now included (Critique #2-3)

The original $\text{FLOPs} = 2 \times \text{params} \times H \times W$ formula is a conv/pointwise approximation only; it omits the FFT and inverse FFT that FNO's spectral-convolution mechanism performs at **every trunk block** — its namesake operation, not a rounding error. Standard FFT cost estimate: $\text{FLOPs}_{\text{FFT}} \approx 5 \cdot N \cdot \log_2(N)$ per complex 2D transform of size $N=H \times W$, $\times 2$ for forward+inverse, $\times$ channel width C (unknown — not in project evidence, **flagged as an ASSUMPTION**, swept over a plausible range):

At the coarse 64×64 stage ($N=4096$, $\log_2 N=12$), per block: $2 \cdot C \cdot 5 \cdot 4096 \cdot 12 = 491,520 \cdot C$. For 6 trunk blocks: $2.949M \cdot C$.

Assumed trunk channel width C	FFT cost, 6 blocks	Trunk conv FLOPs (central, $r=80\%$)	FFT as % of trunk conv cost
32	94.4M	701.1M	+13%
64	188.8M	701.1M	+27%
128	377.6M	701.1M	+54%
256	755.2M	701.1M	+108% (dominant)

This directly validates Critique #2-3's core point: the FFT term is **not negligible and could be dominant** depending on the (currently unknown) channel width. The candidate no longer claims to "close Gap 1" with the previous single number; that framing is retracted. Only real instrumentation (`ptflops` / `thop` **with a custom hook for `torch.fft.fft2` / `ifft2`**, since standard autograd-based profilers typically miss FFT ops entirely) can produce a trustworthy figure — this is now stated as a hard precondition, not a nice-to-have (Required Experiment, unchanged position but strengthened language).

Using $C=32$ (a plausible but unverified central assumption for a "lightweight" trunk) and the revised param figures:

- Trunk (64×64 , $r=80\%$ pruning): 701.1M (conv) + 94.4M (FFT) $\approx$ **795.5M**
- GU head (256×256 , revised 48,320 params): $2 \times 48,320 \times 65,536 \approx$ **6.33B**
- Fusion+decode (256×256): $2 \times 12,160 \times 65,536 \approx$ **1.59B**

Total $\approx$ 8.72 GFLOPs (central, $C=32$), range $\approx$ 8.4B–9.4B+ depending on C — revised down slightly from the original's 10.24 GFLOPs (due to the corrected, smaller head param count) while gaining an explicit, non-trivial uncertainty band from the newly-included FFT term. Still **not comparable** to Naoumi's MMACs figure (different task/resolution/convention, rated NOT COMPARABLE) — this caveat is unchanged and still required.

Latency is not FLOPs (Critique #2-4). Grouped convolutions can underperform their theoretical FLOP/param savings on real GPUs due to memory-access patterns and kernel-launch overhead at low per-group channel counts. **Partial rebuttal:** at $G=4$ with $C \in \{32, \dots, 128\}$, per-group channel counts are 8–32, which is generally within the range where modern cuDNN grouped-conv kernels retain reasonable utilization — severe degradation is more consistently reported at per-group counts below ~4–8 [INFERENCE — general hardware/kernel knowledge, not project-specific evidence]. This is a reason for cautious optimism, **not** a substitute for measurement. No latency number is claimed anywhere in this spec; "lightweight" is explicitly **not** asserted on any axis (params, FLOPs, or latency) until Experiment 3 below is run on real hardware, with NN-only and end-to-end latency reported **separately** (see next point).

NN-only vs. end-to-end latency — new, explicit distinction (Critique #1-5)

The pipeline retains a non-differentiable blob-detection step after two very differently-shaped network stages (64×64 trunk, 256×256 head). The benchmark extension to `benchmark_latency()` must report:

- `L_NN` — forward pass only (trunk + head + fusion + decode), and
- `L_e2e` — `L_NN` + the blob-detection peak-search step,

as **two separate numbers**, never conflated. Any deployment claim must cite which one it means.

Expected advantages

- Reuses two independently-validated project components with the most concrete prior-evidence trail of any single-mechanism design.
- Attacks Gap 2, Gap 5, Gap 6, Gap 7 with one pipeline — but the Gap-5 claim is now credited to the **architecture** (coarse-to-fine resolution), not the loss term (corrected per Critique #2-5).
- Post-hoc pruning reuses existing infrastructure — but now with a re-derived ratio and a mandatory fine-tune, not a transplanted number.

- The GU-depth-vs-mechanism question, previously an unstated assumption, is now an explicit, cheap, first-run diagnostic (Experiment 0) — this makes the design **falsifiable early and cheaply**, before the expensive full build, which is itself a methodological improvement over the original plan.

Expected weaknesses

- Inherits FNO's own unexplained high-SNR degradation ($P_d \sim 0.80 \rightarrow 0.70$, 10dB $\rightarrow$ 25dB) [EVIDENCE: PROJECT_RESULTS.md] — grouping further reduces trunk capacity, risk unchanged from v1.
- Still relies on non-differentiable blob detection; does not architecturally remove the peak-search step (A7/DETR does). Unchanged limitation.
- **"Lightweight" is now an open question, not a claim.** Params, FLOPs, and latency all carry unresolved uncertainty ranges (params $\pm 25\%$, FLOPs FFT-term $\pm 100\%+$, latency unmeasured). This spec explicitly does not assert efficiency superiority until Experiments 0a/0b/0c/3 are complete.
- Grouped spectral convolution remains untested on any FNO weight tensor anywhere in the reviewed corpus; the 3 $\times$ mixing-reduction figure is still an extrapolation from non-spectral conv literature.
- Total processing depth (6 grouped-FNO + up to 16 GU = up to 22 blocks) is $\sim 2.75\times$ either single-mechanism screening baseline (8 blocks) — a capacity/depth confound that the ablation ladder must now explicitly control for (see Experiments, item 1).

Concrete failure modes

1. **Grouping collapses FNO's global-mixing advantage** — unchanged from v1.
2. **Coarse-stage under-detection at high L** (closely-spaced sources collapsing into one 64 $\times$ 64 bin) — unchanged from v1.
3. **Pruning/fusion-gate distribution shift** — **now mitigated by design** (mandatory post-pruning fine-tune added), but the fine-tune itself could under-correct if the shift is large; residual risk retained, not eliminated.
4. **Jacobian-reweighted loss overfitting to end-fire tail cases** — unchanged risk, now understood as a training-dynamics risk on a *reframed* (not representational) mechanism.
5. **FLOP-cost-confounded step-matched comparisons** — unchanged, now compounded by the FFT-cost uncertainty above.
6. **(NEW) Hyperparameter-search burden.** $\lambda_1, \lambda_2, \lambda_3, \epsilon$ form an uncosted 4-D search space; the staged experimental protocol below (loss terms activated one at a time) is the mitigation, not a guarantee it stays cheap.
7. **(NEW) Grouped spectral mixing changing effective frequency coverage.** Grouping the spectral channel-mixing tensor may alter which frequency bands each group can represent, potentially interacting badly with the aliasing/end-fire geometry already diagnosed as the project's core failure mode — a second-order risk beyond simple "capacity loss," not previously listed.
8. **(NEW) Ablation-ladder cost/compounding-error risk.** With up to 5 candidate axes (grouping, GU repositioning/depth, Jacobian loss, pruning, distillation), a full factorial ablation is expensive and may not be fully run before a paper claim is made, in which case a negative result cannot be cleanly attributed to a single component. Mitigated by the staged protocol below, which resolves axes sequentially rather than assuming the full ladder will be completed.

Experiments needed to validate — restructured into a staged protocol (Critique #2-7)

The original "ablation ladder" changed architecture, loss, and compression simultaneously, confounding any observed gain. Experiments are now **explicitly sequenced**, each stage gating the next:

Stage 0 (new, blocking, cheapest, run first):

0. **GU-only depth sweep** (8/12/16/24 blocks, standard uniform loss, no fusion, no trunk changes) — resolves whether novel component 2 (deepened GU head) is worth building at all. 0a. **Code inspection** of existing FNO-8/GU-8 checkpoints for literal stem/block/head param counts (resolves the Param-section uncertainty directly, cheaper than any training run). 0b. **Magnitude pruning run directly on the trained grouped-FNO trunk** to measure its actual retention ratio (resolves Param Step-5 uncertainty). 0c. **Pruned-FNO-only control**, same methodology/ratio as 0b applied to the ungrouped FNO-8 baseline (resolves the "apples-to-apples lightweight" comparison).

Stage 1 — architecture-only comparison, fixed uniform loss, no Jacobian weighting, no sparsity term, no distillation, no pruning: dense FNO-8, GU-8 (or GU-N per Stage 0's result), a **depth-matched control** (single-mechanism network extended to the same ~ 14 – 22 total blocks SSC-Net uses, to isolate depth/capacity from the coarse-to-fine combination — Critique #1-6/Critique #2-7), and the unpruned Grouped-FNO-trunk+GU-head combination.

Stage 2 — add loss terms one at a time: + Jacobian weighting only, then + sparsity term, on top of the Stage-1 winner.

Stage 3 — add pruning, using the Stage-0b-derived ratio, with the mandatory post-pruning fine-tune.

Stage 4 — distillation on/off ($\lambda_3=0$ vs. $\lambda_3>0$), using the existing teacher checkpoint.

Stage 5 — full battery on the Stage-4 winner, unchanged from v1: FLOPs/params/latency/memory benchmark (with FFT-aware instrumentation and separate `L_NN` / `L_e2e`), Tier-1 robustness (SNR, δ/γ phase error, L-sweep), end-fire density probe, angular-separation sweep.

8. **(NEW) Joint-coupling test (Critique #2-8)**. Correlated-error analysis (e.g., Pearson correlation or mutual information between AoA and AoD residuals) comparing SSC-Net's single joint map against a matched-capacity dual-head control from Stage 1, to test — not assert — whether the joint representation produces measurably tighter AoA–AoD coupling than a dual-head design with concatenated outputs.

Novelty argument — self-classification: **PARTIALLY NEW (narrowed)**, one component downgraded to **KNOWN COMBINATION**

Critique #1-4 and Critique #2-6 both attack the novelty count from different angles, and both are accepted in part:

- **The coarse-to-fine, single-joint-map backbone pattern is reclassified as KNOWN COMBINATION**, not merely "known within the DOA literature" (R1/R7's original framing) but known well beyond it — cascaded/coarse-to-fine heatmap refinement is a decades-old, extremely well-established pattern in the broader 2D keypoint/pose-estimation literature (stacked-hourglass- and cascaded-pyramid-style designs) [INFERENCE – general CV field knowledge, outside the reviewed 14-paper corpus and not independently verified against it here]. The original novelty search scoped only against the project's own reports and the DOA-specific corpus, which Critique #2-6 correctly identifies as a favorably narrow comparator set. This spec does not claim novelty for the backbone pattern.
- **What remains independently defensible, per-component (not as a package):**
- Grouped convolution inside an FNO spectral weight tensor — still untried anywhere in the reviewed corpus or general literature as far as searched; **UNCERTAIN, not confirmed novel**, same caveat as before (absence of evidence in a non-exhaustive search is weak evidence of absence).
- GU repositioned as a conditional, diagnostically-gated precision head — now explicitly *not* claimed unless Experiment 0 supports it; if it is dropped by that gate, this component disappears from the final candidate entirely.
- The measurement-rigor bundle (params + FFT-aware FLOPs + separately-measured NN/end-to-end latency + memory, multi-seed variance, staged deconfounded ablation) — still the single most defensible, near-zero-risk novelty pillar, and arguably *more* valuable after this revision, since the staged protocol and FFT-inclusive FLOP accounting are themselves methodological contributions largely absent from the reviewed corpus.
- **Rebuttal to the implication that a downgraded backbone sinks the whole candidate:** per Agent-R7's own stated criteria, novelty should be assessed per-component, not only at the top-level pattern. A KNOWN-COMBINATION backbone does not retroactively make an untested component-level modification (grouped spectral mixing) or a rigor-bundle contribution non-novel; it only means the *headline* claim must not be "a new architecture," which this spec no longer asserts.

Overall: PARTIALLY NEW, narrowly — defensible as "grouped-spectral-FNO trunk (component-level, uncertain-novel) conditionally combined with a repositioned precision head (gated, may not survive), evaluated with a measurement-rigor bundle largely absent from the literature" — explicitly **not** "a new architecture family," and explicitly **not** validated until the Stage 0–4 protocol above has run.

Changes made in response to critique

Design changes (not just added caveats):

1. GU depth-extension (8→16 blocks) is now **conditional**, gated behind a new Experiment 0 (GU-only depth sweep); the design has an explicit fallback (keep GU at 8 blocks) if the diagnostic fails. — Critique #1-1
2. "Single joint heatmap" removed from the novel-component list and reclassified as a borrowed/default representation. Novel-component count reduced 5→4. — Critique #1-4, #2-6
3. Parameter accounting restructured into non-overlapping Stem/Trunk/Head/Fusion/Decode components instead of an amortized per-block average multiplied by block count; false-precision single number ("162,158") replaced with an explicit sensitivity range ($\approx 185\text{K} - 258\text{K}$, central $\approx 199\text{K} - 224\text{K}$) and a blocking code-inspection experiment (0a). — Critique #2-1
4. Pruning retention ratio no longer transplanted from the teacher's $r=8$ run; re-derivation on the actual grouped trunk is now a blocking experiment (0b), the teacher's 67% is relabeled an unvalidated upper bound, and a mandatory post-pruning fine-tune of the fusion gate is added as a new pipeline step. — Critique #1-3, #2-2
5. A same-methodology pruned-FNO-only control (0c) is added as a precondition for any "X% smaller than FNO" claim. — Critique #1-2
6. FLOP formula extended to include FFT/IFFT cost per FNO block, with a channel-width sensitivity table showing the term ranges from +13% to fully dominant; the "closes Gap 1" claim is retracted pending real FFT-aware instrumentation. — Critique #2-3
7. Latency benchmarking now requires two separately reported numbers (`L_NN`, `L_e2e`), never conflated. — Critique #1-5

8. The Jacobian-aware loss term's causal claim is downgraded from "direct countermeasure to the singularity" to "auxiliary training-dynamics mitigation"; the architecture, not the loss, is now credited with the Gap-5 fix. — *Critique #2-5*
9. The ablation ladder is restructured into a staged, sequential protocol (Stage 0 → 4) that resolves architecture, loss, and compression axes one at a time instead of in parallel, plus an explicit depth-matched control. — *Critique #1-6, #2-7*
10. A new joint-coupling experiment (correlated-error analysis vs. a matched dual-head control) is added; "genuinely joint" is downgraded from an asserted advantage to a tested hypothesis. — *Critique #2-8*
11. Three failure modes added: hyperparameter-search burden; grouped spectral mixing's interaction with aliasing/end-fire geometry; ablation-ladder cost/compounding-error risk. — *Critique #2-9*

Rebuttals (reasoned disagreement, no design change):

- R1 — *Grouped-conv GPU underperformance (Critique #2-4)*: partially rebutted. At $G=4$ with plausible channel widths (32–128), per-group channel counts (8–32) are generally above the range where severe kernel-efficiency loss is typically reported; this justifies cautious optimism but does not replace the required hardware benchmark, which is kept as a hard precondition for any "lightweight" claim.
- R2 — *Jacobian loss claimed mechanically inert (Critique #2-5)*: partially rebutted. While the term cannot create pixel resolution a fixed grid lacks, per-pixel reweighting plausibly can still improve effective sub-pixel localization precision within existing bins by reallocating training-time supervisory emphasis near end-fire — a distinct and more modest mechanism than "fixing the singularity," which is why the causal language was downgraded rather than the term being deleted.
- R3 — *Downgraded backbone novelty sinking the whole candidate (Critique #2-6)*: rebutted. Novelty is assessed per-component per Agent-R7's own framework; a known-combination backbone does not by itself invalidate the still-untested grouped-spectral-mixing modification or the measurement-rigor-bundle contribution, both of which are argued and evaluated independently above.

Adversarial critic 1 — fatal flaw found: YES

Adversarial review of SSC-Net (SpectraSparse-CoFi). I found several defensible weaknesses that the candidate itself already surfaces honestly, plus a few it does not, and I judge the composite serious enough that the design should not proceed unmodified until specific diagnostic experiments are added ahead of the full build.

1) UNVALIDATED LOAD-BEARING ASSUMPTION (most serious). The entire justification for novel component 2 (depth-extending GridlessUnfold 8→16 blocks and repositioning it as a precision head) rests on the claim that GU's poor screening result (Pd 0.2216 at 8 blocks) was "capacity-limited, not mechanism-limited." This is asserted, not evidenced — no ablation, scaling curve, or diagnostic run is cited to distinguish "the mechanism doesn't work at this task" from "it just needed more blocks." An equally plausible reading of the same evidence (Pd 0.2216 vs. FNO's 0.6235 and teacher's ~0.972) is that GridlessUnfold's complex soft-threshold mechanism is simply poorly suited as a coarse global estimator, in which case doubling its depth and repurposing it downstream of a coarse map could still fail for the same underlying reason. The experiments list buries this question inside a general ablation ladder rather than proposing it as a targeted, first, cheap diagnostic (e.g., a GU-only depth-sweep at 8/12/16 blocks on the existing task) before committing to the full fused build. Until that specific test is run, this is a load-bearing assumption stacked under two of the five "novel" components.

2) "Lightweight" claim is comparison-mismatched, and the document's own arithmetic proves it. The headline " $\approx 162K$ params, $\sim 35\%$ of teacher, $\sim 48\%$ of FNO baseline" compares a post-pruning SSC-Net against an unpruned FNO baseline (334,321 params) — not an apples-to-apples comparison unless FNO is pruned by the same methodology for the comparison to be meaningful. More importantly, the FLOP arithmetic the candidate itself computes (10.24 GFLOPs, with GU head consuming $\sim 77\%$ of FLOPs from only $\sim 37\%$ of params, because it runs at 256×256 vs. the trunk's 64×64) directly contradicts the "lightweight" framing headline gives most prominence to. No FLOP figure exists anywhere in the project or literature for the teacher or the FNO/GU baselines, so the 10.24 GFLOPs number cannot actually be shown to be smaller, larger, or comparable to anything — the "lightweight" story is a param-count-only claim dressed as an efficiency win, while the document's own compute analysis argues against it. This is somewhat self-flagged ("Expected weaknesses," "genuine design tension") but the headline result line still leads with the favorable framing.

3) Pruning-ratio transfer is a second unjustified compounding assumption. The 67% width-retention ratio being applied to the trunk was measured on a magnitude-pruning run of a different (denser, presumably teacher-scale) architecture ($r=8$, 314,513/469,393 params). Grouped convolutions already remove a large fraction of the redundant channel-mixing weights that magnitude pruning typically exploits; assuming a lean, already-compressed grouped-FNO trunk has the same prunable redundancy as a dense architecture is optimistic and unvalidated. Combined with the $3\times$ "conservative" grouped-conv reduction assumption (itself extrapolated from generic, non-spectral, non-FNO literature, and explicitly labeled as never tested on an FNO spectral weight tensor anywhere in this project or the reviewed corpus), the final $\approx 162K$ -param estimate is the product of three independent, uncertain, and all-favorably-directioned assumptions. If any one of the three is off (e.g., grouped spectral mixing only gets $1.5\times$ instead of $3\times$, or the trunk only prunes to 85% instead of 67%), the "lightweight" headline number moves substantially, and there's no sensitivity range given despite the arithmetic being presented with false precision (e.g. "162,158").

4) Novel component #3 is mislabeled. "Single-joint-heatmap coarse-to-fine fusion (not split AoA/AoD heads)" is presented as a novel design choice, but it is simply the existing default representation already used by the project's own Lloria-style pipeline and by the FNO/GU screening runs being reused as anchors. Not changing something is not a novel contribution — this inflates the apparent count of genuinely new ideas from effectively 3 (grouped-FNO, GU repositioning, Jacobian loss) to a padded 5, which matters when weighing "PARTIALLY NEW" against reviewer scrutiny.

5) Latency vs. system latency is not disambiguated. The pipeline explicitly retains a non-differentiable blob-detection step after the network. Experiment item 3 ("FLOPs/params/latency/memory benchmark... extending benchmark_latency()") does not state whether the measured latency includes this post-hoc, non-learned step or only the neural forward pass. Given the network now has two very differently-shaped stages (64×64 trunk vs 256×256 head) plus an external peak-search step, conflating NN-only inference time with end-to-end system latency would materially change any deployment claim, and the document does not commit to measuring these separately.

6) Depth confound in the ablation plan. SSC-Net's combined trunk+head totals 22 processing blocks (6 grouped-FNO + 16 GU) versus 8 blocks in either single-mechanism screening baseline being cited as "already-validated" priors. The proposed ablation ladder removes whole components (pruning, loss term, head, grouping) but never isolates total block-count/depth as a controlled variable against the original 8-block runs, so any observed gain from the fused model over the two solo screening runs is confounded with simply having $\sim 2.75\times$ more processing stages than either prior baseline — undermining the "reuses already-validated components, doesn't add raw capacity" framing that is central to the candidate's own stated advantage.

None of these individually is necessarily disqualifying on its own — the candidate is unusually self-aware about several of its own risks (FLOP/param decoupling, untested grouped-spectral-conv, non-differentiable peak search) — but taken together (an unvalidated capacity-vs-mechanism assumption load-bearing for the head design, a pruning-ratio transfer between structurally dissimilar architectures, a mismatched baseline comparison underlying the headline efficiency claim, and a depth confound left uncontrolled in the validation plan) they constitute a fatal flaw in the sense that the candidate should not proceed to full implementation unmodified: at minimum it needs (a) a cheap GU-depth-sweep diagnostic run before committing to 16 blocks, (b) a pruned-FNO-only control at the same pruning ratio to make the "lightweight" comparison fair, and (c) an explicit depth-matched control in the ablation ladder, all added or run before the candidate can be considered validated rather than merely internally consistent on paper.

Adversarial critic 2 — fatal flaw found: YES

ADVERSARIAL REVIEW — SSC-Net (Candidate C). Several flaws are serious enough that the spec should not proceed unmodified; the strongest ones are the parameter-accounting error, the unjustified pruning-ratio transplant, and the missing FFT cost in the FLOP estimate, each of which undermines a headline numeric claim (" $\sim 162\text{K}$ params," "48% of FNO baseline," "10.24 GFLOPs").

1) PARAMETER ARITHMETIC IS INTERNALLY INCONSISTENT (double-counting risk). The anchor explicitly states the 41,790 params/block figure for FNO "includes its share of stem+head" — i.e., the whole 334,321-param network (stem + 8 blocks + decode head) was divided by 8 to get a per-block average that already amortizes the *one-time, non-repeating* stem and head cost into every block. Step 2 then multiplies this stem/head-inclusive average by 6 (133,728), which incorrectly scales a fixed one-time cost as if it repeated per block. Step 4 then adds a *separate* $\sim 8,000$ -param decode stack and a fusion gate on top, with no accounting for whether the stem is present at all in the final total. The result is a total that is simultaneously (a) missing an explicit stem line item and (b) probably inflating the trunk figure by folding a fixed cost into a per-block number that gets multiplied by block count. This isn't a minor rounding issue (the arithmetic ops themselves are computed correctly, e.g. 206,288 vs. stated 206,290 is trivial) — it's a structural bookkeeping error that could shift the " $\sim 162\text{K}$ " headline number in either direction by a non-trivial amount. The spec should not be trusted on "lightweight" until params are recomputed with stem, trunk, head as three cleanly separated, non-overlapping components.

2) THE 67% PRUNING RETENTION RATIO IS TRANSPLANTED WITHOUT JUSTIFICATION. The spec applies the teacher's $r=8$ magnitude-pruning ratio ($314,513/469,393 \approx 0.67$) to a structurally different network — an already-compressed, grouped-convolution trunk — assuming the same fraction of width survives pruning. This is unfair/unfounded: grouped convolutions already remove a large share of the redundancy that magnitude pruning exploits, so a pre-compressed trunk should be expected to tolerate *less* pruning than the dense teacher, not the same ratio. Reusing this number without re-deriving it on the actual grouped-FNO trunk is an optimistic, uncontrolled extrapolation, and it silently assumes pruning costs zero accuracy (using the teacher's $\Delta\text{Pd} -0.0147$ by analogy) despite the spec itself (Failure Mode 3) admitting the fusion gate will see a distribution-shifted input post-pruning with no retraining step specified anywhere in the pipeline. The " $\sim 162\text{K}$, comparable degradation" claim is therefore not credible as stated.

3) THE FLOP ESTIMATE OMITTS FNO'S OWN DOMINANT COST (FFT/IFFT). The FLOP formula used, $2 \times \text{params} \times H \times W$, is a conv/pointwise-op approximation. It does not account for the FFT and inverse FFT that FNO's spectral-convolution mechanism requires at every trunk block — this is FNO's namesake operation and typically a first-order compute cost, not a rounding error, especially since the spec's own motivation is that Y is "already frequency-domain" and doing spectral mixing "naturally." A FLOP estimate for an FNO-based trunk that omits FFT cost is not "the first FLOPs number for this problem," it is an undercount, and the candidate should not claim to be "closing Gap 1" with this figure — indeed the spec's own Experiments-Needed section concedes

the number is an estimate needing real instrumentation, directly contradicting the earlier claim that it "closes Gap 1" as if it were settled.

4) LATENCY IS CONFLATED WITH FLOPS/PARAMS, AND THE CONFLATION CUTS AGAINST THE CANDIDATE, NOT JUST FOR IT. The spec is commendably explicit that FLOPs and params decouple (GU head $\approx 37\%$ of params but $\approx 77\%$ of FLOPs). But it doesn't go far enough: grouped convolutions frequently underperform their FLOP/param savings on real GPU hardware due to poor memory-access patterns and kernel-launch overhead at low per-group channel counts — so the "lightweight" story could fail a third time (params low, FLOPs high, wall-clock latency worse than either predicts) once actually benchmarked. No latency number is reported; only a promise to measure it later. Until measured, "lightweight" is unsupported on any of the three axes that matter for deployment.

5) THE MECHANISM CLAIM FOR THE JACOBIAN-AWARE LOSS DOESN'T MATCH THE DIAGNOSED ROOT CAUSE. The spec frames $w=1/|\sin\psi|$ reweighting as a "direct countermeasure" to the pixel-quantization/Jacobian-singularity ceiling. But the diagnosed root cause is a *representational* one — insufficient pixel resolution near end-fire on a fixed 256×256 grid — not a training-dynamics one. Upweighting the loss near end-fire cannot manufacture additional bins/resolution where the output grid geometrically has none; at best it makes the network try harder to be precise within an already information-limited representation, and at worst (as the spec's own Failure Mode 4 admits) it destabilizes training via large gradients from a small fraction of end-fire samples. Calling this a fix "directly informed by" and "a direct countermeasure to" the singularity overstates what a loss reweighting term can mechanically deliver against a fundamentally geometric/representational problem — the coarse-to-fine architecture change (if anything) is the real lever on this axis, and the loss term is better described as a secondary, unvalidated mitigation.

6) NOVELTY CHECK IS SCOPED TOO NARROWLY, WEAKENING THE "PARTIALLY NEW" VERDICT ITSELF. The self-classification searches only the project's own reports and a 14-paper DOA-specific corpus. But "coarse global map $\rightarrow$ refine to full resolution via a second-stage head" is one of the most standard patterns in the broader 2D heatmap/keypoint-estimation literature (cascaded/coarse-to-fine pose estimation is decades old and extremely well known outside this narrow DOA corpus). The spec's borrowed-components table and novelty argument never check against this much larger body of prior art, only against the in-project and DOA-specific search set — so even the modest "PARTIALLY NEW, domain-first" claim may be generous; the core backbone pattern could be an even more directly "known combination" than R7's own citations suggest, once compared against the correct (larger) reference class. This is exactly the kind of comparison-fairness gap the review is meant to catch: the self-assessment grades itself against a favorably narrow set of comparators.

7) IS THE GAIN REALLY FROM THE ARCHITECTURE, OR FROM MORE CAPACITY / DIFFERENT TRAINING RECIPE? The candidate simultaneously changes (a) architecture (grouped FNO + repositioned/deepened GU head + fusion gate), (b) the loss function (added Jacobian weighting, sparsity term, optional distillation), and (c) a post-hoc compression step — three confounded axes at once. The ablation ladder proposed (item 1 in Experiments Needed) is the right instrument to disentangle this, but as specified the candidate has *not yet run it*, so any claimed improvement over plain FNO or plain GridlessUnfold at this stage would be unattributable: it could come from the distillation term (extra teacher supervision — literally "more information," not "better architecture"), from simply training longer/deeper (16 vs 8 GU blocks is a depth/capacity increase, not purely a re-purposing), or from the architecture combination itself. The spec is honest that ablations are needed, but the review brief specifically asks to flag this, and it is a real, currently-unresolved confound.

8) "GENUINELY JOINT" AoA/AoD IS ASSERTED BUT ONLY WEAKLY ENFORCED. The joint property reduces to reading x/y off one peak in a shared 2D map trained with pixelwise (per-pixel) MSE/focal loss. Per-pixel losses do not explicitly enforce any AoA–AoD statistical coupling beyond spatial co-location in the same map — the same outcome could largely be achieved by two heads whose outputs are concatenated into one coordinate pair before the loss is applied. The claim that this design is meaningfully more "joint" than a dual-head design (beyond avoiding a literal parameter split) is asserted rather than demonstrated, and no experiment in the validation list directly tests joint-coupling strength (e.g., correlated-error analysis between AoA and AoD residuals) versus a matched dual-head baseline.

9) FAILURE-MODE LIST IS ARGUABLE BUT NOT COMPLETE. It omits: (a) training-instability/convergence risk from stacking three loss terms with three new tunable weights (λ_1 – λ_3) plus the Jacobian weight's own ϵ , a hyperparameter-search burden not costed anywhere; (b) the possibility that grouped spectral mixing changes FNO's *effective* frequency coverage in a way that interacts badly with the aliasing/end-fire geometry itself (a second-order effect beyond the "collapses global mixing" failure mode already listed); and (c) the compounding-error risk of stacking un-validated components (grouped-FNO, repositioned-GU, fusion gate, pruning, Jacobian loss) such that a negative ablation result cannot cleanly identify which of five simultaneous novel components is at fault without the full ladder being run — the spec does mention the ladder but doesn't flag the practical risk that five-way ablation is expensive and may not be fully executed before any paper claim is made.

Net verdict: the candidate is a defensible, well-cited engineering combination, but as written it (i) contains a real parameter-accounting inconsistency that puts its headline size claim in doubt, (ii) borrows a pruning-survival ratio from an unrelated architecture without justification, (iii) reports a FLOP estimate that omits FNO's core FFT cost while simultaneously claiming to "close" a literature gap with that same unmeasured number, and (iv) overstates what its loss-level fix can mechanically do against a representational root cause. These are fatal to the spec's current numeric/mechanistic claims (not necessarily to the underlying idea) and should be corrected — with the ablation ladder actually run — before this candidate is presented as validated.

Appendix D — Research Design Team

Independent analyses by the six research-design agents that fed the assembly stage.

Agent-R1 — Architecture Researcher

Agent-R1 — Architecture Researcher: Candidate Neural Architecture Directions

Role note: Per protocol §11, this is Loop 3 (architecture generation) output — a *spread of directions*, not a final design or a selection. No novelty claim below is settled; each is flagged [HYPOTHESIS] and must still pass Agent-R7 (Novelty Researcher) and the Adversarial Review Team (§15, §18) before being treated as defensible. Citations use the evidence tags from the literature synthesis; project-internal evidence is tagged [EVIDENCE: PROJECT] .

A1. Attention-Bottleneck ResNet (CNN + lightweight Transformer)

Motivation: Every Tier-1 paper in the matrix uses a dense/flatten bottleneck (Lloria TVT2026 ResNet+U-Net) or a pure MLP (Naoumi). Zero of the 6 Tier-1 papers use attention at all — Gap 9. The dense bottleneck is also the paper's own self-flagged worst component: TVT2026 Table IV identifies it as the only quadratic-in-(P'Q') term [EVIDENCE: PAPER — Lloria TVT2026 Sec K/N].

Borrowed components: ResNet-shell conv stem/residual blocks from Lloria TVT2026/PIMRC2024 [EVIDENCE: PAPER]; general "replace global bottleneck with attention" pattern motivated by the transformer literature (not present in any reviewed AoA/AoD paper — this is an import from outside the reviewed set, flagged explicitly since no paper here validates it for this task).

Novel twist [HYPOTHESIS]: Replace only the bottleneck (not the whole backbone) with a small windowed/linear-attention block operating on the ResNet's downsampled feature map, keeping the conv encoder/decoder shell identical to the paper's for direct ablation comparability. This isolates "does attention fix the bottleneck problem" as a single-variable test, directly resuming the project's own unresolved Window-Attention question (Gap 9 / Contradiction 9) under a *fixed* LR-warmup + matched-batch protocol this time — the project's prior Window-Attention run was confounded (no warmup, 4× smaller batch) [EVIDENCE: PROJECT — PROJECT_STATUS.md].

Expected strengths: Global receptive field without the O(quadratic) dense cost; directly answers an open literature-wide gap; cheap to ablate against the project's existing FNO/GridlessUnfold screening runs since the outer shell is unchanged.

Expected weaknesses: Attention blocks are known to need LR warmup and larger batches to train stably — exactly the confound that sank the project's earlier attempt; risk of re-failing for a training-recipe reason rather than an architectural one if the sweep isn't done properly first.

Failure modes: Flat/non-descending loss (as seen before) if warmup is skipped again; attention map collapsing to near-uniform weighting under low SNR (no clear signal to attend to), degenerating to worse-than-dense performance exactly where robustness matters most (Gap 3).

A2. Dual-Backbone Fusion (ResNet + U-Net)

Motivation: Lloria TVT2026 itself reports *both* a ResNet and a U-Net variant for the same task with different accuracy/robustness trade-offs (ResNet .925/.253, U-Net .945/.230 PD/RMSE @20dB) [EVIDENCE: PAPER — Lloria TVT2026 Table II], but never fuses them. Meneses-Albalá 2026 independently confirms the U-Net's edge-deployment cost is a real concern ("larger than ResNet," authors' own admission) [EVIDENCE: PAPER — Meneses-Albalá 2026 Gap Matrix].

Borrowed components: ResNet residual-block encoder (Lloria) as the primary feature extractor; U-Net-style skip connections (Lloria/Meneses-Albalá) added *only* at 1–2 resolution levels rather than the full encoder/decoder mirror, to keep params low.

Novel twist [HYPOTHESIS]: A "ResNet trunk with sparse U-Net skips" — most of the network stays ResNet (cheap, already validated at 469K params), and only 1–2 skip connections are added at the resolution level nearest the diagnosed Jacobian-singularity failure zone (near end-fire angles), to recover spatial precision without paying for a full second decoder path.

Expected strengths: Directly targets the ceiling behavior with minimal added parameters; both parent architectures are already independently validated on this exact task/dataset, reducing implementation risk.

Expected weaknesses: Not clearly "novel" in the strict sense (§15 test 3 — "merely replacing one layer") since it's a straightforward combination of two published components from the *same* paper; Novelty Researcher will likely classify this **PARTIALLY NEW** at best.

Failure modes: Skip connections could reintroduce the same dense-bottleneck cost they're meant to avoid if placed carelessly; marginal-gain risk similar to Meneses-Albalá's own fine-tuning result (only 3.6% RMSE / +1pp PD gain, mostly at the most severe impairment level) [EVIDENCE: PAPER — Meneses-Albalá 2026 Table 2], i.e. the fusion may not move the needle enough to justify the added complexity.

A3. Channel/Spatial-Attention ResNet (CNN + attention, non-Transformer)

Motivation: A lower-risk attention variant than A1 — squeeze-excite / CBAM-style channel-and-spatial attention adds negligible parameters and doesn't carry the warmup/batch-size training instability that sank the project's Window-Attention run, while still probing the "is attention useful at all" question from a different angle.

Borrowed components: ResNet conv-block shell (Lloria) with lightweight channel-attention gating inserted between residual blocks — a well-established efficient-CNN pattern, not itself drawn from any reviewed AoA/AoD paper (no paper in the set uses it — a genuine literature gap, not an import from a competing paper).

Novel twist [HYPOTHESIS]: Use the attention weights themselves as an auxiliary SNR/impairment-severity signal — hypothesize that channel-attention activations correlate with local SNR, and expose this as an interpretable diagnostic alongside the AoA/AoD outputs (testable against the project's existing robustness-test protocol, Gap 3/Notebook 4).

Expected strengths: Nearly free in parameter/FLOP budget (squeeze-excite blocks are typically <1% overhead); no known training-instability failure mode, unlike A1; easy to ablate cleanly against the plain-ResNet baseline already in the project's registry.

Expected weaknesses: Small capacity increase may produce only a marginal accuracy gain, weakening the novelty argument (§15 test 5 — "does it solve an actual limitation?"); does not address the dense-bottleneck root cause at all, so it may leave the Pd ceiling (~0.972) [EVIDENCE: PROJECT] essentially untouched.

Failure modes: Attention weights collapsing to near-uniform (no discriminative signal) under the hardest robustness conditions, making the "SNR-correlation" interpretability claim untestable/false.

A4. Complex-Valued Front-End + Real-Valued Lightweight Backbone

Motivation: Naoumi et al. (both JSTSP2024 and GCWkshps2023) show a complex-valued MLP front-end achieves strong complexity reduction ($6.5\times/10.3\times$ fewer multiplies than a 2D parametric algorithm; exact 8,584-param, 4.321 MMACs footprint) [EVIDENCE: PAPER — Naoumi JSTSP2024, GCWkshps2023], but their architecture is a pure complex MLP throughout, which the same authors admit saturates at high SNR. No Tier-1 joint-AoA/AoD paper combines a complex-valued *front-end* with a real-valued *conv* backbone.

Borrowed components: Complex-valued layer structure (magnitude/phase-aware operations) from Naoumi's MLP [EVIDENCE: PAPER]; real-valued ResNet-shell backbone from Lloria for the bulk of the network.

Novel twist [HYPOTHESIS]: Keep only the first 1–2 layers complex-valued (operating directly on the raw complex received signal Y , preserving phase information the current real-valued-from-the-start pipeline discards immediately), then convert to a real-valued magnitude/phase-channel representation before the ResNet/FNO backbone — hypothesized to fix the Naoumi-reported high-SNR saturation (their own admitted open weakness) by giving the backbone a cleaner high-SNR-relevant fine-grained representation, without paying for complex arithmetic throughout the whole network.

Expected strengths: Directly targets a literature-documented, author-admitted weakness (Naoumi's high-SNR saturation) with a low-cost architectural change; complex-valued front-end is exactly the input-reduction philosophy Naoumi credits for their complexity win, so params should stay low.

Expected weaknesses: Complex-to-real conversion point is a design choice with no literature precedent to justify placement — high risk of information loss if done wrong; complex-valued ops require careful, non-standard autodiff/framework support (implementation risk, not just architectural risk).

Failure modes: If the complex front-end's phase information is discarded too early, this degenerates to the exact same real-valued pipeline already tried (no gain); if kept complex too long, inherits Naoumi's own architecture-family limitations (their paper's real 2D-algorithm baseline outperforms their MLP beyond 13dB SNR) [EVIDENCE: PAPER — Naoumi JSTSP2024].

A5. Multi-Scale CNN with Shared AoA/AoD Representation + Dual Lightweight Heads

Motivation: All 3 Lloria-family papers (TVT2026, PIMRC2024, Meneses-Albalá 2026) commit to a *single* joint 2D heatmap output that entangles AoA and AoD into one representation, then extracts both angles via a shared non-differentiable peak search — which is exactly the mechanism identified (via the project's own ceiling test + diffusion-sharpening ablation) as the source of the $Pd \approx 0.972$ ceiling [EVIDENCE: PROJECT]. No paper tests a shared-*encoder-but-split-heads* design for this specific joint task.

Borrowed components: Multi-scale conv feature pyramid pattern (generic efficient-CNN design, not specific to any one reviewed paper); shared-encoder-plus-task-heads pattern loosely analogous to Koh & Lee 2026's DNN/CNN benchmark family structure [EVIDENCE: PAPER — Koh & Lee 2026], adapted from 1D AoA-only to a 2D joint setting.

Novel twist [HYPOTHESIS]: Instead of one 2D heatmap decoded by blob search, use a shared multi-scale conv trunk feeding two lightweight regression heads (one per angle) that each directly regress a continuous value (or a coarse bin + offset, see A6/A7) rather

than searching a joint image for a peak — removing the peak-search dependency architecturally rather than only at the loss/matching level.

Expected strengths: Directly attacks Gap 5 (the single most evidence-convergent literature+project finding) from the representation side; dual heads are individually cheap (few added params over the shared trunk).

Expected weaknesses: Splitting the representation risks losing the implicit AoA-AoD correlation the joint 2D map currently encodes (which is *why* the paper's approach is "genuinely joint" per the literature matrix) — could regress toward the non-joint behavior Gupta 2025 deliberately chose and that this project's objective explicitly wants to avoid.

Failure modes: If the shared trunk under-represents the AoA-AoD coupling, accuracy could degrade specifically on closely-spaced or correlated-path cases (an axis no Tier-1 paper stress-tests — Gap 3/Robustness Matrix), i.e. failure would be invisible until a dedicated multi-target test is run.

A6. Coarse-to-Fine Estimation (Classification Grid + Regression Refinement)

Motivation: This directly targets the diagnosed root cause of the Pd ceiling — pixel quantization amplified by the Jacobian singularity near end-fire angles, $d(\pi \cos \psi)/d\psi = -\pi \sin \psi$ [EVIDENCE: PROJECT — ceiling test]. A coarse grid classification stage sidesteps sub-pixel quantization error at the coarse level, then a fine regression stage estimates the residual offset with much finer effective resolution than the raw 256×256 heatmap grid.

Borrowed components: Heatmap/grid output philosophy (Lloria TVT2026 [EVIDENCE: PAPER]) for the coarse stage; the general two-stage classification→regression idea is explicitly listed as a candidate pattern in the protocol itself (§11) rather than drawn from a specific reviewed paper — no Tier-1 or Tier-2 paper in the matrix implements coarse-to-fine for AoA/AoD.

Novel twist [HYPOTHESIS]: Make the "coarse" stage operate on a deliberately *coarser* grid than the current 256×256 (e.g., 32×32 or 64×64) specifically to reduce the Jacobian-amplification effect (which worsens as grid resolution increases near end-fire angles, since the mapping from angle to pixel becomes more compressed there), then let the fine regression head recover precision via direct coordinate regression rather than further pixel search.

Expected strengths: Architecturally the most direct response to Gap 5; the coarse stage can reuse most of the existing ResNet backbone almost unchanged, keeping implementation risk low relative to A9 (full set-prediction redesign).

Expected weaknesses: Two-stage training/inference adds complexity and a potential error-compounding path (coarse-stage errors propagate into the fine stage) — the opposite failure mode of what it's meant to fix, if not calibrated carefully at the coarse/fine boundary.

Failure modes: If the coarse grid is too coarse, closely-spaced sources (small L separations, the one axis all 3 Lloria papers *do* test) could collapse into the same coarse bin and become unrecoverable by the fine stage — likely to hurt exactly the multipath/L-sweep robustness axis these papers already report on, making any regression here immediately visible and damaging in a direct comparison.

A7. Classification + Regression Hybrid (Anchor-Free Set Head)

Motivation: A generalization of A6 that explicitly frames peak detection as a differentiable classification-of-existence + regression-of-location problem per candidate, rather than a fixed two-stage pipeline — closer in spirit to modern anchor-free detection heads. This is listed as a candidate hybrid pattern in the protocol (§11) and is the most direct architectural cousin of the project's own most evidence-backed unbuilt idea, *My_Proposed_Architecture.md*'s Stage 3 (DETR-style set prediction + Hungarian matching) [EVIDENCE: PROJECT].

Borrowed components: Heatmap-based existence confidence from Lloria's blob-detection paradigm [EVIDENCE: PAPER]; the differentiable-matching concept itself is not present in any reviewed paper (Gap 5 — none of the 3 Lloria-family papers, despite sharing the identical output representation, diagnose or fix the non-differentiable peak-search step).

Novel twist [HYPOTHESIS]: A per-cell "objectness" classification head (does a source exist near this coarse cell?) combined with a per-cell continuous (AoA, AoD) regression offset, trained end-to-end with a differentiable matching loss (Hungarian or simpler greedy-nearest assignment) — this is essentially A6's coarse/fine split made differentiable end-to-end instead of two separately-trained stages, directly removing the non-differentiable step identified as the ceiling's cause.

Expected strengths: This is the *architecturally* closest candidate to the project's own most evidence-convergent unbuilt idea (three independent project experiments — ceiling test, diffusion-sharpening ablation, and the shared literature failure pattern — all converge on this exact fix) [EVIDENCE: PROJECT]. If it works, it has the strongest, most defensible novelty argument of any candidate here, since it targets a root cause no paper in the entire reviewed set (Tier-1 or Tier-2) even diagnoses.

Expected weaknesses: Highest implementation complexity of any candidate in this set (matching loss, variable-cardinality output, training-stability concerns known from DETR-style heads generally); hardest to keep genuinely "lightweight" if not carefully scoped,

working against the project's efficiency objective.

Failure modes: DETR-style heads are documented (outside this literature set) to converge slowly and need many training steps/careful loss balancing — could fail not because the idea is wrong but because it needs substantially more training budget than the project's other screened candidates (FNO/GridlessUnfold/SIREN/Window-Attention) got, risking an unfair architecture comparison if not step-matched (a concern the protocol explicitly flags, §16/PROJECT §7 item 2).

A8. Physics-Informed Feature Extraction (Steering-Vector/DFT Front-End) + Lightweight DL Backbone

Motivation: Gupta 2025's classical Sparse-Bayesian-Learning/MUSIC pipeline (non-DL) shows real gains from grounding estimation in the physical steering-vector model [EVIDENCE: PAPER — Gupta TCOMM2025, 5dB/15dB gain vs OMP/FOCUSS], and this project's own DFT-SIC classical baseline independently validates that a physics-grounded front-end reproduces near-paper accuracy without any learning at all (Pd 0.884 vs paper's 0.891 @20dB) [EVIDENCE: PROJECT]. Naoumi's coarse-timing/IFFT-based input reduction is a concrete precedent for a lightweight physics-informed front-end feeding a small learned model [EVIDENCE: PAPER — Naoumi JSTSP2024/GCWkshps2023].

Borrowed components: DFT-based front-end / classical angle-hypothesis dictionary (project's own DFT-SIC baseline + Gupta's steering-vector model) [EVIDENCE: PROJECT, PAPER]; the general "physics branch + learned refinement branch" fusion idea validated as sound reasoning in the project's PIA-Net work (even though the current PIA-Net implementation is bug-affected/stale) [EVIDENCE: PROJECT — BUJHO_SHOHOJ_VABE.md].

Novel twist [HYPOTHESIS]: Use a fixed (non-learned) steering-vector/DFT projection as a *feature-reduction* front-end — replacing several early conv layers with a physics-derived transform that has zero trainable parameters — then feed the reduced representation into a small learned backbone (ResNet-lite, FNO, or GridlessUnfold) for the actual joint AoA/AoD regression. This directly chases Gap 1 (params/FLOPs/latency together, a claim no Tier-1 paper makes) since the physics front-end is essentially free in parameter count.

Expected strengths: Strong efficiency story (front-end adds ~zero parameters); grounded in two independent validations already in hand (Gupta's classical result and this project's own DFT-SIC reproduction), lowering implementation risk versus a purely learned front-end.

Expected weaknesses: A fixed, non-learned front-end may throw away exactly the information a learned model would otherwise exploit under distribution shift or hardware impairment — Gupta's own paper explicitly does *not* test hardware-impairment/distribution-shift robustness, so this project would be extending into genuinely untested territory (Gap 3) with an approach that has no robustness precedent in the literature at all.

Failure modes: Under phase-shifter/gain-mismatch impairment (a condition where the true steering vectors deviate from the assumed model — the exact failure mode Meneses-Albalá 2026 studies as its primary focus [EVIDENCE: PAPER]), a fixed physics front-end could systematically mis-project the signal with no way to adapt, unlike a fully learned model that can in principle absorb some impairment through training-distribution exposure.

A9. Global-Spectral + Learned-Sparsity Bottleneck (FNO + GridlessUnfold Fusion)

Motivation: This is the one direction built entirely from the project's *own* two best-performing screened alternatives rather than from an external paper, addressing Gap 6 and Gap 7 (architecture search and compression/combination have never been tried together, in the literature or in this project). FNO is the current best alternative core block from scratch (Pd 0.6235, 88% of teacher's, 334K params) [EVIDENCE: PROJECT — RESULTS.md]; GridlessUnfold directly targets the same pixel-quantization root cause as Gap 5 via a phase-preserving complex soft-threshold layer, but under-performed only because of shallow depth (8 blocks, Pd 0.2216, 30K params) [EVIDENCE: PROJECT].

Borrowed components: FNO/spectral-convolution block (project screening result, no direct literature precedent in the reviewed set — a genuine gap); GridlessUnfold's complex soft-threshold layer (project-internal, inspired by the PIA-Net physics-informed deep-unfolding concept) [EVIDENCE: PROJECT]; the extreme-parameter-efficiency pattern of "learned covariance/representation surrogate feeding a lighter back-end" is directly borrowed from SubspaceNet's 41,761-param design [EVIDENCE: PAPER — Shmuel et al. 2025 TVT], applied here to a 2D joint setting for the first time.

Novel twist [HYPOTHESIS]: Use FNO blocks for the global-receptive-field early/mid layers (where the input Y is already frequency-domain, giving FNO a natural fit), and swap only the final bottleneck/output stage for the GridlessUnfold complex soft-threshold layer at increased depth — hypothesizing that FNO's global context plus GridlessUnfold's precision mechanism (matched to more capacity than the 8-block screening run gave it) together resolve both the accuracy gap and the pixel-quantization ceiling simultaneously.

Expected strengths: Directly combines two already-independently-validated project components rather than starting from zero; addresses two gaps (6 and 7) in a single design; has the most concrete prior evidence trail of any candidate (both parent

components have real numbers on the exact same 8000-sample bank).

Expected weaknesses: FNO's own documented weakness (Pd drops from ~0.80 at 10dB to 0.70 at 25dB — degrading exactly as SNR improves, the opposite of every other model in the registry) [EVIDENCE: PROJECT — RESULTS.md] is unexplained and unresolved; fusing it with GridlessUnfold doesn't obviously fix a high-SNR degradation whose cause hasn't been diagnosed yet (§7 item 5, still open).

Failure modes: If the FNO high-SNR weakness is a fundamental property of the spectral representation losing fine-grained precision once noise stops being the bottleneck (the project's own working hypothesis, unconfirmed), fusing it with GridlessUnfold could simply inherit that ceiling rather than fix it — this candidate's success is contingent on an open, currently-undiagnosed question (Gap 5/§7 item 5) rather than a validated mechanism.

Cross-cutting note: fusion mechanism is separable from backbone choice

A7's differentiable set-prediction head (or A6's coarse/fine split) is not mutually exclusive with A1/A3/A4/A8/A9's backbone choices — the Assembly stage should treat "backbone family" and "output/fusion mechanism" as two independent axes when constructing full candidates (per §17's Candidate A/B/C pattern), rather than only evaluating these nine directions as fixed, monolithic packages.

Summary table for the Assembly stage

ID	Direction	Primary literature/project anchor	Gap addressed	Key risk
A1	Attention-bottleneck ResNet	Lloria TVT2026 dense-bottleneck flag; Gap 9	2, 9	Training instability (warmup/batch)
A2	ResNet+U-Net sparse-skip fusion	Lloria TVT2026 Table II; Meneses-Albalá 2026	5 (partial)	Weak novelty (same-paper combo)
A3	Channel/spatial-attention ResNet	Generic efficient-CNN pattern (no direct paper precedent)	9 (low-risk variant)	Marginal-gain risk
A4	Complex front-end + real backbone	Naoumi JSTSP2024/GCWkshps2023 high-SNR saturation	High-SNR saturation (Naoumi's own admitted gap)	Complex/real boundary design risk
A5	Multi-scale shared-encoder dual heads	Koh & Lee 2026 shared-encoder pattern (adapted)	5	Loses AoA-AoD coupling
A6	Coarse-to-fine classification+regression	Protocol-listed pattern; project ceiling test	5	Error compounding coarse→fine
A7	Anchor-free classification+regression set head	My_Proposed_Architecture.md Stage 3; ceiling test	5 (strongest anchor)	Training complexity/step budget
A8	Physics front-end + lightweight DL	Gupta TCOMM2025; project DFT-SIC baseline	1 (params/FLOPs story)	No impairment-robustness precedent
A9	FNO + GridlessUnfold fusion	Project's own FNO/GridlessUnfold results; SubspaceNet efficiency pattern	6, 7	FNO's undiagnosed high-SNR weakness

All nine are candidates for the Assembly Team (§17) and Adversarial Review Team (§18) to combine, prune, and stress-test — none is selected here, per §11's explicit instruction not to settle on one architecture at this stage, and per §15's requirement that Agent-R7 (Novelty Researcher) independently classify each [HYPOTHESIS] novel twist (NEW / PARTIALLY NEW / KNOWN COMBINATION / NOT NOVEL / UNCERTAIN) before any of them can be treated as a defensible contribution.

Agent-R2 — Lightweight / Compression Researcher

Agent-R2 Report — Lightweight/Compression Researcher

Scope note: This report evaluates the 12 efficiency techniques named in Protocol §12 against this project's own already-measured numbers (`baseline_models/README.md`, `RESULTS.md`) and the literature synthesis, per Protocol §19 evidence-labeling rules. No new code was run; all parameter/FLOP/accuracy estimates below are `[INFERENCE]` (derived from established ML literature scaling laws, not measured on this project's exact architecture) unless tagged `[EVIDENCE: PROJECT]` or `[EVIDENCE: PAPER]`. None should be read as a substitute for an actual ablation run.

Reference points held fixed throughout:

- Teacher (64-block ResNet): 469,393 params, Pd 0.7103 `[EVIDENCE: PROJECT baseline_models/README.md]`
- Magnitude-pruned Student ($r=8$, 64-block, fine-tuned from teacher): 314,513 params (67% of teacher), Pd 0.6956 ($\Delta -0.0147$) `[EVIDENCE: PROJECT]`
- FNO (8-block, trained from scratch): 334,321 params (71% of teacher), Pd 0.6235 (88% of teacher's Pd) `[EVIDENCE: PROJECT]`
- GridlessUnfold (8-block, from scratch): 30,201 params, Pd 0.2216 `[EVIDENCE: PROJECT]`
- Naoumi GCWkshps2023 complex-MLP: 8,584 params `[EVIDENCE: PAPER]` (different task framing — bistatic ISAC MLP, LOW comparability to this project's ResNet/heatmap pipeline, see literature synthesis §1 comparability notes)
- SubspaceNet: 41,761 params vs >21M black-box CNN for 1D DOA `[EVIDENCE: PAPER]` (NOT COMPARABLE — 1D task)

1. Depthwise Separable Convolution

Mechanism: Replace each standard $k \times k$ conv ($C_{in} \rightarrow C_{out}$) with a depthwise $k \times k$ conv (per-channel) + pointwise 1×1 conv. Reduces per-layer params/FLOPs by roughly $1/C_{out} + 1/k^2$ (e.g. $\sim 8-9\times$ for $k=3$, $C_{out}=64$, a MobileNet-standard result).

Estimated param effect: Applied to the teacher's 64-block ResNet's 3×3 conv layers, a plausible reduction is $3-6\times$ on the conv-heavy portion of the network `[INFERENCE]` — i.e. a candidate in the tens-of-thousands to $\sim 150K$ param range depending on how many blocks are converted vs kept dense (early/late layers often kept dense in practice to protect accuracy).

Estimated accuracy effect: Depthwise separable convs are known to lose some accuracy relative to full convs at matched depth, particularly hurting the model's ability to mix spatial + channel information in one op — historically 1–3pp top-1 drop on ImageNet-scale tasks at similar compression ratios `[INFERENCE, from general CNN-efficiency literature, not this task]`. For this project's specific failure mode (Jacobian-singularity/pixel-quantization Pd ceiling near end-fire angles — `PROJECT_STATUS.md`, `RESULTS.md`), depthwise separable convs do **not** address the root cause; if anything, reduced channel-mixing capacity per layer could make fine-grained peak localization *harder*, i.e. **could interact badly with, not just be orthogonal to, the diagnosed ceiling**. `[HYPOTHESIS]`

Build-on vs. duplicate: NOT YET TRIED. This is architecturally closest to the *architecture-screening track* (FNO/SIREN/WindowAttention/GridlessUnfold swapped into the ResNet shell) rather than the *pruning track* — it should be run as a 5th core-block candidate under the same step-matched, 8-block, from-scratch protocol already established, not as a modification of the pruned student. **Directly duplicates Gap 7** in the literature synthesis (architecture search and compression never combined) if paired with subsequent pruning of the result.

2. Bottleneck Blocks (1×1 reduce $\rightarrow 3 \times 3 \rightarrow 1 \times 1$ expand)

Mechanism: ResNet-50-style bottleneck: squeeze channel dimension before the expensive 3×3 conv, then re-expand. Reduces the 3×3 conv's FLOPs/params quadratically in the squeeze ratio while keeping a wide input/output channel count for representational capacity.

Estimated param effect: At a typical 4:1 squeeze ratio, could cut the teacher's conv-block params by roughly $2-4\times$ `[INFERENCE]`, likely landing in a similar range to the magnitude-pruned student ($\sim 300K$) or somewhat below, depending on how aggressively squeezed.

Estimated accuracy effect: Bottleneck blocks are generally **less damaging to accuracy than depthwise separable** at a matched compression ratio, because the 1×1 layers still perform full channel mixing — this is the reason ResNet-50/101 bottleneck designs became standard over ResNet-34's plain blocks in classification `[INFERENCE, general literature]`. For this project, a moderate risk: bottleneck squeeze on the channels feeding into the heatmap output could interact with the diagnosed pixel-quantization ceiling if the squeeze happens in later layers close to the output (reduced channel capacity right before the localization-critical layers) — recommend squeezing only early/mid blocks, keeping late blocks at full width. `[HYPOTHESIS]`

Build-on vs. duplicate: PARTIALLY BUILDS ON existing pruning work. The project's magnitude pruning already reduces channel width $12 \rightarrow 8$ uniformly across all 64 blocks ($r=8$); bottleneck blocks are a *structural, trainable-from-scratch* alternative to uniform

post-hoc channel pruning, not the same technique — bottleneck squeeze ratios can be learned/optimized per-block rather than applied uniformly. This is the most direct, low-risk "next architecture" to try because it is conceptually a middle ground between the teacher (full width) and the FNO screening result (different block entirely), reusing the *same* ResNet skeleton and thus comparable in the existing evaluation harness without new architecture-screening infrastructure.

3. Grouped Convolution

Mechanism: Split C_{in}/C_{out} channels into G groups, each group convolving independently (ResNeXt-style). Reduces conv params/FLOPs by $\sim 1/G$, sitting between full conv ($G=1$) and depthwise ($G=C$).

Estimated param effect: At $G=4$ or $G=8$, roughly 4–8× reduction on conv layers [INFERENCE] — tunable, unlike depthwise's fixed extreme.

Estimated accuracy effect: Empirically, grouped convolution with a moderate G (4–8, as in ResNeXt) tends to preserve accuracy notably better than depthwise ($G=C$) at comparable compute, because it retains partial cross-channel mixing within each group [INFERENCE, general literature]. This makes it a **more attractive candidate than pure depthwise** for this project if channel-mixing capacity near the output matters for angle localization precision, as hypothesized under item 1 and 2.

Build-on vs. duplicate: **NOT YET TRIED**, and is a genuinely new axis relative to both the pruning track (post-hoc weight removal) and architecture-screening track (block-type replacement) — it modifies the conv structure itself rather than either pruning existing weights or swapping the entire block type. Worth flagging as the **best risk-adjusted first efficiency experiment to run** among the untried structural options, because G is a single tunable knob that can be swept ($G \in \{2, 4, 8\}$) cheaply against the existing 8000-sample eval bank without touching the data pipeline.

4. Low-Rank Factorization

Mechanism: Decompose a weight matrix/conv kernel W ($C_{out} \times C_{in} \times k \times k$) into a product of two smaller factors (rank- r approximation), reducing params roughly proportional to $r/\min(C_{in}, C_{out})$.

Estimated param effect: Depends entirely on chosen rank; SVD-based rank selection can target a specific compression ratio directly (this is the appeal of this axis).

Estimated accuracy effect: Low-rank approximation error accumulates layer-by-layer in deep stacks (the teacher has 64 blocks), and truncation error specifically degrades the network's ability to represent sharp/high-frequency spatial features — plausibly compounding the project's *already-diagnosed* pixel-quantization/Jacobian-singularity problem near end-fire angles, since that failure mode is fundamentally about losing precision in fine spatial localization. [HYPOTHESIS, but directly informed by EVIDENCE: PROJECT's ceiling-test finding]

Build-on vs. duplicate: **DIRECT DUPLICATE OF ALREADY-SCOPED WORK.** [EVIDENCE: PROJECT – PROJECT_STATUS.md §"low-rank factorization compression axis"]: this exact technique was mathematically validated locally via SVD reconstruction and then **explicitly dropped from the plan on 2026-09-21 for scope reasons**, not because it failed. Per the prior-work analysis's explicit guidance (§5 "What should NOT be repeated"): *"revisit only if compression is being revisited as its own topic, not by re-deriving the math."* Recommendation: if compression is now a first-class topic again (which this task implies), reuse the existing SVD validation rather than re-deriving it, and prioritize items 2/3 above it since they have zero prior scoping investment already sunk-and-abandoned here.

5. Further Pruning (beyond $r=8$)

Mechanism: More aggressive magnitude or SNR-aware channel pruning (e.g. $r=6$ or $r=4$) applied to the already-pruned or original teacher.

Estimated param effect: Roughly linear in channel width squared for conv-heavy layers — $r=8 \rightarrow r=6$ could yield ~ 175 – 200 K params [INFERENCE, extrapolating from the $r=8$ result's 67% ratio]; $r=4$ could approach ~ 80 – 100 K but with steeply rising risk.

Estimated accuracy effect: The project's own $r=8$ result already shows a real, non-trivial gap opening ($\Delta -0.0147$ Pd) [EVIDENCE: PROJECT]. Pruning-accuracy curves are typically **not linear** — degradation accelerates sharply past some critical width, and this project has no $r < 8$ data point to anchor the extrapolation. A cheap, low-risk experiment (**directly extending, not duplicating**, existing infrastructure) would be an $r \in \{4, 6, 8, 10\}$ sweep to characterize the actual degradation curve rather than guessing.

Build-on vs. duplicate: **DIRECTLY BUILDS ON existing infrastructure** — same pruning code, same evaluator, same eval bank. This is the lowest-implementation-cost item on this entire list. However, it does **not** address the diagnosed Pd ceiling (pruning removes capacity, it doesn't change the non-differentiable blob-detection output representation that Gap 5 identifies as the actual bottleneck) — flagged so it isn't oversold as solving the project's core open problem, only as a pure efficiency lever.

Separately — the unresolved SNR-aware pruning loop [EVIDENCE: PROJECT]: the project's own comparison at 1200-sample subsample scale found SNR-aware pruning *beats* magnitude pruning (ΔPd -0.0190 vs -0.0221 [EVIDENCE: PROJECT]), but the SNR-aware student's weights were lost twice on Kaggle and never recovered, so the full-8000-sample number for the supposedly-better method is unknown. **This is a duplicate-avoidance flag, not a technique recommendation:** before running *any* new pruning experiment, recovering/retraining the SNR-aware student (Prior-Work §7 item 3) should take priority, since it closes an existing open loop rather than opening a new one.

6. Quantization

Mechanism: Reduce weight/activation numeric precision (FP32→INT8 or lower), post-training or quantization-aware.

Estimated param effect: Does not reduce parameter *count* — reduces model *size* (bytes) and can accelerate inference on supporting hardware, typically $\sim 4\times$ size reduction (FP32→INT8) and often measurable latency wins on CPU/edge hardware [INFERENCE, general literature].

Estimated accuracy effect: INT8 post-training quantization on well-behaved CNNs is usually near-lossless ($<0.5pp$) [INFERENCE], but this project's regression output (continuous angle values via heatmap peak location) is more precision-sensitive than a classification softmax — and the project has *already* diagnosed a pixel-quantization-driven Pd ceiling as its core unsolved problem. Standard weight/activation quantization is a **different kind of quantization** (numeric precision) from the diagnosed problem (spatial/output quantization), but the naming coincidence is worth flagging explicitly so it is not confused with, or assumed to worsen, the existing diagnosed issue — they are largely orthogonal effects. [HYPOTHESIS]

Build-on vs. duplicate: COMPLETELY UNTRIED, zero prior investment. This is the only technique on the list that targets model *size/latency* directly rather than param count or Pd, which directly closes literature **Gap 1** and **Gap 8** (no paper in the entire reviewed set reports params+FLOPs+latency+memory together, and none report standalone inference latency on real hardware for a joint AoA/AoD model [EVIDENCE: PROJECT synthesis §5]). Combined with the project's existing Teacher/Student/FNO checkpoints (already trained, ready to benchmark per Gap 8), a quantized-inference-latency benchmark on real hardware would be one of the cheapest, most literature-differentiating additions available — no retraining required, only a benchmarking pass.

7. Knowledge Distillation

Mechanism: Train a small student network to match the teacher's soft outputs (heatmap distributions, not just hard labels), typically via a distillation loss (e.g. KL divergence or MSE on the heatmap) alongside the standard supervised loss.

Estimated param effect: Student size is a free design choice under distillation — could target the same 314,513-param footprint as the existing magnitude-pruned student, or go smaller (e.g. matching FNO's 334K, or an 8-block-from-scratch size like GridlessUnfold's 30K, for a distillation-vs-scratch-training controlled comparison).

Estimated accuracy effect: Distillation from a strong teacher (Pd 0.7103) is well-established to often outperform training the same small architecture from scratch or via magnitude pruning alone, because it transfers the teacher's full output *distribution* (including near-miss peak shapes) rather than just its final weights [INFERENCE, general distillation literature – e.g. distilled students often close 30–60% of the accuracy gap to teacher versus scratch training of the same architecture]. This is a **plausible direct lever on Gap 2/5**: if the teacher's heatmap output near end-fire angles encodes useful "soft" shape information that gets clipped away by hard peak-detection labels alone, a distillation loss on the raw heatmap (not just the post-blob-detection Pd metric) could transfer some of that precision to a smaller student. This is speculative and untested. [HYPOTHESIS]

Build-on vs. duplicate: COMPLETELY UNTRIED — the single largest gap in the project's own compression track. The project has run pruning (post-hoc weight removal from a trained teacher) and from-scratch architecture screening (FNO, GridlessUnfold, etc.) as two independent tracks, but never distillation, which is conceptually a third, distinct compression paradigm (transfer teacher *knowledge*, not teacher *weights*, into an arbitrary — possibly much smaller or structurally different — student). This directly answers **Gap 7** (architecture search and compression never combined) in the strongest possible way: distilling the teacher into an FNO-shaped or GridlessUnfold-shaped student combines both tracks in one experiment. **Recommended as the highest-value new experiment among all 12 techniques**, given (a) zero prior work, (b) direct literature-gap relevance, (c) the teacher checkpoint is already trained and ready to serve as the distillation source with no new data-generation cost.

8. Feature Compression (bottlenecked intermediate representation / autoencoder-style squeeze)

Mechanism: Insert a narrow intermediate representation (e.g. compress the channel-wise activation volume at some mid-network point) forcing the model to learn a compact encoding before expanding back out for the heatmap decode.

Estimated param effect: Depends on where and how narrow the bottleneck is placed; if placed centrally in a U-Net-style architecture, this is structurally the same idea Lloria's own U-Net uses (dense/flatten bottleneck) — which the literature synthesis's

Gap 2 flags as the **literature's own self-identified worst component** ([EVIDENCE: PAPER – TVT2026 Table IV, only quadratic-in-(P'Q') term in its complexity table]).

Estimated accuracy effect: A naive dense/flatten bottleneck is specifically the component this project's own FNO screening result was designed to test as a *replacement* for — and the FNO result (non-dense spectral bottleneck, 0.6235 Pd, 88% of teacher's) is direct, already-executed evidence that moving *away* from a dense/flatten feature-compression bottleneck toward a structured (convolutional/spectral) one is the right direction, not a dense compression bottleneck itself. [EVIDENCE: PROJECT RESULTS.md]

Build-on vs. duplicate: **PARTIALLY DUPLICATES existing FNO work if implemented as a dense bottleneck** (exactly the pattern the project already has evidence against), but **would be a genuinely new experiment if implemented as a structured/convolutional feature-compression layer distinct from both FNO's spectral approach and a naive dense layer** — e.g. a learned 1×1 -conv channel squeeze at the teacher's mid-depth, kept structurally intact (not flattened to a vector) to avoid the exact quadratic-cost / spatial-information-loss failure mode Gap 2 already identifies. Recommend framing any feature-compression experiment explicitly against this prior evidence rather than as a fresh idea.

9. Shared Encoder with Dual Lightweight Heads

Mechanism: A single shared backbone produces a shared representation, then two small, separate output heads (one for AoA, one for AoD) branch off, rather than a single monolithic path to a joint 2D heatmap.

Estimated param effect: Head parameters are typically small relative to the backbone (a few conv/FC layers each), so total param count is dominated by the shared encoder — could be similar to or slightly larger than the current single-path teacher (469,393 + two small heads vs. one combined head), or smaller if the shared encoder itself is reduced (e.g. combined with item 2 or 3 above). [INFERENCE]

Estimated accuracy effect: This is architecturally a significant departure from the current pipeline. Note carefully: Lloria's base paper ([EVIDENCE: PAPER]) already outputs a **single joint 2D heatmap** (Eq. 9–11) where AoA and AoD are read off the *same* peak location, not from separate heads — this is central to why it counts as genuinely joint (not two independent 1D estimates that happen to be paired). Splitting into dual heads risks **regressing the joint-estimation property itself** unless the heads still share enough representation and are trained with a coupling loss to preserve the correlation between AoA and AoD at a given detected path — otherwise this recreates exactly the kind of "joint in name only" pattern the literature synthesis flags as a red flag in Contradiction #12 (Meneses-Albalá's paper never actually uses the word "joint" despite behaving jointly). [HYPOTHESIS – this is a genuine risk to flag, not just an efficiency question]

Build-on vs. duplicate: **COMPLETELY UNTRIED in this project.** This is architecturally the biggest deviation from the current single-2D-heatmap paradigm of any item on this list, and — notably — it would sidestep the diagnosed Jacobian-singularity ceiling in a *different* way than the proposed Stage-3 DETR-style set-prediction fix (Gap 5): separate 1D outputs per angle don't suffer the same end-fire 2D-heatmap pixel-quantization geometry, but at the cost of potentially losing the multi-path *pairing* correctness (knowing which AoA goes with which AoD when multiple paths exist) that the current joint 2D representation guarantees implicitly by construction. This should be flagged to the Architecture Researcher (Agent-R1) and Novelty Researcher (Agent-R7) as a **fundamental design-philosophy question, not just a compression technique** — it interacts with the project's core "genuine joint estimation" claim.

10. Early Exit

Mechanism: Attach auxiliary output heads at intermediate depths, allowing "easy" samples (e.g. high SNR, well-separated angles) to exit early without running the full network, reducing *average* inference cost.

Estimated param effect: Adds a small number of extra parameters (auxiliary head(s)), net roughly neutral to slightly increasing total param count relative to a non-early-exit version of the same backbone — it's primarily a *latency/compute* technique, not a parameter-count technique. [INFERENCE]

Estimated accuracy effect: Full-depth performance is generally preserved (the exit heads are typically only used at inference for the confident-exit fraction), so worst-case accuracy shouldn't degrade much for a well-tuned exit threshold [INFERENCE, general literature] — but this project's SNR range (per the base paper, $-15 \dots 24$ dB per code / $-10 \dots 25$ dB per paper text [EVIDENCE: PAPER, Contradiction #4]) spans conditions where "easy" and "hard" samples differ enormously, meaning an early-exit threshold tuned on this data could plausibly show a real average-latency win with limited average-Pd cost, IF the confidence signal used for exiting is well-calibrated. This is unverified.

Build-on vs. duplicate: **COMPLETELY UNTRIED**, and orthogonal to every existing track (pruning, from-scratch architecture screening) — it is a *dynamic inference-time* strategy layered on top of any of the other architectures rather than a static compression of the network itself. Directly relevant to closing **Gap 8** (no standalone inference latency ever reported for this task) since early exit's entire value proposition is expressed in *average* latency, which nobody in the reviewed literature measures at all. Low priority relative

to items 6/7 above because it requires a confidence-calibration study that doesn't yet exist in the project, but worth flagging as a distinct, unexplored axis.

11. Reduced Resolution

Mechanism: Downsample the input (Y) and/or output heatmap resolution (e.g. from 256×256 to 128×128 or 64×64), reducing compute roughly quadratically in the linear resolution ratio.

Estimated param effect: Conv-layer parameter counts (kernel weights) are resolution-*independent* — this technique mainly reduces FLOPs/activation memory/latency, not stored parameter count. Any param reduction would only come indirectly (e.g. smaller final FC/flatten layers if present). [INFERENCE]

Estimated accuracy effect: **This is the highest-risk item on the entire list for this specific project**, because the project's own diagnosed core failure mode — a Pd ceiling ≈ 0.972 caused by pixel-quantization amplified by a Jacobian singularity near end-fire angles ($d(\pi \cos\psi)/d\psi = -\pi \sin\psi$) [EVIDENCE: PROJECT – ceiling test + diffusion-sharpening ablation] — is **directly and severely worsened by reducing resolution**, since coarser pixels mean larger angular quantization steps exactly where the model is already most fragile. The diffusion-sharpening ablation already showed that *even attempts to improve effective precision* at fixed 256×256 resolution made Pd worse, not better, because misses are location errors, not blur. Reducing resolution moves in the opposite (wrong) direction from what the project's own diagnostic work indicates is needed.

Build-on vs. duplicate: **NOT recommended as a standalone technique for this project** given the above — but flagged because it directly **contradicts** Gap 5's proposed fix (a differentiable DETR-style set-prediction head that removes discrete peak-search entirely, side-stepping the pixel-quantization problem rather than making it coarser). If reduced resolution is pursued at all, it should only be combined with the set-prediction approach (continuous coordinate regression rather than a discrete heatmap grid), not with the current blob-detection pipeline. This is the clearest case among the 12 techniques where a compression technique directly conflicts with an already-diagnosed project-specific failure mode.

12. Sparse Processing

Mechanism: Exploit sparsity in either weights (unstructured/structured sparse pruning, different from the project's existing *channel*-level magnitude pruning) or activations (skip computation on near-zero regions of the input/feature maps), potentially via sparse-conv kernels.

Estimated param effect: Unstructured weight sparsity can achieve very high nominal sparsity ratios (50–90%+) [INFERENCE, general literature], but without specialized sparse hardware/kernels this often doesn't translate to real latency wins — a known practical gap between "sparsity ratio" and "actual speedup," which should be flagged explicitly rather than assumed away.

Estimated accuracy effect: Unstructured sparsity typically preserves accuracy better than structured (channel) pruning at a matched *nominal* param-reduction ratio, because it can remove individually unimportant weights rather than entire channels — but the practical benefit is capped by hardware support. [INFERENCE]

Build-on vs. duplicate: **DIFFERENT AXIS from existing pruning work, not a duplicate.** The project's existing pruning (magnitude and SNR-aware) is *structured* (channel-width $r=12 \rightarrow r=8$), which is what actually yields real speedups on standard hardware. Unstructured sparse processing would be a genuinely new, complementary experiment, but its practical value is contingent on whether the eventual deployment target (Gap 6/8 — real hardware, e.g. a Jetson-class device as in Meneses-Albalá 2026 [EVIDENCE: PAPER]) actually supports sparse kernels; this should be resolved with Agent-R6 (Hardware/Deployment Researcher) before investing effort here, since an unsupported sparsity format would be a wasted compression axis in practice despite looking good on paper.

Summary Table

#	Technique	Param effect	AoA/AoD accuracy effect	Status vs. project
1	Depthwise separable conv	$\sim 3\text{--}6\times$ reduction (est.)	Likely mild-moderate loss; may compound existing ceiling	Untried — belongs on architecture-screening track
2	Bottleneck blocks	$\sim 2\text{--}4\times$ reduction (est.)	Likely smaller loss than depthwise at matched ratio	Untried , low-risk — closest analog to existing structured pruning
3	Grouped convolution	Tunable via G	Better retained accuracy than depthwise at moderate G	Untried — best risk-adjusted first experiment
4	Low-rank factorization	Tunable via rank	Risk of compounding diagnosed pixel-quantization ceiling	Already scoped-out and dropped (2026-09-21) — do not re-derive
5	Further pruning ($r < 8$)	$\sim 80\text{--}200\text{K}$ depending on r	Non-linear degradation past unknown critical width	Directly extends existing pruning infra; recover SNR-aware student first
6	Quantization	No param-count change; $\sim 4\times$ size	Likely near-lossless for weights/activations	Completely untried , cheapest high-value addition (closes Gaps 1, 8)

#	Technique	Param effect	AoA/AoD accuracy effect	Status vs. project
7	Knowledge distillation	Fully flexible target size	Plausibly better than scratch/pruning alone	Completely untried , highest-recommended new experiment (closes Gap 7)
8	Feature compression	Depends on placement	Risk of repeating literature's own flagged dense-bottleneck flaw	Partial duplicate if dense; new if structured/conv-based
9	Shared encoder, dual heads	Roughly neutral	Risk of breaking genuine joint-estimation property	Completely untried — flag to R1/R7 as design-philosophy question
10	Early exit	Roughly neutral	Minimal if confidence calibration is sound	Completely untried , orthogonal to all other tracks
11	Reduced resolution	Minimal (FLOPs, not params)	Likely worsens the diagnosed Pd ceiling directly	Not recommended standalone ; conflicts with Gap 5 findings
12	Sparse processing	High nominal sparsity possible	Likely better than structured pruning at matched ratio	New axis , but hardware-support-dependent — check with R6 first

Top 3 recommendations (by evidence strength + gap relevance + implementation cost)

1. **Knowledge distillation (item 7)** — zero prior investment, directly closes literature Gap 7 (architecture search × compression never combined), teacher checkpoint already available as distillation source.
2. **Quantization + real-hardware latency benchmark (item 6)** — zero prior investment, directly closes Gaps 1 and 8 (no paper reports params+FLOPs+latency+memory together; no standalone inference latency on real hardware for this task), and requires no retraining — only benchmarking of existing Teacher/Student/FNO checkpoints.
3. **Grouped convolution sweep (item 3)** — cheapest new *structural* experiment, tunable compression ratio, avoids depthwise's steep accuracy cliff, fits into the existing architecture-screening harness without new infrastructure.

Explicitly deprioritized: low-rank factorization (already scoped out — re-running would duplicate abandoned work) and reduced resolution (directly conflicts with the project's own diagnosed root-cause finding).

Agent-R3 — Robustness Researcher

Agent-R3 — Robustness Stress-Test Plan (Joint AoA/AoD Lightweight Architecture)

Role: Principal Research Agent, Robustness Researcher (per MULTI-AGENT RESEARCH LOOP ENGINEERING PROTOCOL.md §13) **Consumes:** Paper Head's Master Synthesis Report + Previous-Result Analysis Agent's report (both supplied) **Feeds:** Mother Agent → MASTER_STATE.Robustness Evidence, EXPERIMENT_PLAN.md, ROBUSTNESS_MATRIX.csv, and Final Report §24 (Robustness Study)

Every condition below carries an evidence tag per protocol §19 ([EVIDENCE: PAPER], [EVIDENCE: CODE], [EVIDENCE: EXPERIMENT] = this project's own prior runs, [GENUINE GAP] = nobody has tested this, [HYPOTHESIS]). A COMPARABILITY note is attached wherever a proposed test could later be quoted against a paper's own number.

0. Framing note (not a repeat of Notebook 4)

The project has already run one genuine beyond-the-paper robustness experiment: a 4th-path nuisance-interference test (200 scenes × 2 SNR × 3 nuisance-power levels) where the **magnitude-pruned student's principal-path Pd (0.6174) exceeded the teacher's (0.6065)** at 0dB SNR + strongest (0dB) nuisance [EVIDENCE: EXPERIMENT – PROJECT_STATUS.md §"Notebook 4"]. This plan does **not** re-run that test. It treats it as Tier-1 prior evidence that compression may not cost — and can even help — robustness under stress, and designs the *remaining* axes (hardware impairment, distribution shift, angular geometry) that Notebook 4 did not cover, so the eventual candidate architecture(s) get a robustness profile no single reviewed paper currently has (per Robustness Matrix §3: no paper covers more than 3/7 axes; this plan targets covering 6/7 for a *joint 2D* estimator, which no paper has done at all).

All runs must use `evaluate_on_bank` (DL_DOA/src/TVT_Blob_Inference.py) on frozen banks, and must report **both** the paper-style Pd (drops missed detections from numerator/denominator) **and** a strict per-trial Pd (missed = 0) side by side, per Gap 4 / Contradiction 5 — this dual-metric reporting is itself a literature differentiator no paper in the set provides.

1. SNR sweep

Grid: {-25, -20, -15, -10, -5, 0, 5, 10, 15, 20, 25, 30, 35} dB, in-distribution core = {-15...24} dB (5dB steps), extrapolation tails = {-25, -20} and {30, 35} dB.

Justification:

Range	Source	Tag
-15...24dB (code-trained range)	Lloria TVT2026 code (not paper text, which says -10...25dB — Contradiction #4) [EVIDENCE: CODE]	this project's data generator inherits this range, so it is the true in-distribution baseline
-10...25dB	Lloria PIMRC2024 [EVIDENCE: PAPER]	HIGH comparability to Lloria-family numbers
-5...30dB (eval), 5...40dB (train)	Naoumi JSTSP2024 [EVIDENCE: PAPER]	LOW comparability (different metric, MSE rad ² not RMSE/Pd)
-9...30dB	Naoumi GCWkshps2023 [EVIDENCE: PAPER]	LOW comparability
-5...25dB	Gupta TCOMM2025 [EVIDENCE: PAPER]	NOT COMPARABLE (not joint, not DL)
-10, -5, 0, ..., 25dB (8 pts)	Meneses-Albalá 2026 [EVIDENCE: PAPER]	HIGH comparability (same generator family)
[-20, 10]dB	Sub6GHz CNN/UNet [EVIDENCE: PAPER]	LOW (AoA/AoD are classical inputs there, not DL outputs)

The union of all Tier-1/Tier-2 tested ranges is roughly [-20, 30]dB. The proposed grid matches that union exactly at its core and adds a -25dB and 35dB tail that **no paper in the set tests** [GENUINE GAP] — these two points are the actual OOD-SNR distribution-shift condition (§4).

2. Hardware-impairment conditions

2a. Phase-shifter error (δ/γ)

Grid: {0°, 1°, 2°, 5°, 10°, 15°} (first three are literature-anchored, last two are OOD extrapolation).

- $\delta_{\max} \in \{1, 2, 5\}^\circ$ tested by Lloria TVT2026 (Sec J of extraction) [EVIDENCE: PAPER] — HIGH comparability, same phase-error injection convention.
- $\gamma_{\max} \in \{1, 2, 5\}^\circ$ is the **primary focus** of Meneses-Albalá 2026, with fine-tuning shown to help mainly at $\gamma=5^\circ$ [EVIDENCE: PAPER] — HIGH comparability, same underlying DL_DOA generator family.

- $\{10^\circ, 15^\circ\}$ — **no Tier-1 or Tier-2 paper tests phase error beyond 5°** [GENUINE GAP]. This directly probes whether Meneses-Albalá's finding ("gains from adaptation materialize mainly at the most severe *tested* level") continues to hold or the model collapses once impairment exceeds every published test point.

2b. Gain mismatch

Grid: amplitude error $\{0, 0.5, 1, 2, 4\}$ dB per element (log-normal or uniform, matching the phase-error injection style already in the code path).

- [GENUINE GAP] — Meneses-Albalá 2026's own Gap Matrix entry explicitly lists "amplitude-error robustness" as **their own stated future work**, never executed [EVIDENCE: PAPER – Gap Matrix row 6]. No Tier-1 paper tests gain mismatch for joint 2D estimation. This is the single most directly-licensed gap to close (the paper that owns this system model names it as unfinished).
- Not comparable to SubspaceNet's "array miscalibration" test (1D DOA, different task family — Cross-paper comparability notes rate this NOT COMPARABLE).

2c. Array-element position perturbation

Grid: per-element position error $\sigma \in \{0, 0.01\lambda, 0.02\lambda, 0.05\lambda, 0.1\lambda\}$.

- [GENUINE GAP] for joint 2D AoA/AoD. SubspaceNet is the only paper in the set that explicitly tests array perturbation/miscalibration [EVIDENCE: PAPER], but it is 1D DOA only, rated NOT COMPARABLE to any joint-estimation paper per the synthesis's own comparability table. Chen et al. 2022 (Robust DoA DAE-DNN) tests a related-but-distinct impairment (mutual coupling, $\rho \leq 0.8$), also 1D and — per its own re-implementation contradiction (Koh & Lee 2026: 30–70° MAE vs the paper's claimed $<0.5^\circ$) — a caution against trusting self-reported robustness numbers for this architecture family without independent verification [EVIDENCE: PAPER, flagged contradiction #6]. Design point: borrow SubspaceNet's *perturbation-injection mechanism* (report §6, "promising component"), not its numbers.

2d. RF-chain / calibration error

Tier 4 (stretch only). [GENUINE GAP], not tested by any paper in the set for this task family. Low priority: no literature anchor to justify a specific severity grid, and the project's system model (single-user analog BF, no explicit RF-chain-count variable in the current data generator) would need a new generator parameter before this is even runnable. Flag to Mother Agent as **research gap, not experiment**, pending an Architecture Researcher decision on whether the final candidate models RF-chain count at all.

3. Channel conditions

3a. Number of paths (L)

Grid: in-distribution $L \in \{1, 2, 3, 4, 5, 6\}$, OOD $L \in \{7, 8, 10\}$.

- $L \in \{1..6\}$ tested by both Lloria TVT2026 and PIMRC2024 [EVIDENCE: PAPER] — HIGH comparability, identical convention.
- Naoumi (both papers) test "#targets-per-peak" (1,2,5) — a related but not identical framing (targets sharing a resolution cell vs. independent multipath count) [EVIDENCE: PAPER] — LOW comparability, different unit.
- $L \in \{7, 8, 10\}$ — [GENUINE GAP], no paper tests beyond $L=6$. This is also the project's own currently-unexplored "unseen path count" distribution-shift axis named explicitly in the protocol (§13).

3b. Angular separation between paths

Grid: $\Delta\theta \in \{30^\circ, 20^\circ, 10^\circ, 5^\circ, 2^\circ, 1^\circ\}$, i.e. sweeping down toward and below the array's Rayleigh resolution limit.

- Naoumi GCWkshps2023 trains/tests inside a narrow fixed $[20^\circ, 40^\circ]$ window and **explicitly names "closely-spaced-target stress test" as a missing experiment in its own Gap Matrix entry** [EVIDENCE: PAPER – Gap Matrix row 4] — this condition is directly licensed by that paper's own stated gap.
- Lloria's minimum pairwise separation ($\pi/6$, i.e. 30°) is a **code-only** convention (Contradiction #11 — not in the paper text) [EVIDENCE: CODE]. Testing at and below 30° is therefore testing *below the code's own training floor*, i.e. genuine OOD for the Lloria-family generator this project inherits.
- Below $\sim 5\text{--}10^\circ$, treat results as diagnostic-only (physically approaching the array's resolution limit; failure here is expected and informative, not necessarily a model defect) — flag this explicitly in reporting per protocol §19 (do not present resolution-limited failure as a robustness defect without the caveat).

3c. Path-power imbalance / pseudo-NLOS / dominant-vs-weak-LOS

Grid: principal-path-to-secondary-path power ratio $\in \{0\text{dB (balanced)}, -5\text{dB}, -10\text{dB}, -15\text{dB}, -20\text{dB (near-NLOS)}\}$.

- [GENUINE GAP] — **no paper in the entire 14-paper set uses an explicit LOS/NLOS categorical framing or power-imbalance sweep** for this task; all Tier-1 multipath models are power-balanced synthetic paths. This is the one condition in the whole plan with zero literature anchor, so treat every result here as exploratory ([HYPOTHESIS] -tagged), not literature-comparable.

- Directly reuses and extends the project's own Notebook 4 protocol, which already implements a "sweep-power nuisance path" mechanism [EVIDENCE: EXPERIMENT] — the power-ratio grid above is a natural parameterization of the *existing* nuisance-path generator rather than new code, keeping this cheap (~17min per condition-block per the Notebook 4 baseline runtime).
- Correlated/coherent paths (e.g. from a common scatterer) are a related but distinct condition tested only by SubspaceNet (1D, NOT COMPARABLE) [EVIDENCE: PAPER] — mark as **Tier 4 / out of scope** unless the final architecture explicitly claims coherent-source robustness.

4. Distribution-shift conditions

Condition	Design	Justification	Tag
Unseen SNR	Train on code-inherited [-15,24]dB; evaluate only at {-25,-20} and {30,35}dB, never train on these	Gap 3: "zero papers in the Tier-1 set test genuine OOD robustness" [EVIDENCE: PAPER — Sec J audits] + this is the protocol's own explicit distribution-shift axis (§13)	[GENUINE GAP]
Unseen impairment level	Train with $\delta/\gamma \leq 5^\circ$ (matching Lloria/Meneses' tested ceiling); evaluate only at {10°,15°}	Meneses-Albalá 2026 is the closest approach (fine-tunes across severities up to 5°) but never generalizes past its own trained ceiling [EVIDENCE: PAPER]	[GENUINE GAP]
Unseen path count	Train $L \leq 6$; evaluate $L \in \{7,8,10\}$	No paper trains/tests beyond $L=6$	[GENUINE GAP]
Unseen angular range / end-fire density probe	Train on the code's actual angle distribution (Uniform[0,π] per Contradiction #1 resolution [EVIDENCE: CODE]); evaluate with a densified sample near end-fire ($\theta, \varphi \rightarrow 0^\circ$ and 180°)	Directly targets this project's own diagnosed Jacobian-singularity Pd ceiling (~0.972) — a project-only finding absent from every Lloria-family paper's limitations section despite sharing the identical output representation [EVIDENCE: EXPERIMENT — ceiling test + diffusion-sharpening ablation] (Gap 5). This is the highest-value distribution-shift test in the plan because it is the one condition where the project already has a <i>mechanistic</i> hypothesis (pixel quantization near end-fire, not blur) to confirm or falsify against the eventual candidate	[EVIDENCE: EXPERIMENT] + [HYPOTHESIS]
Unseen channel realization (coherent/correlated paths)	Optional, stretch only	SubspaceNet tests this for 1D DOA (NOT COMPARABLE) [EVIDENCE: PAPER]	[GENUINE GAP], Tier 4

5. Priority tiers (given single-RTX-5090 local compute + occasional Kaggle, and the Kaggle weight-loss incident already logged in prior-work §3 — every Tier 2+ run must be checkpointed with explicit "Save Version → Save & Run All" if run on Kaggle)

Tier 1 — cheap, literature-anchored, must run for every candidate + baseline registry (Teacher, magnitude-pruned Student, FNO, eventual final candidate):

1. SNR sweep, in-distribution core [-15,24]dB (5dB steps) — reproduces/extends Lloria & Meneses tables directly, HIGH comparability.
2. Phase error $\delta/\gamma \in \{0,1,2,5\}^\circ$ — reproduces Lloria & Meneses directly, HIGH comparability.
3. Path count $L \in \{1..6\}$ — reproduces Lloria directly, HIGH comparability.

Tier 2 — moderate cost, project-novel, builds on existing Notebook 4 machinery:

4. Angular separation sweep $\Delta\theta \in \{30^\circ \dots 1^\circ\}$ — licensed by Naoumi's own named gap.
5. Path-power-imbalance / pseudo-NLOS sweep — extends Notebook 4's existing nuisance-path generator.
6. Unseen-SNR extrapolation tails {-25,-20,30,35}dB — cheap, reuses Tier-1 infrastructure, first genuine OOD test in the whole evidence base.

Tier 3 — genuine literature gaps, higher implementation cost (new impairment-injection code):

7. Gain mismatch {0,0.5,1,2,4}dB — directly closes Meneses-Albalá's own named future-work item.
8. Array-element-position perturbation $\sigma \in \{0.01\lambda \dots 0.1\lambda\}$ — adapts SubspaceNet's mechanism to the joint-2D setting.
9. Unseen path count $L \in \{7,8,10\}$ and unseen impairment level {10°,15°}.
10. End-fire density probe — directly tests the project's own Jacobian-singularity hypothesis against the new candidate; run this whenever Stage-3 (differentiable set-prediction head, per `My_Proposed_Architecture.md`) is implemented, since that is the component specifically designed to remove this failure mode.

Tier 4 — optional/stretch, no literature anchor or out of current system-model scope:

11. RF-chain/calibration error, coherent/correlated paths, mobility/Doppler (explicitly out of scope: current system model is static estimation, Doppler is Lim et al. 2021's axis on a completely different task — LSTM beam tracking, not comparable per synthesis).

6. Reporting requirements for every condition (protocol §19 / §23 compliance)

- Every number must carry an evidence tag; nothing labeled a "robustness result" unless it came from an actual run ([EVIDENCE: EXPERIMENT]), never inferred or assumed.
 - Every cross-paper comparison must carry a COMPARABILITY level (HIGH/MEDIUM/LOW/NOT COMPARABLE), inherited from the synthesis's own comparability table (§1 above) — never silently upgraded.
 - Both paper-style and strict per-trial Pd reported at every condition (Gap 4).
 - Catastrophic-degradation check (protocol §24 "Robustness" selection criterion): flag any condition where Pd drops by more than, e.g., 50% relative to the matched in-distribution point — this threshold should be confirmed with the Mother Agent/Critical Reviewer before use, since no paper in the set defines a formal catastrophic-failure threshold [ASSUMPTION – needs Mother Agent sign-off] .
 - Genuine-gap conditions (§2b, §2c, §3c, §4 rows marked [GENUINE GAP]) must be explicitly labeled as such in the Final Report's Robustness Study section — these are the plan's actual novelty contribution and must not be quietly presented as "expected" literature reproductions.
-

7. Open items to hand back to Mother Agent

1. The catastrophic-degradation threshold (§6) is an assumption, not literature-derived — needs explicit adjudication.
2. Whether Tier 4 conditions (RF-chain mismatch, coherent paths, mobility) are in scope depends on the Architecture Researcher's (Agent-R1) final system-model decision — flagging as a cross-agent dependency, not resolving unilaterally, per protocol §21 (feed-forward rule).
3. The end-fire density probe (§4) should be sequenced *after* Stage-3 differentiable set-prediction is implemented, since it is the condition most likely to distinguish that architectural change from a capacity-only improvement — recommend the Experimental Design Researcher (Agent-R8) place it in the ablation plan alongside a "Stage-3 removed" variant.

Agent-R4 — Signal Processing Researcher

Agent-R4 (Signal Processing Researcher) — Front-End Compression Assessment

Framing

The question is the **inverse** of what the literature actually contains. Every classical+DL hybrid found in the synthesis (§2, Architecture Component Matrix) chains in one direction only: **DL front end** → **classical back end**.

- **SubspaceNet** (Shmuel et al. 2025): CNN-DCNN autoencoder learns a denoised/regularized surrogate covariance matrix from raw signal, then hands it to an *unmodified* classical subspace algorithm (MUSIC / Root-MUSIC / ESPRIT / MVDR) for the actual angle extraction. Result: 41,761 params (vs >21M for a black-box CNN doing the whole task) and ~60× RMSPE improvement over classical Root-MUSIC alone (12.48°→0.20°, M=2 coherent). The DL only fixes what breaks the classical method's assumptions (coherence, low SNR, miscalibration); the classical algorithm still does the estimation.
- **Koh & Lee 2026's** DAE+MUSIC entry in their 11-model BLE benchmark is the same pattern: learned denoising autoencoder front end, classical MUSIC back end.

Nobody in the reviewed set runs classical processing **first** to shrink a downstream *learned* estimator. That is a genuine, citable gap (extends Gap 6/7 in the synthesis) — the ideas below have no direct precedent to validate against, only adjacent evidence to reason from. I flag this explicitly rather than overstate confidence.

The strongest adjacent evidence for feasibility is actually **this project's own baseline**, not a paper: the classical 2D-DFT+SIC estimator reproduces **Pd 0.884 vs the 64-block ResNet's 0.925 at 20 dB — 95.6% of the DL model's accuracy, with zero learned parameters** (PROJECT_STATUS.md §3). That ~4pp gap is the *entire* burden currently placed on a 469,393-parameter network. If a classical stage can already close most of the distance, the DL backbone's job can shrink from "estimate the angles" to "correct the residual" — a strictly easier, lower-capacity task.

Candidate 1 — Beamspace/DFT-codebook pre-reduction + residual-correction backbone

Mechanism: Project the raw sensor-domain input onto a coarse DFT beamspace codebook (the same P×Q angular grid Lloria's ResNet/U-Net already outputs over) *before* the network, producing a low-resolution coarse heatmap via classical DFT+SIC (already implemented and validated in this project). Feed that coarse heatmap — not the raw antenna-domain tensor — into a much shallower backbone (e.g., 8–16 ResNet blocks instead of 64) whose only job is residual sharpening/de-aliasing around the classical peaks, not full-image angle detection from scratch.

Precedent: Naoumi et al. (GCWkshps 2023/JSTSP 2024) already do input-side classical reduction — a coarse-timing IFFT front end that cut multiplications 6.5×/10.3× vs. the full 2D parametric algorithm at Nr=8/16 — though that comparison is DL-vs-classical-cost, not a literal chained hybrid. Sub6GHz CNN/UNet (arXiv 2025) also feeds classically-pre-estimated AoA/AoD as *input* to a CNN/UNet (for a different downstream task, SE gain, not angle refinement) — validating architecturally that DL backbones can profitably consume classical-domain features rather than raw IQ.

Why it should shrink the backbone specifically: The dense/flatten bottleneck the synthesis's Gap 2 already flags as the *literature's own self-identified worst component* (Lloria's Table IV quadratic-in-(P'Q') term) exists partly because the network must learn a global mapping from raw-antenna to angle-space. Pre-placing the signal in angle-space via DFT removes that mapping burden entirely — directly attacking the same bottleneck Gap 2 targets, but from the input side rather than the architecture side. This is complementary to, not competing with, the FNO-bottleneck result (Gap 2/PROJECT_RESULTS.md).

Risk: The DFT-SIC classical stage's own O(PQ) cost must be counted in any total-latency claim — Gap 1 already notes *no* Tier-1 paper reports full pipeline params+FLOPs+latency together, so this would need fresh measurement, not an assumed win.

Candidate 2 — SubspaceNet-pattern learned covariance-correction + classical 2D-MUSIC back end, re-purposed for compression instead of accuracy

Mechanism: Instead of inverting SubspaceNet's direction, keep its direction (small NN → classical back end) but change the *goal*: SubspaceNet used a tiny NN (41.7K params) to make classical MUSIC accurate enough to compete with a black-box CNN. Here, use the identical pattern — small learned covariance-correction network feeding a **2D-MUSIC** (joint AoA/AoD extension) back end — as a wholesale *replacement* for Lloria's 469K-parameter heatmap network, since 2D-MUSIC's continuous spectral peak search is not confined to a 256×256 pixel grid.

Why this matters beyond just "smaller": This directly attacks **Gap 5**, the project's own strongest evidence-convergent finding — the diagnosed Pd ceiling (~0.972) caused by Jacobian singularity/pixel quantization in the heatmap+blob-detection representation, independently confirmed by the ceiling test and the diffusion-sharpening ablation (which ruled out blur, confirming *location* error is

the mechanism). A classical subspace back end has no pixel grid to quantize onto — it removes the failure mode's root cause architecturally, as an alternative (or complement) to the project's own not-yet-built DETR-style set-prediction Stage 3.

Precedent and its limits: SubspaceNet's 41,761-param model beats classical Root-MUSIC alone by $\sim 60\times$ RMSPE and is dramatically smaller than a 21M-param black-box CNN — direct, verified evidence this pattern scales down parameter count while *improving* accuracy over the pure-classical method, which is exactly the "shrink without sacrificing accuracy" ask. But SubspaceNet is **1D DOA only** (single ULA) — extending the joint-2D-covariance-correction + 2D-MUSIC pipeline to Lloria's joint AoA/AoD problem is unbuilt anywhere in the reviewed literature (this is literally Gap 6, unmodified). Also relevant: Gupta et al. 2025's MUSIC-based Bayesian pipeline is explicitly **not** joint (AoD eliminated by a ZF beamformer) and reports **no hardware-impairment/domain-shift test at all** — so a joint-2D-MUSIC extension inherits an untested robustness gap that would need to be closed before trusting it under Lloria-style phase-error conditions.

Estimated payoff: If SubspaceNet's ratio holds even loosely (41.7K vs teacher's 469K is an $\sim 11\times$ reduction while *improving* on the classical baseline), this is the single most aggressive shrink candidate of the three — but it is also the least de-risked, since nothing in the literature validates it for 2 angles simultaneously.

Candidate 3 — Classical coarse-support localization (ESPRIT / 2D-MUSIC) gating a cropped, narrow-field backbone

Mechanism: Run a cheap classical coarse localization pass (ESPRIT, or a low-resolution 2D-MUSIC scan) to identify the small angular sub-region(s) likely containing the true peaks — feasible because the project's own data generator already enforces a known minimum pairwise separation ($\pi/6$, code-only per Contradiction #11) and $L \in \{1..6\}$ well-separated sources. Crop the DL backbone's receptive field to operate only on that sub-region (e.g., a 32×32 window instead of the full 256×256 grid) rather than convolving over the whole angular space.

Precedent: This is architecturally closest to BeamSeek's single-RF-chain beam-switching constraint, which narrows the estimation problem before any learned component runs — though BeamSeek is 1D-azimuth-only and its own paper reports **no latency numbers at all** (Contradiction #7 explicitly flags a prior pipeline stage as having fabricated BeamSeek latency figures — excluded here). Naoumi's narrow $[20^\circ, 40^\circ]$ training/test window (GCWkshps 2023) is an *unintentional* version of the same idea (fixed rather than adaptive cropping), and its own authors flag that the well-separated-targets assumption breaks under limited temporal resolution — a direct warning sign for Candidate 3's closely-spaced-source failure mode.

Risk (most severe of the three): Coarse localization errors are compounding, not just costly — if ESPRIT misplaces the candidate window, the cropped backbone never sees the true peak at all (worse than a full-field miss, which a full-field model can sometimes still partially correct). Given the project's own diagnosed Jacobian-singularity problem is *worst near end-fire angles* (Gap 5), a coarse classical localization stage is likely to be *least reliable* exactly where the DL backbone most needs help — this candidate's risk profile is inversely correlated with where it would add value, which is a real concern, not a minor caveat.

Recommendation ranking

Candidate	Precedent strength	Novelty vs. literature	Directly attacks a named project gap	Risk
1. Beamspace/DFT pre-reduction + residual backbone	Medium (Naoumi, Sub6GHz UNet — adjacent, not identical)	Low-medium	Gap 2 (bottleneck), Gap 1 (report full pipeline cost)	Low — builds on already-validated project DFT-SIC baseline
2. SubspaceNet-pattern covariance-correction + 2D-MUSIC	High for the <i>pattern</i> (SubspaceNet verified numbers), zero for the <i>2D extension</i>	High (Gap 6, literally unaddressed anywhere)	Gap 5 (Pd ceiling root cause), Gap 6	Medium — pattern proven at 1D, unproven at 2D+joint AoA/AoD
3. ESPRIT/MUSIC coarse-crop gating	Weak/indirect (BeamSeek, Naoumi's narrow-window as unintentional analog)	Medium	Partial (compute reduction) but works against Gap 5's own failure geometry	High — compounding localization error, worst exactly where it's needed most

My assessment as Agent-R4: Candidate 1 is the safest near-term win — it reuses an already-validated project artifact (DFT-SIC at 95.6% of teacher Pd) and directly targets the literature's own self-flagged bottleneck component (Gap 2), with low implementation risk. Candidate 2 is the highest-ceiling idea and the most scientifically interesting because it could resolve Gap 5 architecturally rather than by patching around it (DETR-style set prediction), but it requires validating SubspaceNet's pattern at 2D joint-angle scale first — nobody has done this, so it should be scoped as an experiment, not assumed to work. Candidate 3 I would deprioritize or drop: its failure mode is anti-correlated with exactly the region (end-fire angles) where the project's evidence says help is most needed.

None of these should be read as "classical replaces DL" — consistent with the brief, all three keep DL as the accuracy-critical component and use classical processing only to reduce what the learned model has to do from scratch.

Agent-R7 — Novelty / Gaps Researcher

Agent-R7 — Novelty/Gaps Assessment (Skeptical Pass)

0. Method note and a hard caveat before any verdict

I ran the classification below against the 14-paper synthesis **and** four targeted supplementary web searches (DETR/Hungarian-matching + DOA; joint AoA/AoD lightweight DL 2025–2026; transformer/attention + DOA; classical-subspace + DL hybrids). The supplementary searches surfaced material the 14-paper corpus does **not** cover, some of it directly on-topic. This changes several verdicts from what the synthesis alone would suggest — the corpus's "literature-wide gap" framing (esp. for attention/Transformer use) is partly an artifact of a narrow paper-selection scope, not a true gap in the field. Details in §4. Every "NEW" verdict below is conditional on those items being checked by a dedicated literature agent before anyone writes "no prior work does X."

1. Direct answers to the framing questions

Has "a lightweight hybrid architecture for joint AoA/AoD combining known building blocks" already been published, in spirit? Yes, effectively. Naoumi et al. (GCWkshps 2023 / JSTSP 2024) already ship a genuinely lightweight (8,584-param) joint AoA/AoD estimator built from a known block (complex-valued MLP) plus a known classical algorithm for input reduction. Meneses-Albalá 2026 already does an edge-oriented, adapted variant of Lloria's joint 2D architecture on real hardware. So the bare claim "we built something lightweight that jointly estimates AoA/AoD by combining known parts" is **not new as a genre** — it's the default move this sub-field has already made twice.

Is combining two known architectures automatically novel? No. A reviewer's first question for any "Block A + Block B" paper is: what does the combination do that neither block does alone, and is there an ablation proving the combination — not just the union — is necessary? Simple concatenation/grafting of known components (e.g., "ResNet backbone + attention bottleneck", "pruning + a different backbone") is standard engineering iteration, not a contribution, unless it is (a) motivated by a *specific, diagnosed* failure mode of the existing components, (b) shown via ablation that the combination fixes that failure mode where either component alone does not, or (c) paired with a measurement/rigor contribution nobody else has made. Absent one of these, "combining known blocks" reads as re-labeling.

What would make a contribution here scientifically meaningful?

1. Tie the architectural choice to a *specific diagnosed root cause* already evidenced by this project's own experiments (e.g., the Jacobian-singularity/pixel-quantization ceiling), not a generic "let's try attention/FNO" swap.
2. Report a measurement axis genuinely absent from the field (params+FLOPs+**real hardware inference latency**+memory together, for the *joint 2D* task) — this is thin but real, low-effort novelty.
3. Run an evaluation protocol (OOD, multi-seed variance, strict-PD) that the field's self-reported numbers don't support, and show the architecture's ranking changes under it — that's a claim about *evidence quality*, which is defensible even when the architecture itself is unoriginal.
4. Where a combination is used, ablate it against each block alone on the identical task/data, so "hybrid beats components" is demonstrated, not asserted.

2. Candidate novelty angles — classification

#	Angle	Verdict	Justification
A	Generic "lightweight hybrid architecture for joint AoA/AoD, combining known blocks" (as literally stated in the brief)	KNOWN COMBINATION	Naoumi (8,584-param complex MLP, already lightweight, already joint) and Meneses2026 (edge-adapted joint 2D net) already occupy this exact genre. Standalone, this is not a defensible thesis contribution.
B	Params + FLOPs + real-hardware standalone inference latency + memory, reported together for a joint AoA/AoD model, using the project's inference-time/pilot-rate latency framing (Gap 1 + Gap 8)	PARTIALLY NEW	No Tier-1/2 paper reports all four together for the joint 2D task. But Meneses2026 already did real-edge-hardware profiling (Jetson) for this paper family — for training/fine-tuning <i>energy</i> , not clean inference <i>latency</i> . The novel slice is narrow (inference latency specifically, on a genuinely from-scratch-lightweight model) and is a measurement contribution, not an architectural one — fine as a supporting result, thin as a headline claim.
C	Replace literature's dense/flatten bottleneck (TVT2026's own self-flagged quadratic term) with FNO/spectral or attention block (Gap 2, Gap 9)	PARTIALLY NEW, conditional on §4	The specific weakness (dense bottleneck) and its fix direction (non-dense) are real and unaddressed within the 14-paper corpus, and the project already has an executed FNO result to build on. But §4 found multiple published lightweight/attention-based 2D-DOA and DOA architectures outside this corpus (Triple-Attention 2D-DOA for L-shaped arrays, DACL-Net, two-stage transformer DOA) — "attention works for compact angle estimation" is not a field-wide open question, only an open question <i>within this narrow joint-AoA/AoD sub-literature</i> . Frame the contribution as "first systematic dense-vs-non-dense bottleneck comparison for <i>joint 2D AoA/AoD specifically</i> ," not "first use of attention for angle estimation."

#	Angle	Verdict	Justification
D	Differentiable DETR-style set prediction + Hungarian matching to remove the non-differentiable peak-search step diagnosed as the Jacobian-singularity/quantization ceiling cause (Gap 5)	UNCERTAIN, most promising but unverified	This is the single most evidence-backed angle <i>inside</i> the project (three converging internal experiments point at the same root cause), and nothing in the 14-paper corpus or my supplementary search found DETR/set-prediction applied to DOA/AoA estimation specifically. But my search was not exhaustive — DETR-style set prediction is an extremely popular pattern across detection domains since 2020, and radar/array literature adopts CV techniques quickly; absence-of-evidence here is weak evidence of absence. Do not claim novelty on this angle without a dedicated, deeper search (e.g. "query-based DOA", "set prediction radar angle estimation", "learned NMS direction finding") before committing. If confirmed absent, this becomes the strongest angle in the set because it targets a mechanism, not a block swap.
E	SubspaceNet-style learned-covariance-surrogate + classical back-end, extended to joint 2D and combined with the project's Gridless-Unfold layer (Gap 6)	KNOWN COMBINATION at the pattern level, PARTIALLY NEW at the task level	"Model-based deep learning" (learned front end feeding a classical resolver) is an established paradigm (SubspaceNet is one instance of it; my search found others, e.g. deep-unrolling+GNN hybrids for DOA). Applying an established paradigm to a new task variant (2D instead of 1D) is a domain-transfer contribution, not a conceptual one — legitimate as an engineering result but weak as a "novelty" headline unless paired with the Gap 5 mechanism story.
F	Combine architecture search/screening with pruning/compression (Gap 7)	NOT NOVEL as a general technique; PARTIALLY NEW only as domain-first	NAS+compression combinations are mature, widely published in general efficient-DL literature. The only possible novelty is "first time applied to joint AoA/AoD," which is a thin, non-mechanistic claim.
G	Resolve the SIREN/Window-Attention LR/warmup confound (Gap 9) and report whether attention is viable for this bottleneck	UNCERTAIN (contingent, not a claim yet)	This is a diagnostic ablation, not a contribution by itself. It only becomes a novelty-relevant result if it flips the current near-zero-Pd outcome into a positive one and that positive result is then shown to beat the literature's dense/flatten baseline specifically for <i>joint 2D AoA/AoD</i> (per angle C's caveat — attention-for-DOA broadly is already known elsewhere).
H	Strict per-trial PD (missed detections counted as failures) reported alongside the field's own inflated convention (Gap 4)	PARTIALLY NEW	Real and easy to establish as absent from the corpus, but it is evaluation-hygiene, not an architectural or algorithmic contribution — useful as a credibility differentiator in a paper, not sufficient as its central claim.
I	Multi-seed / cross-training-run statistical significance reporting (Gap 10)	PARTIALLY NEW	Same category as H — genuinely absent from this literature, standard practice in ML broadly, so it reads as "finally doing due diligence" rather than a discovery.
J	Classical front-end (2D-DFT/MUSIC-style) feeding a lightweight DL refinement stage for joint 2D AoA/AoD (the "classical-front-end+lightweight-DL gap" named in the brief)	UNCERTAIN	Within the 14-paper corpus this direction (classical→DL, as opposed to SubspaceNet's DL→classical) is untested for the joint 2D task. But my supplementary search surfaced an explicit hybrid pattern in this exact space ("DNN-based beamformer + MUSIC subspace tracking," alternating DL/classical roles across time), which is close enough in spirit that a dedicated search specifically for "classical-front-end, DL-refinement, joint AoA AoD" is needed before treating this as open.

3. Prior art surfaced outside the given corpus — must be checked before any novelty claim is finalized

The literature-gathering stage scoped itself to 14 papers built around the Lloria/Naoumi/Gupta lineage. A quick supplementary search found several concretely on-topic items **not in that corpus**, which the R7 role is obligated to flag rather than let a downstream "novelty" claim rest on an incomplete search:

- **"Joint Single-Shot AoD/AoA Estimation in mmW Systems and Analysis Under Hardware Impairments" (IEEE, 2023, doc 10279062)** — joint AoA/AoD **and** explicitly studies hardware impairments. This directly overlaps Gap 3 (robustness/OOD) and possibly Gap 1 (the project's "nobody reports X" claims). Architecture/params unknown from title alone — **must be pulled and read** before any claim that "no paper tests joint AoA/AoD under hardware impairment" survives.
- **"Deep Machine Learning-Based AoD Map and AoA Map Construction for Wireless Networks" (IEEE, 2024, doc 10683141)** — another joint AoD/AoA paper missed by the corpus; unknown whether lightweight.
- **"SABER: Symbolic Regression-based Angle of Arrival and Beam Pattern Estimator" (arXiv 2510.26340, Oct 2025)** — very recent, and symbolic regression is by construction far lighter than any NN discussed in the synthesis. Directly threatens any "ours is the lightweight one" framing if not distinguished (different technique family, but same efficiency pitch).
- **Cluster of attention/transformer-based DOA and 2D-DOA papers** (DACL-Net; "Efficient 2D-DOA Estimation Based on Triple Attention Mechanism for L-Shaped Array"; "Two-Stage Transformer Framework for Sparse-Array DOA Estimation"; "Gridless coherent polarization-DOA estimation with dilated neighborhood attention and TCN") — none are joint AoA/AoD, but all directly undercut the synthesis's framing of "zero Tier-1 papers use attention" as evidence of a *field-wide* gap (Gap 9). It is a gap only in this specific narrow sub-literature. The Gridless+attention item is particularly relevant to Gap 6 (Gridless-Unfold combination ideas) and should be pulled.
- **"Hybrid deep unrolling and graph neural networks for super-resolution DOA in physics-informed antenna arrays" (Frontiers, 2026)** — relevant context for the PIA-Net concept: physics-informed deep-unfolding for DOA is not a blue-sky technique family; it already has other instances outside this project.
- Two entries titled close to "Deep Learning Based AoA and AoD Estimation for Millimeter Wave MIMO Systems" / "Deep-Learning-Based AoA and AoD Estimation in Analog Millimeter Wave MIMO Systems" (ACM 2024 / ResearchGate) may be duplicates or close variants of the Lloria paper family already in the corpus — needs a dedup check, not an assumption either way, before being cited as separate prior art.

None of this is definitive — I did not read any of these papers' full text, only titles/abstract snippets — but their existence means the corpus's implicit "we checked the literature" claim is **not exhaustive**, and any of the classifications above (especially C, D, and G, which lean on "the field hasn't tried attention/set-prediction here") could be downgraded further once these are actually read.

4. Bottom line

- The idea as stated in the brief ("lightweight hybrid combining known blocks for joint AoA/AoD") is, by itself, **not novel** — it restates work the field (Naoumi, Meneses) has already done.
- The only angle with a real, mechanism-level (not just block-swap) justification is **Gap 5's differentiable set-prediction head**, because it targets a specific, multiply-confirmed internal root-cause diagnosis rather than a generic architecture change — but it is currently **UNCERTAIN**, not confirmed novel, pending a dedicated DETR/set-prediction-for-DOA literature search.
- The **cheapest, most defensible, least risky** novelty available to this project right now is not architectural at all: it is the *measurement and rigor* bundle (Gaps 1, 4, 8, 10 — real inference latency + strict PD + multi-seed variance, reported together for the joint 2D task, which genuinely nobody in the reviewed set does). This should be treated as a supporting pillar of any paper, not its sole contribution, but it is close to zero-risk.
- Before this project commits to an architecture-novelty story, someone should (a) pull and read the four newly surfaced papers in §4, and (b) run a dedicated search for "query-based / set-prediction DOA" and "DETR angle estimation" beyond what this pass covered — both are cheap and materially change the confidence of the D and C/G verdicts above.

Agent-R8 — Experimental Design Researcher

Agent-R8 — Experimental Design Researcher: Evaluation Matrix for Final Candidate Architecture

Grounded in the Paper-Head literature synthesis (6 Tier-1 + 8 Tier-2 papers) and the Previous-Result Analysis of `D:\ai_ml_project`, and verified directly against this project's code (`DL_D0A/src/TVT_Blob_Inference.py`, `scripts/generate_frozen_banks.py`, `DL_D0A/src/tvt_data_generation_v3.py`, `baseline_models/README.md`, `notebooks/*`).

Verification note on the brief itself: the computed task text asserted FLOPs/latency measurement "does not currently exist in this project's notebooks." Code inspection shows this is *partially* wrong and I'm correcting it rather than passing it upward uncritically (per Mother Agent rule 1–7): `notebooks/proposed_pin_architecture.ipynb` already has a wall-clock `benchmark_latency()` function (`time.perf_counter`, 20 runs, single-sample batch, TF `.count_params()`), used for ResNet/U-Net/PIA-Net. FLOPs/MACs counting, however, is genuinely absent project-wide (no `thop` / `ptflops` / `fvcore`, no manual FLOP-counting code found anywhere). This distinction matters for what's "reuse" vs "new instrumentation" below.

0. What the existing pipeline gives you for free vs. what must be built new

`evaluate_on_bank(model, data_arr, feat_arr, meta_arr, batch_size, max_deg_error=1.0)`, redefined identically across `DLD0A_Architecture_FullEval_Local.ipynb`, `DLD0A_Compression_0{1,2,3}*.ipynb`, `snr.ipynb`, built from the original (non-reimplemented) `TVT_Blob_Inference.py` primitives (`get_blob_detector`, `get_blob_peaks`, `peaks_to_angles`, `prepare_for_metric`, `get_ang_difference`, `filter_angles`):

Given directly by <code>evaluate_on_bank</code> on <code>frozen_banks/eval_bank.npz</code> (reuse as-is)	Confirmed by code
Combined angular RMSE per SNR (8 points, -10...25dB step 5)	<code>get_ang_difference</code> flattens ψ (AoA) and φ (AoD) errors together (<code>H = angle(exp(i*gt)*exp(-i*pred))</code> , <code>.flatten()</code>) — RMSE is AoA+AoD pooled, not per-axis
Pd per SNR, paper-style convention	<code>filter_angles(ang_dif_flat, max_deg_error=1.0)</code> splits good/bad at 1°; trials where <code>peaks_to_angles</code> under-detects (<code>np.isnan(pred).any()</code>) are continued — excluded from both numerator and denominator , reproducing the exact non-standard convention flagged as literature Contradiction 5 / Gap 4
Deterministic, cross-hardware-reproducible (13+ sig-figs, GPU vs CPU)	<code>baseline_models/README.md</code>
Params via <code>model.count_params()</code>	Already used per-model in the registry and in <code>proposed_pin_architecture.ipynb</code>
A crude wall-clock inference latency benchmark	<code>benchmark_latency()</code> in <code>proposed_pin_architecture.ipynb</code> , TF/Keras only, 1-sample batch, mean $\pm$ std over 20 runs

Fixed by construction in `frozen_banks/eval_bank.npz` (per `scripts/generate_frozen_banks.py`): `L=[3]` (single path count, no multipath sweep), `P=Q=16`, `nt=nr=16`, `sigma=0.07`, phase/gain error = 0 (the generator is called with no impairment kwargs). **Only the SNR axis varies.** So every metric that isn't "RMSE/Pd vs SNR at `L=3`, no impairment" needs a **new bank** — not new code, since `validation_data_generator` (imported unmodified from `tvt_data_generation_v3.py`, already supports `error_deg` for both TX and RX beamforming per `create_tx_bf_matrix` / `create_rx_bf_matrix`) already supports it. This is "new instrumentation = a new `generate_*_bank.py` call with different conditions," not new physics code.

Missing from the pipeline entirely (new instrumentation required)	Why
Separate AoA-only RMSE / AoD-only RMSE	<code>get_ang_difference</code> pools both axes before RMSE; needs a variant that keeps ψ/φ separate before the <code>sqrt(mean(x²))</code>
MAE	Trivial add (<code>mean(abs(good_angles))</code>) alongside existing RMSE — same arrays already computed
P95 angular error	Not computed anywhere; must decide whether it's computed over <i>all</i> trials (including currently-discarded misses) or only "good" ones — see §6 design note, this materially changes the number
Strict/per-trial PD (missed detections counted as failures)	Requires not continue-ing NaN trials — a ~3-line change to <code>evaluate_on_bank</code> , but a real code change, and the two PD numbers must be reported side-by-side per Gap 4
False-detection rate	Not measured — current pipeline only checks "is a true angle matched within 1°," never checks for spurious <i>extra</i> peaks (over-detection). Needs new logic against <code>get_blob_peaks</code> ' raw peak count vs <code>L</code>
FLOPs/MACs	No tool anywhere in repo; needs <code>thop</code> / <code>ptflops</code> / <code>fvcore</code> (PyTorch) or manual layer-wise FLOP accounting (TF/Keras) added fresh
Model size on disk (MB)	Trivial (<code>os.path.getsize</code> on <code>.h5</code> / checkpoint) but not currently tabulated anywhere except informally
Peak/steady memory (RAM/VRAM)	Not measured anywhere; needs <code>tf.config.experimental.get_memory_info</code> or <code>torch.cuda.max_memory_allocated</code> instrumentation

Missing from the pipeline entirely (new instrumentation required)	Why
Rigorous end-to-end latency + throughput	Existing <code>benchmark_latency</code> is single-sample, TF-only, no batch-size sweep, no CPU-vs-GPU split, no percentile (p50/p99) reporting
Multipath ($L \neq 3$) sweep	Frozen bank fixes $L=3$; needs a new bank with <code>L_VALUES=[1,2,3,4,5,6]</code> (matches Lloria TVT2026/PIMRC2024's own $L \in \{1..6\}$ sweep for direct comparability)
Angular separation sweep	Not parameterized in current generator calls at all; would need explicit control over path angle spacing at generation time (check whether <code>validation_data_generator</code> exposes this — if not, this is genuinely new generator logic, not just new conditions)
Phase/gain error sweep	Generator supports <code>error_deg</code> (phase) already; gain error is not implemented anywhere found (<code>create_tx_bf_matrix</code> / <code>create_rx_bf_matrix</code> only inject phase, not amplitude, perturbation) — gain-error robustness is a real gap, matches literature Gap: Meneses-Albalá 2026 also flags gain error as future work, so this project would be first to close it
Array perturbation (element position/mutual coupling)	Not implemented; SubspaceNet is the only literature reference point (§3 Robustness Matrix) — would need new steering-vector perturbation code
Distribution shift (OOD SNR/ L /angle-range)	Not implemented as a <i>held-out</i> protocol; Notebook 4's nuisance-interference test is the closest existing precedent (novel vs. literature, reusable methodology) but tests interference power, not train/test distribution mismatch

1. Accuracy metrics

Metric	Definition	Reuse vs. new	Comparable to (from literature matrix)	Comparability level
AoA RMSE (ψ only)	RMSE of matched ψ errors only, degrees	New: split <code>get_ang_difference</code> output by axis before pooling	Lloria TVT2026/PIMRC2024 report combined RMSE, not split — so even after building this, cross-paper comparison stays approximate	LOW vs. Lloria (they don't split either); enables project-internal ablation at HIGH confidence
AoD RMSE (φ only)	Same, φ axis	New (same code path)	Same caveat	Same
Combined RMSE (current convention)	Pooled $\psi + \varphi$ angular error, deg	Reuse directly — <code>evaluate_on_bank</code>	Lloria TVT2026 Table II (ResNet .253, U-Net .230 @20dB); Meneses-Albalá 2026 Table 2 (.27–.28)	HIGH vs. Meneses-Albalá (identical generator lineage, identical convention); MEDIUM vs. Lloria TVT2026 (same family, PH vs. blob detector differs in PIMRC2024 only — TVT2026 also uses blob)
MAE	<code>mean(abs(error))</code> on same matched-angle arrays	New (trivial, same data)	Naoumi (JSTSP2024/GCWkshps2023) report MSE in rad^2 , not MAE in degrees — unit mismatch	NOT COMPARABLE cross-paper (different units/metric family); project-internal only
P95 angular error	95th percentile of <code>\ error\ </code>	New — must decide denominator : all trials (harsher, penalizes misses) vs. matched-only (current pipeline's implicit convention). Recommend reporting both , mirroring the strict-vs-paper-style PD split below, since no paper in the set reports P95 at all — this is a project-original robustness-of-tail-error metric	No paper reports P95	Not comparable to literature; useful as an internal tail-risk indicator, directly relevant to Gap 5 (the Jacobian-singularity end-fire ceiling shows up in tail error, not mean RMSE)

2. Detection metrics

Metric	Definition	Reuse vs. new	Comparable to	Comparability level
PD (paper-style)	Fraction of matched-angle pairs within 1° threshold, computed only over trials where enough peaks were found (current convention)	Reuse directly	Lloria TVT2026 Table II; Meneses-Albalá 2026 Table 2	HIGH vs. Meneses-Albalá; MEDIUM vs. Lloria TVT2026 (same convention family, per Contradiction 5)
PD (strict/per-trial)	Same threshold, but missed-detection trials counted as failures in the denominator	New (remove the <code>continue</code> on NaN trials; ~3-line change)	No paper reports this — it's the literature-gap-closing metric identified in Gap 4	Not comparable to any paper directly, but should be reported <i>alongside</i> paper-style PD on every table so the gap between the two quantifies how much the paper-style convention inflates results
Missed-detection rate	$1 -$ (fraction of GT angles matched at all, regardless of 1° threshold)	New — needs peak-count vs. L bookkeeping before the 1° filter	No direct literature analogue at this granularity	Project-internal
False-detection rate	Fraction of detected peaks beyond L per scene (over-detection)	New — not measured anywhere currently	No paper reports this cleanly either (most fix L a priori and don't test over-detection)	Project-internal, but relevant since a lighter model's blob detector could hallucinate peaks under noise — this is a genuine failure mode the current pipeline is blind to

3. Efficiency metrics

Metric	Definition	Reuse vs. new	Comparable to	Comparability level
Params	<code>model.count_params()</code> (TF) / <code>sum(p.numel())</code> (PyTorch)	Reuse directly — already tabulated in <code>baseline_models/README.md</code> (Teacher 469,393 / Student 314,513 / FNO 334,321)	Naoumi GCWkshps2023 (8,584, exact); SubspaceNet (41,761, exact)	HIGH as a raw number (params are architecture-agnostic), but task framing differs (Naoumi's 8,584-param MLP solves a much lower-dimensional bistatic problem; SubspaceNet's 41,761 is 1D DOA) — report side-by-side but flag task mismatch explicitly, don't claim "beats Naoumi 55×" without that caveat
FLOPs/MACs	Forward-pass multiply-accumulates	New — no tool in repo. Recommend <code>ptflops / thop</code> for a PyTorch reimplement, or manual per-layer accounting for the existing TF/Keras models ($\text{Conv2D FLOPs} = 2 \times k_h \times k_w \times C_{in} \times C_{out} \times H_{out} \times W_{out}$)	Naoumi GCWkshps2023 (4.321 MMACs fwd, exact); Naoumi JSTSP2024 (headline 6.5×/10.3× reduction vs. classical 2D algo, no absolute number); Lloria TVT2026 (asymptotic O(PQ) only, no concrete FLOPs)	HIGH vs. Naoumi GCWkshps2023 numerically once built (both report a single fwd-pass MACs number); this project would be the <i>first</i> to report FLOPs for the Lloria architecture family at all — directly closes Gap 1
Model size (disk, MB)	Serialized weights file size	New (trivial — <code>os.path.getsize()</code>) but not currently tabulated anywhere as a standing metric	No paper reports this	Project-internal, cheap
Peak memory (RAM/VRAM, MB)	Peak allocation during a forward pass at a fixed batch size	New — needs <code>tf.config.experimental.get_memory_info('GPU:0')</code> or <code>torch.cuda.max_memory_allocated()</code> instrumentation, none present today	No paper reports this at all (only Meneses-Albalá reports <i>energy</i> , not memory)	No literature comparison possible; still needed for the "lightweight" claim in the research objective (§0 of the protocol lists memory explicitly)
Inference latency (ms, single-sample)	Wall-clock per forward pass	Partially reuse — <code>benchmark_latency()</code> pattern from proposed <code>pin architecture.ipynb</code> ; needs generalizing to the current baseline registry's models (currently only wired for ResNet/U-Net/PIA-Net in that one notebook) and to whichever framework the final candidate uses	No Tier-1 paper reports concrete latency (all asymptotic-only); Meneses-Albalá 2026 reports training/fine-tuning energy on Jetson Orin Nano , not inference latency	NOT COMPARABLE to any paper's number directly — this is Gap 8: reuse Meneses-Albalá's <i>profiling methodology</i> (real edge hardware) but for genuine standalone inference latency, since no paper (including theirs) reports it
End-to-end latency	Forward pass + blob-detection + peak-to-angle conversion (the <i>whole</i> pipeline a deployed system would pay)	New — <code>benchmark_latency</code> only times the NN forward pass, not <code>get_blob_peaks / peaks_to_angles</code> , which run on CPU via OpenCV and could dominate at small model sizes	No paper measures this either (the blob-detector cost is universally ignored in the literature)	Project-original metric; directly relevant if Stage-3 differentiable set-prediction (<code>My_Proposed_Architecture.md</code> , targeting Gap 5) replaces the non-differentiable blob detector — this metric is exactly what would demonstrate that stage's efficiency payoff, not just its accuracy payoff
Throughput (samples/sec)	Batch-size-swept, steady-state	New — no batch-size sweep exists today (<code>benchmark_latency</code> is fixed at batch=1)	No paper reports this	Project-internal

4. Robustness conditions

Condition	Frozen-bank coverage today	Reuse vs. new	Literature grounding	Comparability if built
SNR sweep	Full: $-10\dots25$ dB, step 5 (8 pts)	Reuse directly	Matches Lloria TVT2026's <i>code</i> range exactly (paper text says $-10\dots25$, code says $-15\dots24$ — Contradiction 4); matches Meneses-Albalá 2026's 8-point grid exactly	HIGH vs. Meneses-Albalá; note the text-vs-code range discrepancy explicitly whenever quoting "matches Lloria's range"
Multipath (L)	Not covered — bank fixes $L=[3]$	New — regenerate bank with $L_VALUES=[1,2,3,4,5,6]$, reusing the same unmodified <code>validation_data_generator</code>	Lloria TVT2026/PIMRC2024 both sweep $L\in\{1..6\}$	HIGH once built (same generator lineage, same L range)
Angular separation	Not parameterized at all in current generator calls	New — check whether <code>validation_data_generator</code> exposes angle-spacing control; if not, this needs new generator logic, not just new call args (flag before assuming it's cheap)	Naoumi JSTSP2024 sweeps #targets-per-peak (proxy for separation); Naoumi GCWkshps2023 explicitly flags well-separated-target assumption as untested at failure — a project result here would directly speak to a literature-acknowledged weakness	MEDIUM (different system framing — bistatic ISAC vs. this project's single-user analog BF, per literature's own LOW-comparability note on Naoumi-vs-Lloria)
Phase error	Generator supports it (<code>error_deg</code> param, TX+RX), frozen bank has it at 0	New bank, existing code — reuse <code>create_tx_bf_matrix / create_rx_bf_matrix</code> unmodified, generate at $\delta_{\max}\in\{1,2,5\}^\circ$	Lloria TVT2026 ($\delta_{\max}\in\{1,2,5\}^\circ$, exact match); Meneses-Albalá 2026 ($\gamma_{\max}\in\{1,2,5\}^\circ$, PRIMARY focus, exact match)	HIGH — literally the same three severity levels as both papers, same generator lineage
Gain error	Not implemented anywhere in the codebase	New — genuinely new physics code (amplitude perturbation on <code>create_tx_bf_matrix / create_rx_bf_matrix</code> , not present)	Meneses-Albalá 2026 explicitly flags gain error as future work (their own gap); no Tier-1 paper tests it	Would be a literature-first if built — no comparability baseline exists, but strengthens the "this project closes documented gaps" narrative
Array perturbation	Not implemented	New — steering-vector/element-position perturbation, no existing code found	SubspaceNet only (1D DOA, not joint 2D)	NOT COMPARABLE (different task per literature's own note) — project-internal robustness signal only
Distribution shift	Not implemented as held-out train/test split; closest precedent is Notebook 4's nuisance-interference test (different axis: interference power, not train/test mismatch)	New protocol — reuse Notebook 4's <i>methodology</i> (200 scenes, sweep design) but redefine the swept variable as unseen SNR/L/angle-range rather than nuisance power	Zero Tier-1 papers test genuine OOD (Gap 3); this is the single most literature-differentiating robustness axis available	Not comparable to any paper (none report it) — but exactly the axis that would make this project's evaluation more rigorous than the entire reviewed literature

5. Ablation plan (remove one component at a time)

Grounded in `My_Proposed_Architecture.md`'s 3-stage design (Stage 1: physics-informed deep-unfolding sparse recovery; Stage 2: subspace attention/transformer; Stage 3: differentiable DETR-style set prediction + Hungarian matching) and the project's own screened core blocks (FNO, GridlessUnfold, SIREN, WindowAttention):

Ablation	What's removed	Predicts / tests	Metrics to watch	Reuse vs. new eval
Full candidate	— (baseline)	—	All of §1–§3	Reuse <code>evaluate_on_bank</code> for accuracy/PD; new for efficiency
– Stage 3 (set-prediction head, revert to blob detector)	Differentiable DETR-style query + Hungarian matching → back to non-learned peak search	Directly tests whether Stage 3 is what fixes the diagnosed Jacobian-singularity Pd ceiling (~ 0.972 , Gap 5) — the single most evidence-convergent unbuilt idea in the project	PD (paper-style + strict), P95 angular error near end-fire angles specifically (not just mean RMSE — the ceiling is a <i>tail</i> effect)	Reuse <code>evaluate_on_bank</code> for the blob-detector variant; the set-prediction variant needs a new eval path since its output isn't a heatmap for <code>get_blob_peaks</code> to consume
– Stage 2 (subspace attention/transformer)	Removes attention block, keep Stage 1+3	Tests whether attention contributes given Gap 9 (zero Tier-1 papers use attention at all) and the project's own unresolved SIREN/WindowAttention confound (§7 item 1 of prior-work analysis — untuned LR/warmup, not a settled negative)	Accuracy metrics + params/FLOPs delta (attention is usually FLOP-heavy relative to its param count)	Reuse accuracy pipeline; FLOPs comparison is new instrumentation either way

Ablation	What's removed	Predicts / tests	Metrics to watch	Reuse vs. new eval
– Stage 1 (physics-informed deep-unfolding front end)	Revert to raw Y as direct backbone input	Tests the PIA-Net dual-branch fusion idea in isolation from the 4 known implementation bugs (BUJHO_SHOHOJ_VABE.md) — only valid once Stage 1 is <i>bug-free and retrained</i> , since all current PIA-Net numbers are known-stale (unresolved item, \$7.4 of prior-work analysis)	RMSE/PD at low SNR specifically (physics priors should help most where noise dominates)	Reuse
– dense/flatten bottleneck → non-dense (FNO/conv) bottleneck	Swaps the literature's self-flagged worst component (TVT2026 Table IV's only quadratic-in-(P'Q') term)	Direct extension of the project's own already-executed FNO screening result (0.6235 mean Pd, 88% of teacher, from scratch, Gap 2)	Params, PD, RMSE — this ablation already has a partial answer in RESULTS.md ; treat as replication/extension, not a fresh unknown	Reuse <code>evaluate_on_bank</code> fully — no new instrumentation needed, this is the cheapest ablation to run
– pruning (compare full-width vs. pruned candidate)	Channel pruning (magnitude or SNR-aware) applied post-hoc to the final candidate	Tests Gap 7 — architecture search and compression have never been combined in this project or the literature; also resolves the open loop of recovering/re-running the SNR-aware student (\$7.3 of prior-work analysis) at full scale	Params, PD, robustness-test delta under Notebook 4's nuisance-interference protocol (the only point in the project where a compressed model beat its teacher)	Reuse <code>evaluate_on_bank</code> + reuse Notebook 4's robustness protocol directly
– learned front end, signal-processing-only front end	Tests the Signal Processing Researcher's (Agent-R4) question: can a classical front end (DFT codebook / covariance features) shrink the DL model while preserving accuracy	Params, FLOPs, PD	New FLOPs instrumentation required regardless of which ablation	

Note on ablation validity: every ablation above must be evaluated on the **identical** `frozen_banks/eval_bank.npz` (or its multipath/phase-error/gain-error extensions once built) with the **identical** `evaluate_on_bank` call, matching the prior-work analysis's explicit warning (§5) not to conflate the stricter per-source Hungarian-matching metric used in early exploratory notebooks with the paper-Table-II-comparable metric used in `baseline_models/`. Ablations involving Stage 3 (set-prediction head) are the one exception requiring a genuinely different eval code path, since its output format isn't a heatmap.

6. Design decisions this task surfaced that need a Mother Agent / Validation Head ruling before implementation

- P95 and MAE denominator:** compute over all trials (including currently-discarded misses, penalizing them at a large fixed error) or only over matched/"good" trials (current implicit convention)? These give materially different numbers. Recommend reporting both, exactly as recommended for PD (paper-style vs. strict), for consistency.
- Strict-PD code change:** removing the `continue` on NaN trials in `evaluate_on_bank` changes a function reused by 5+ existing notebooks. Recommend adding a `strict: bool` parameter rather than mutating shared behavior, so all existing paper-comparable numbers in `baseline_models/README.md` remain reproducible unchanged.
- Angular-separation sweep feasibility:** unconfirmed whether `validation_data_generator` exposes angle-spacing control at all — this should be checked directly (read `tvf_data_generation_v3.py`'s generator signature in full) before this axis is scheduled as "cheap new instrumentation" in any experiment timeline; it may be classical-algorithm-level new work rather than a new bank-generation call.
- Gain-error and array-perturbation robustness** are real new-physics-code items (not present anywhere in the repo), not new-config items like multipath/phase-error — timeline estimates should reflect that difference explicitly.

Files consulted

D:\ai_ml_project\DL_DOA\src\TVT_Blob_Inference.py, D:\ai_ml_project\scripts\generate_frozen_banks.py,
D:\ai_ml_project\DL_DOA\src\tvf_data_generation_v3.py, D:\ai_ml_project\baseline_models\README.md,
D:\ai_ml_project\My_Proposed_Architecture.md, D:\ai_ml_project\notebooks\proposed_pin_architecture.ipynb,
D:\ai_ml_project\notebooks\DLDOA_Architecture_FullEval_Local.ipynb, D:\ai_ml_project\MULTI-AGENT RESEARCH LOOP ENGINEERING PROTOCOL.md (§16).